

사이클별 지표 패널

M02Ch103

각 패널 = 한 지표의 cycle 궤적 (routine 0.5C).

early/late 점선 = 앞·뒤 5포인트 median.

트렌드 = 선형 slope/100cyc + aging 기대방향 대조.

주의: LAM_PE / *_proxy / pattern_score는 가설·proxy이며 절대 LAM%/LLI%가 아닙니다. R_ohmic은 $\sqrt{t}$ 외삽 proxy (0.1s 단일값 ≠ 순수 음).

트렌드 요약 — M02Ch103

- 유효 지표: 358개 · aging 방향 일치 47 · 반대 10

하락 트렌드 (상위)

- 국소 CE(20): early=528.7 → late=273.2 ($\Delta=-255.4$, $-85.37\%/100\text{cyc}$) → decreasing · matches_aging
- 방전후 휴지 완화 τ : early=369.3 → late=291 ($\Delta=-78.3$, $-37.29\text{s}/100\text{cyc}$) → decreasing · context
- 방전후 완화 τ Δ vs기준: early=9.884 → late=-68.41 ($\Delta=-78.3$, $-37.29\text{s}/100\text{cyc}$) → decreasing · context
- 충전 에너지: early=243.6 → late=204.5 ($\Delta=-39.03$, $-14.6\text{Wh}/100\text{cyc}$) → decreasing · matches_aging
- LAM 곡선 proxy: early=-3.11 → late=-17.48 ($\Delta=-14.37$, $-5.717\%/100\text{cyc}$) → decreasing · context

상승 트렌드 (상위)

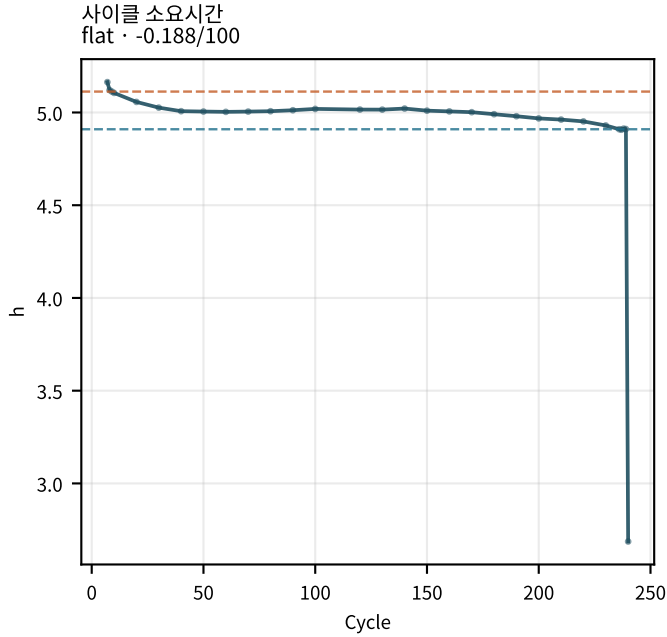
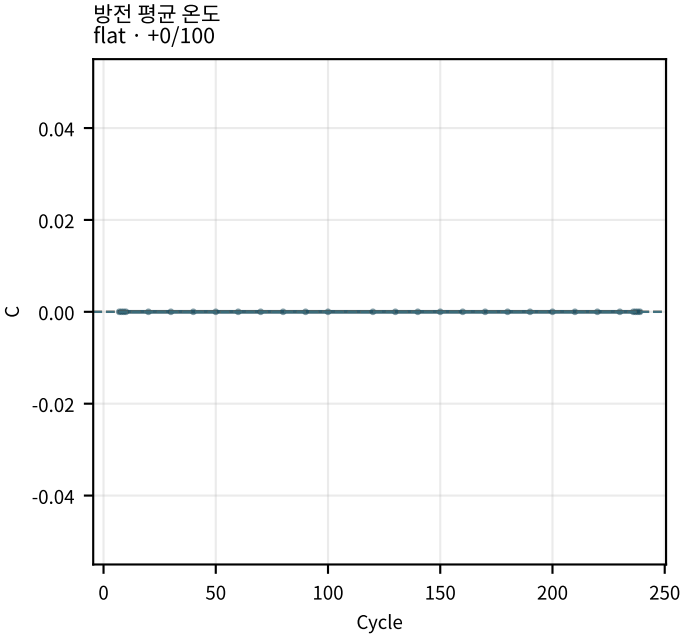
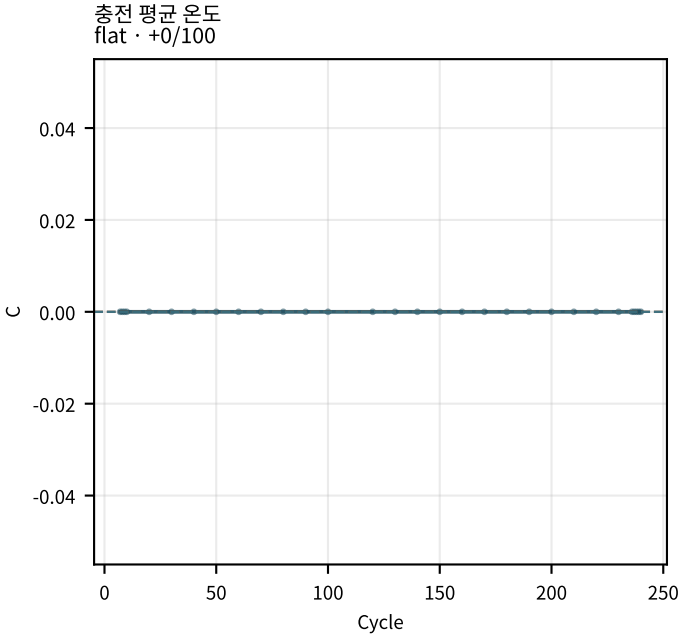
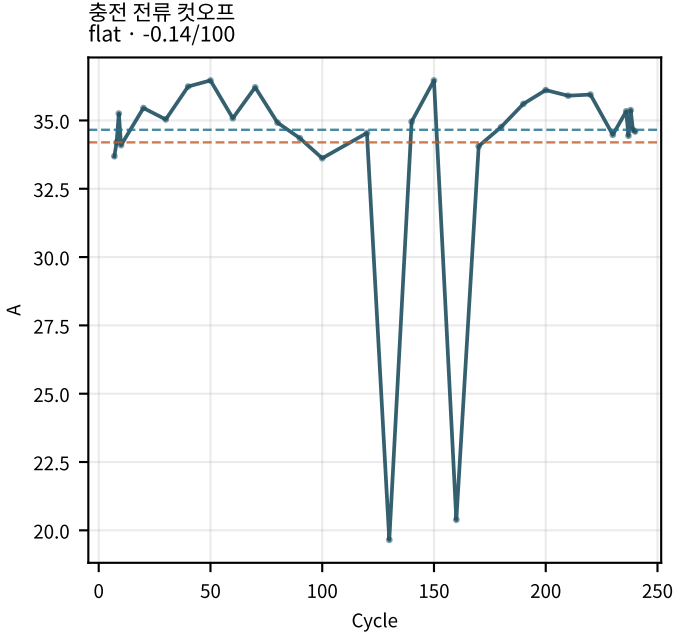
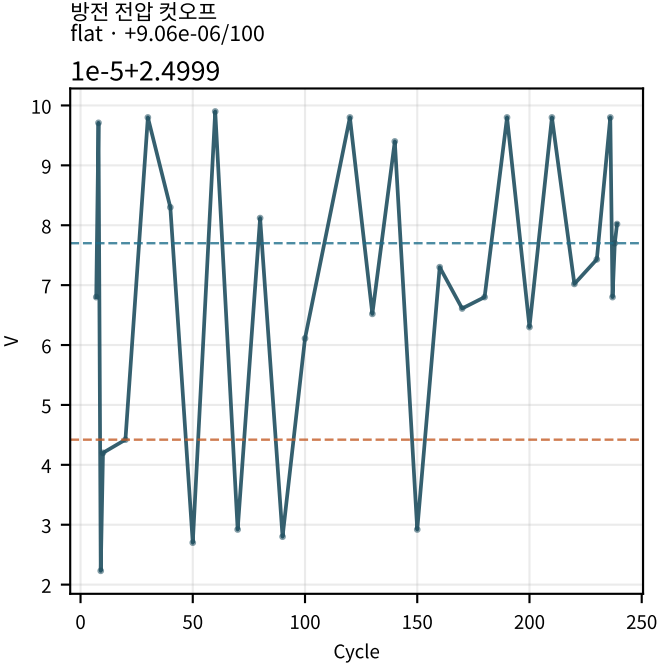
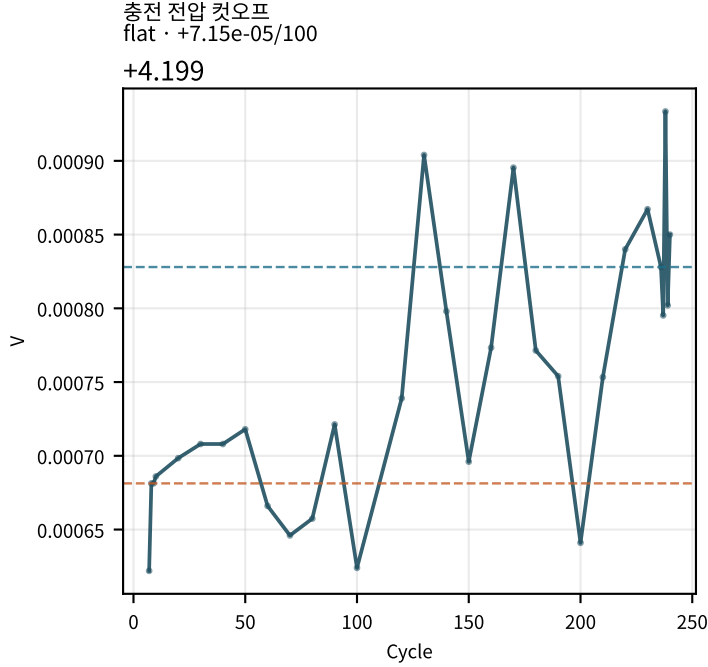
- fit 잔차 max: early=55.92 → late=152.7 ($\Delta=+96.79$, $+42.63\text{mV}/100\text{cyc}$) → increasing · matches_aging
- EoC 방전 10s 증가%: early=28.76 → late=49.05 ($\Delta=+20.29$, $+13.58\%/100\text{cyc}$) → increasing · matches_aging
- EoC 방전 30s 증가%: early=22.07 → late=43.82 ($\Delta=+21.75$, $+12.44\%/100\text{cyc}$) → increasing · matches_aging
- EoC 방전 60s 증가%: early=17.67 → late=38.96 ($\Delta=+21.29$, $+11.32\%/100\text{cyc}$) → increasing · matches_aging
- CV 시간: early=0 → late=0 ($\Delta=+0$, $+8.492\text{s}/100\text{cyc}$) → increasing · matches_aging

CycleDiag : plot_cycle_metrics_panel.py

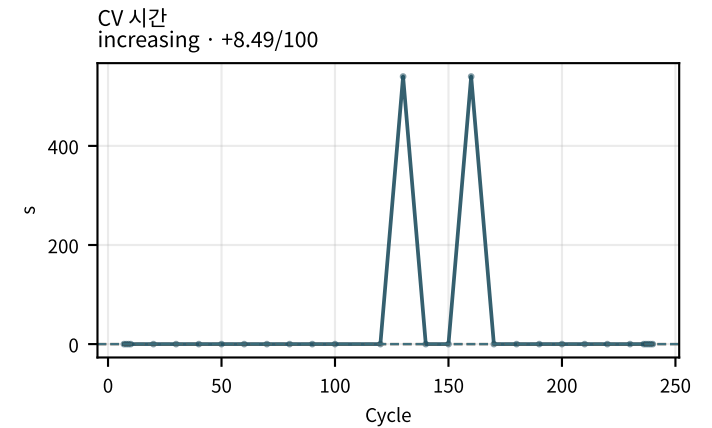
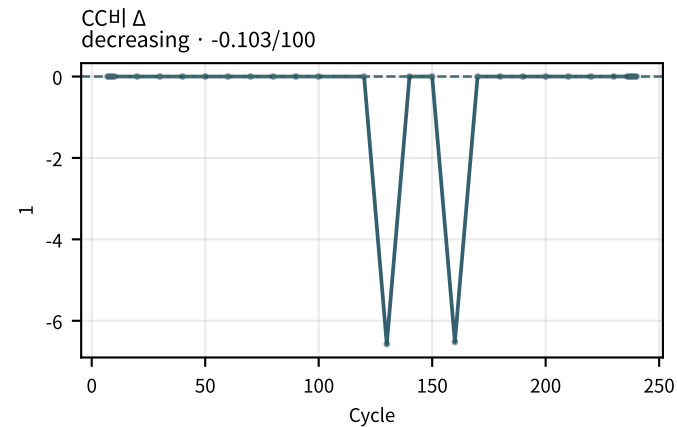
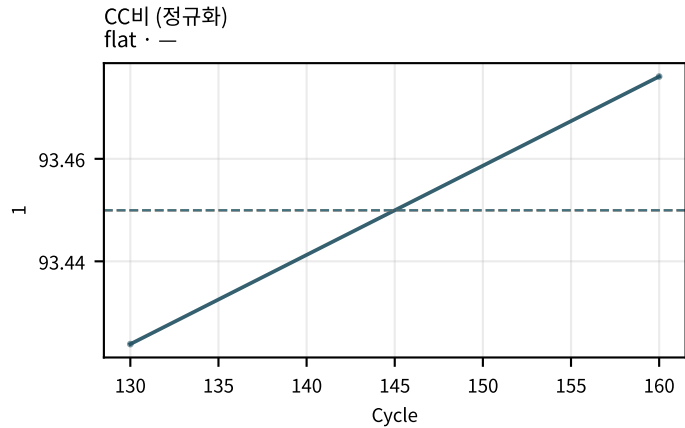
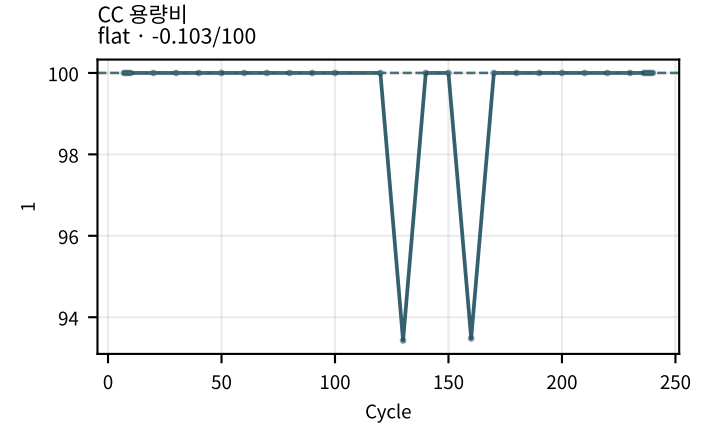
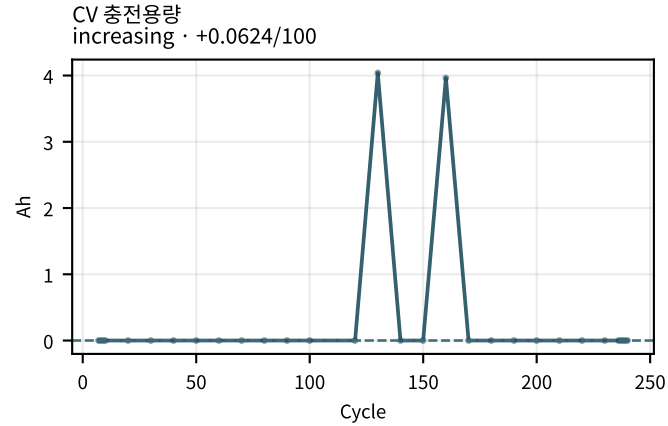
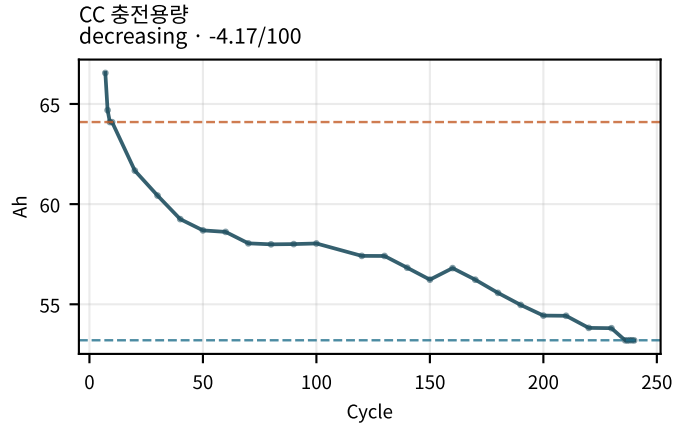
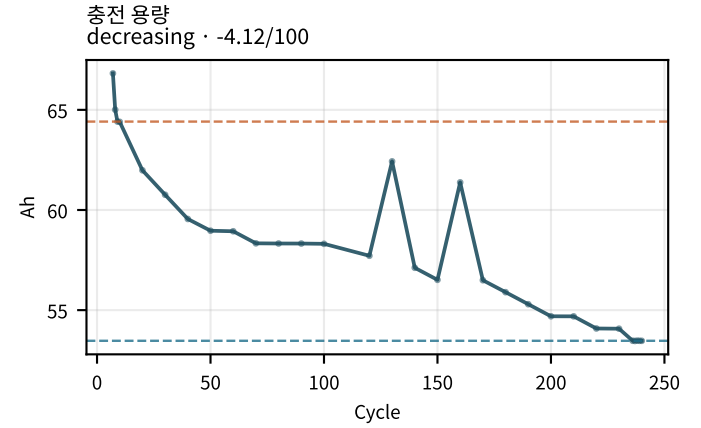
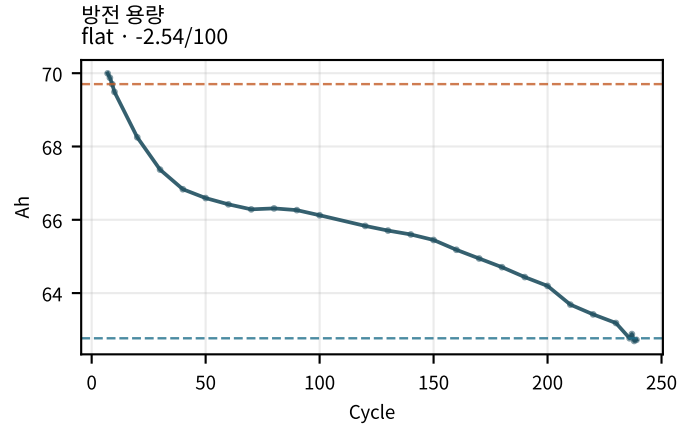
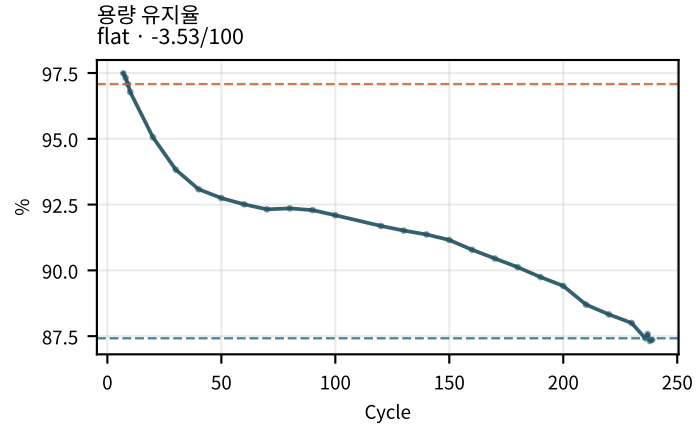
기대aging과 반대

- 쿨롱 비효율: early=-7.902 → late=-17.31 ($\Delta=-9.409$, $-3.596\%/100\text{cyc}$) → decreasing · opposite_aging
- CI /시간: early=-1.548 → late=-3.525 ($\Delta=-1.977$, $-0.7513\%/h/100\text{cyc}$) → decreasing · opposite_aging
- 에너지 손실: early=8.32 → late=-1.791 ($\Delta=-10.11$, $-3.476\text{Wh}/100\text{cyc}$) → decreasing · opposite_aging
- dSoH/dN: early=-0.1665 → late=-0.02386 ($\Delta=+0.1426$, $+0.1581\%/cyc/100\text{cyc}$) → increasing ·

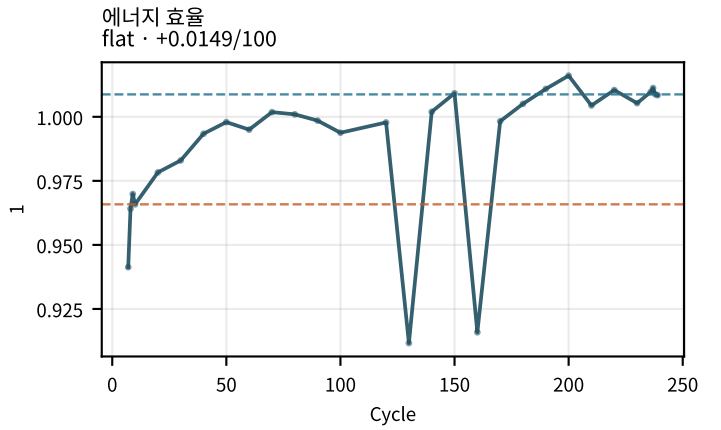
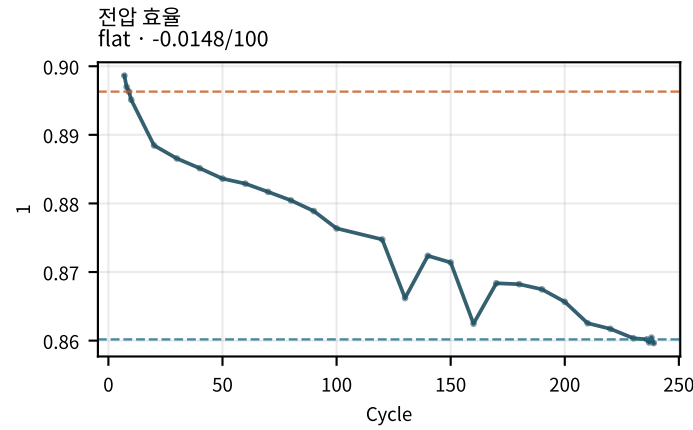
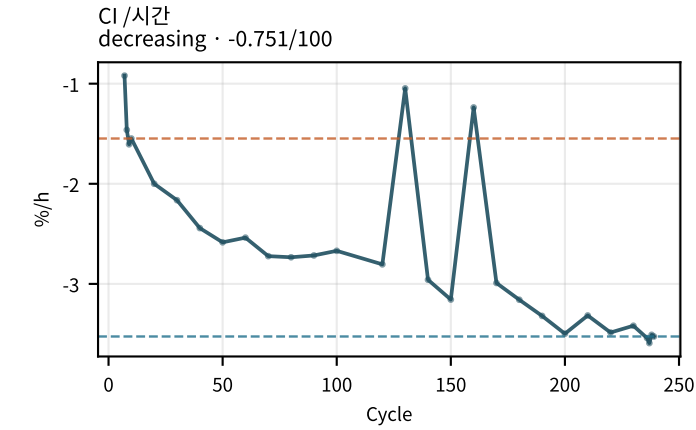
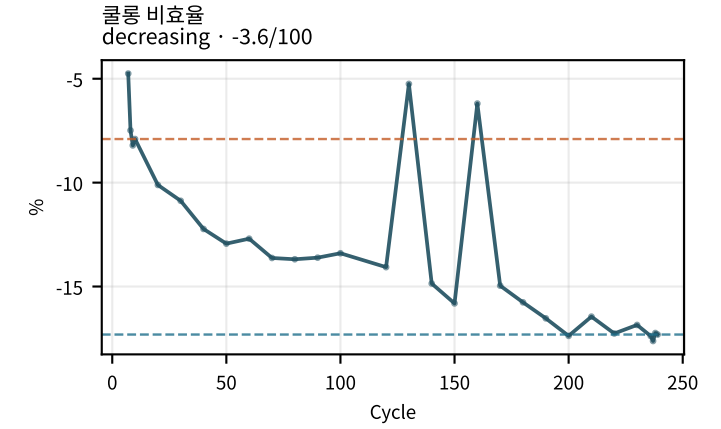
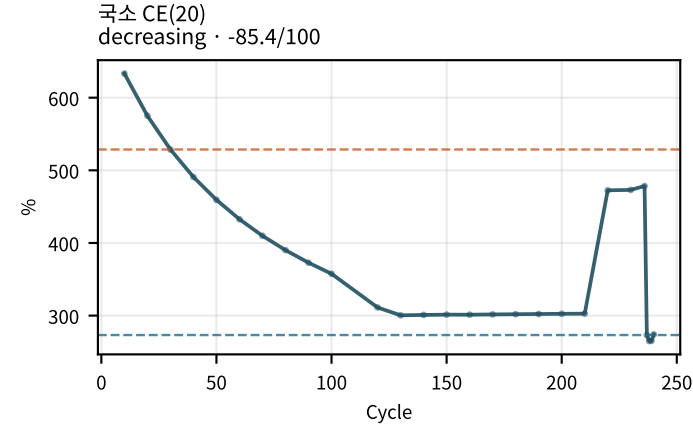
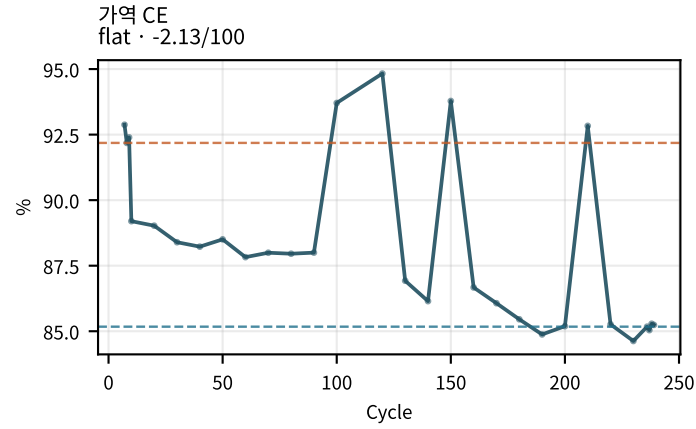
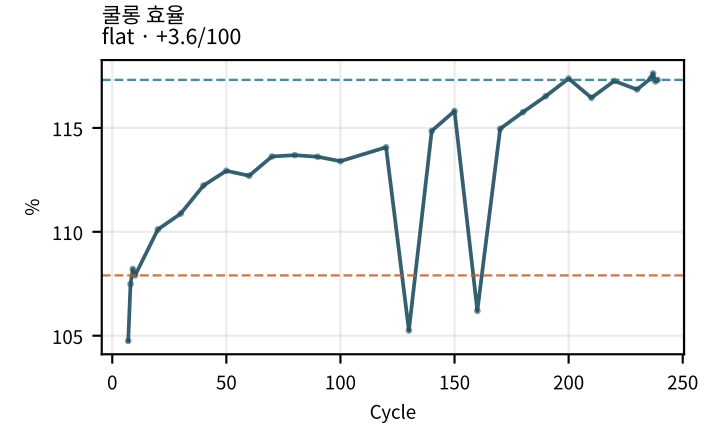
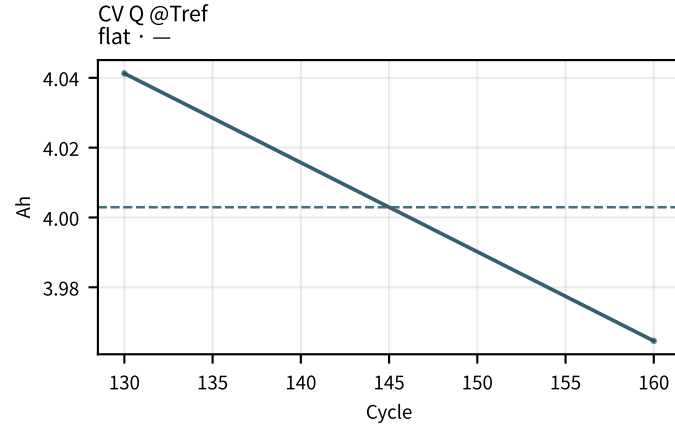
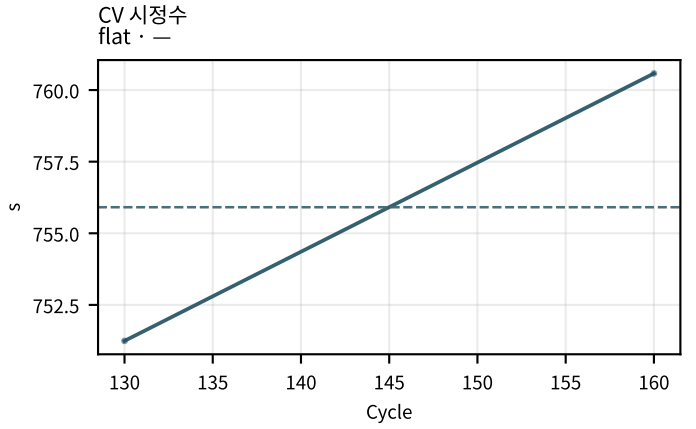
M02Ch103 · 프로토콜 · 온도



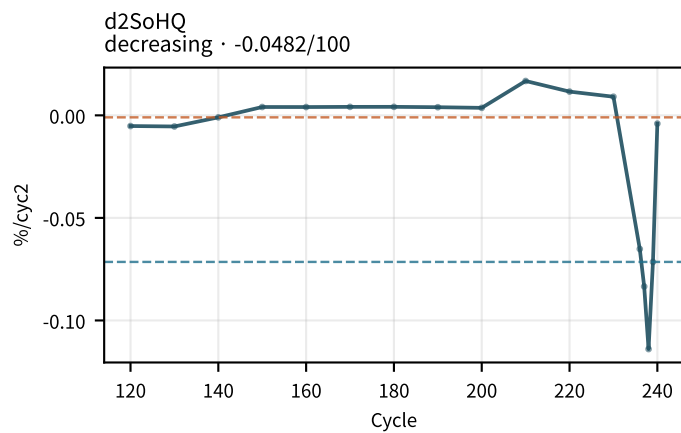
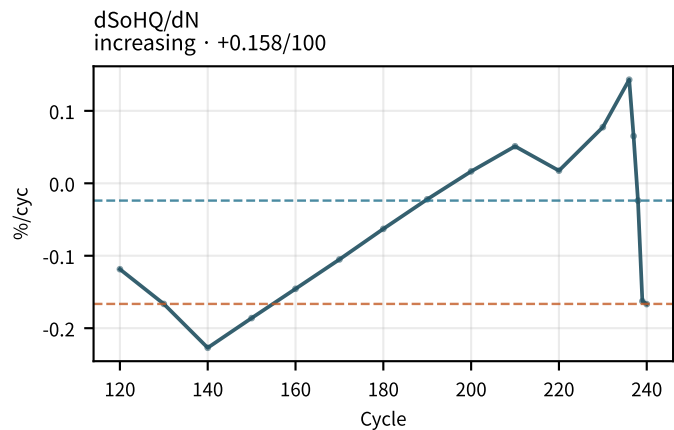
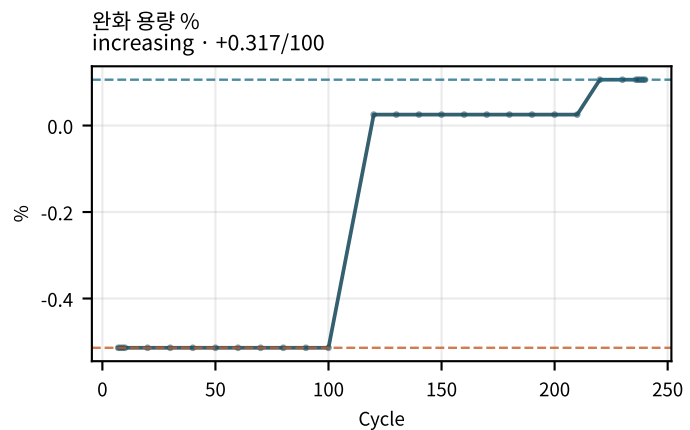
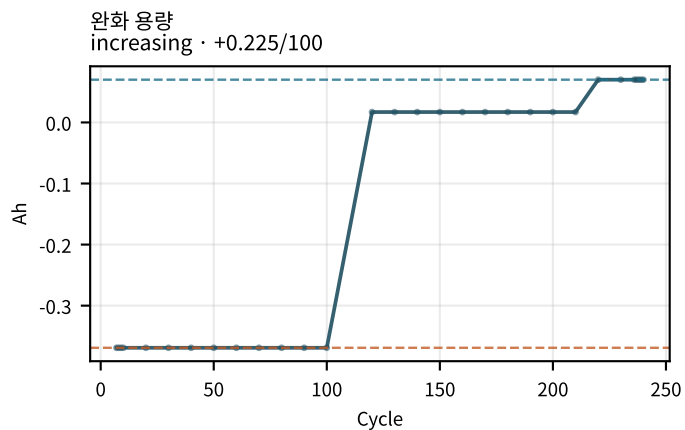
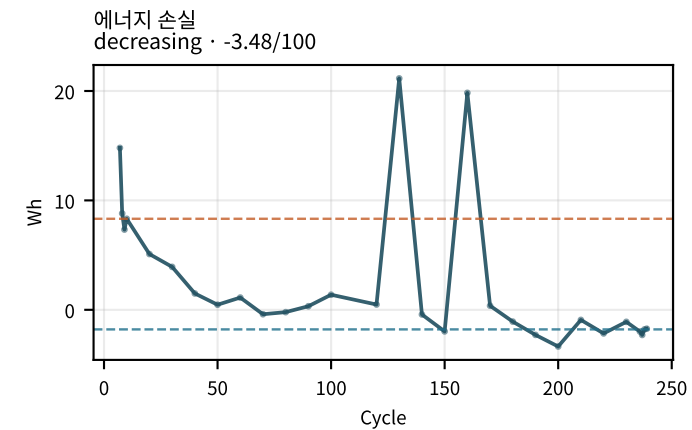
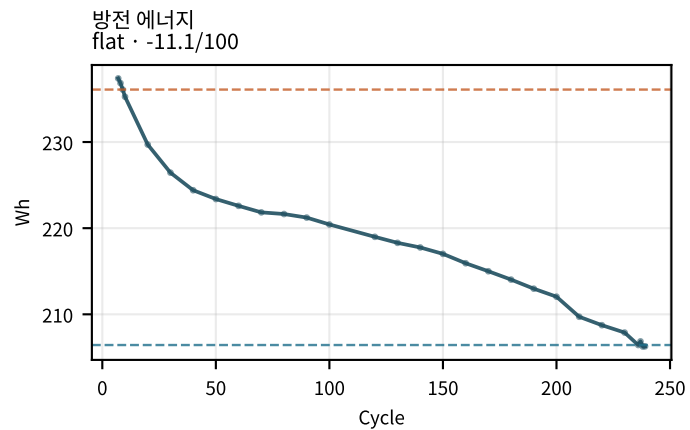
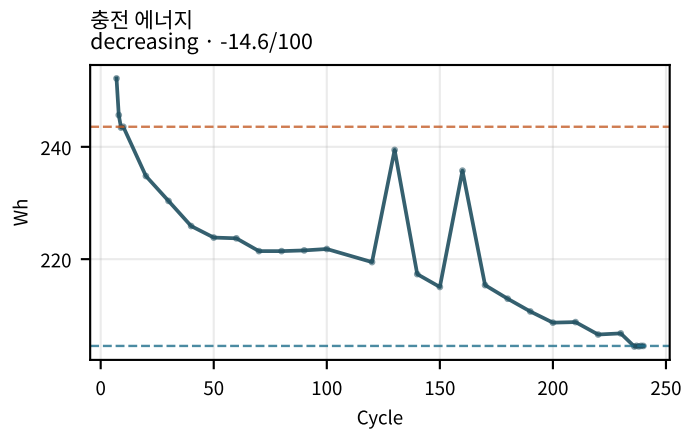
M02Ch103 · 용량 · 효율 · CV



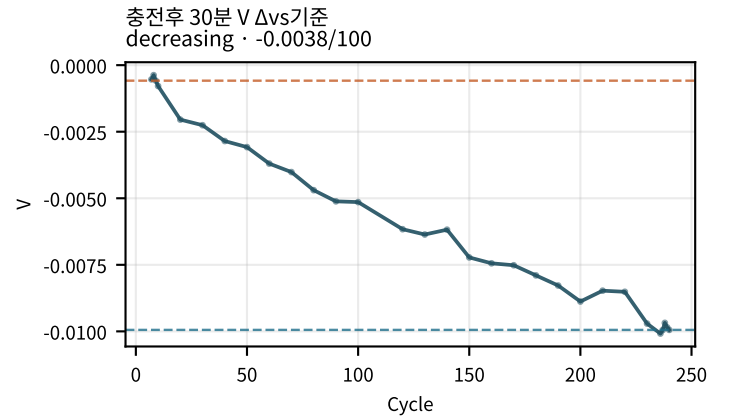
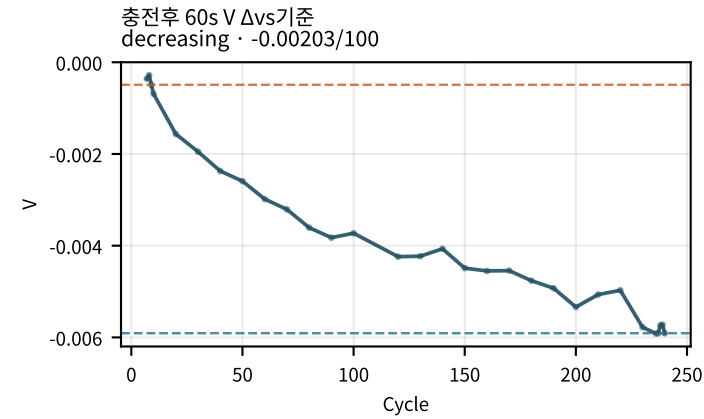
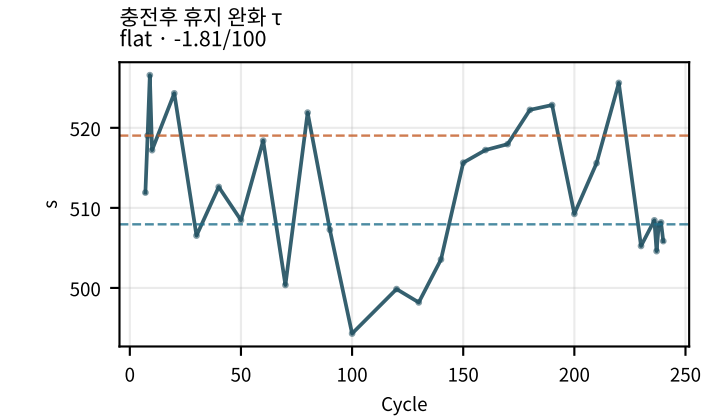
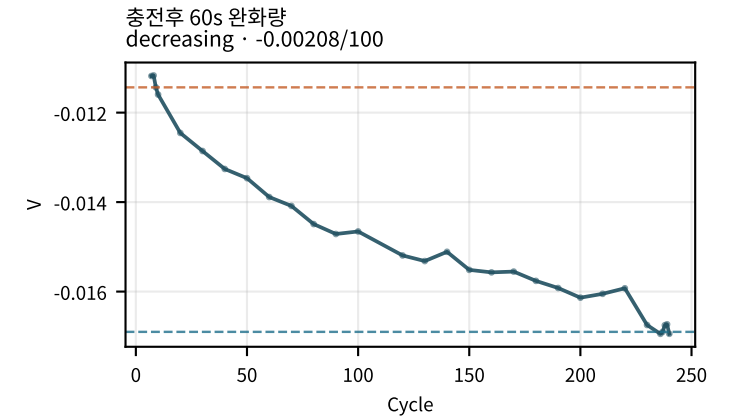
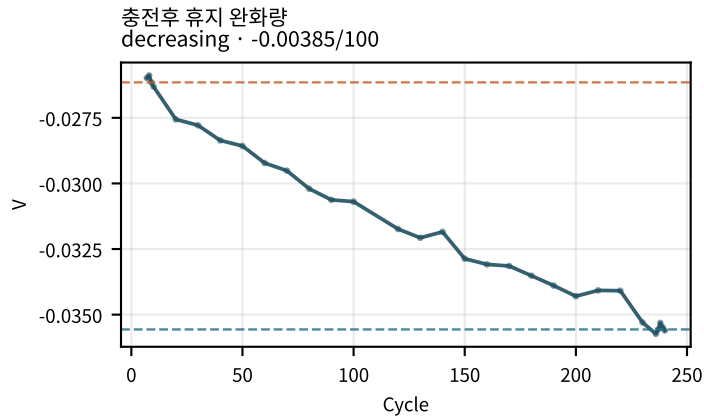
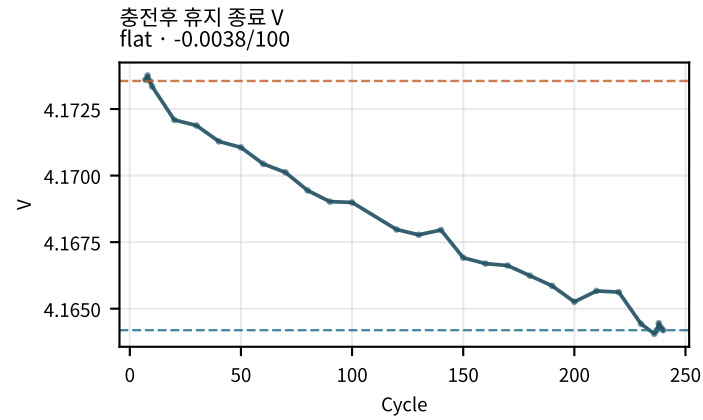
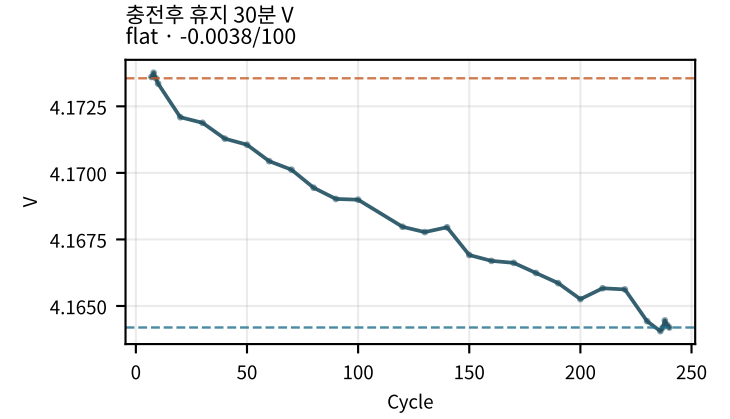
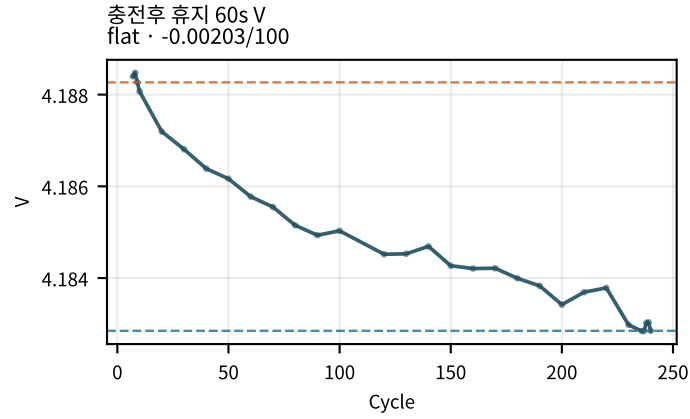
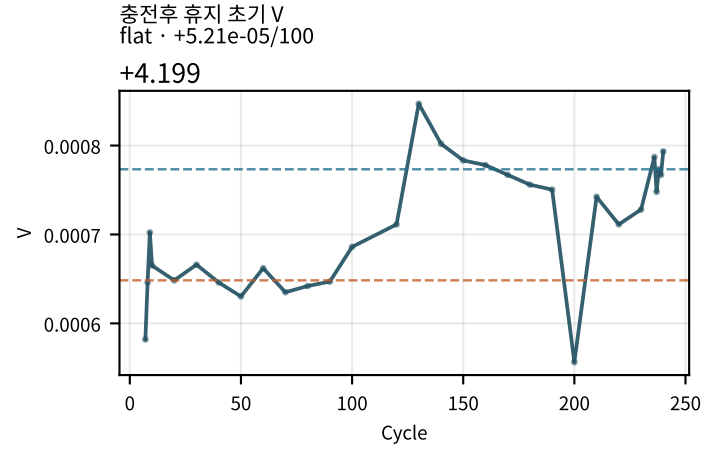
M02Ch103 · 용량 · 효율 · CV



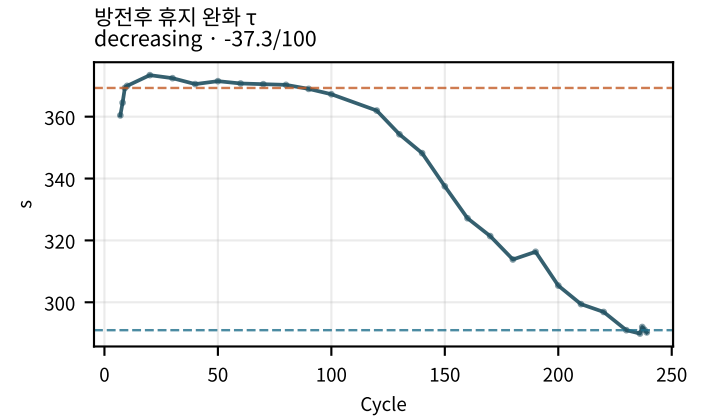
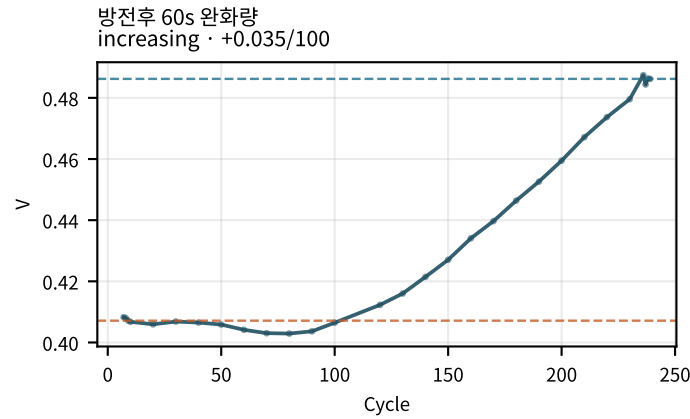
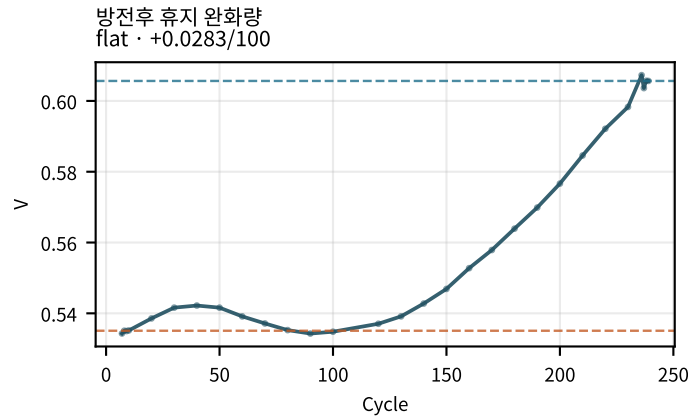
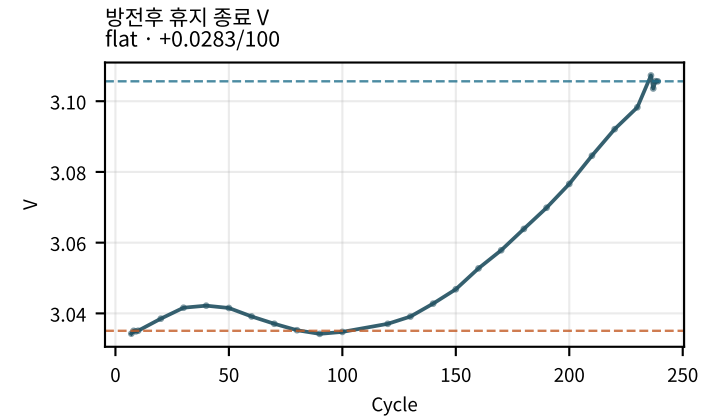
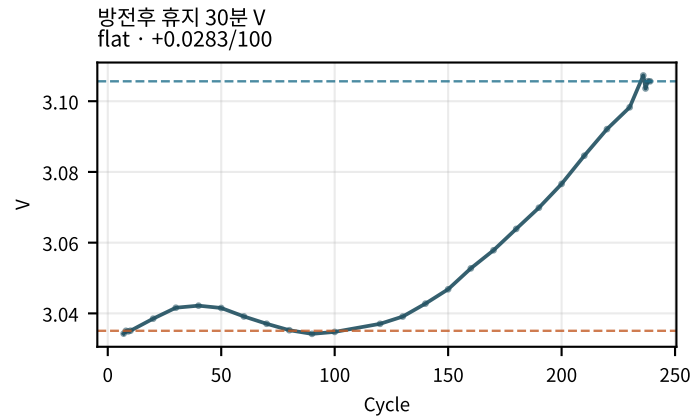
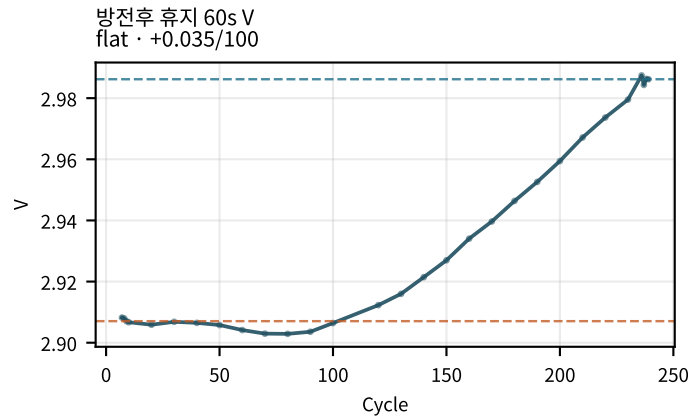
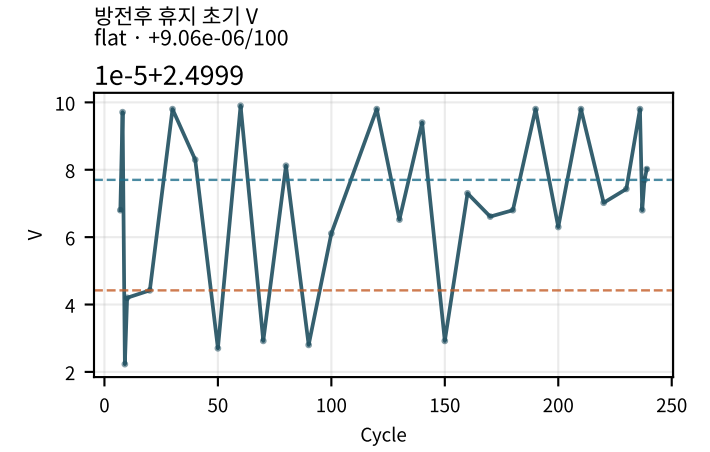
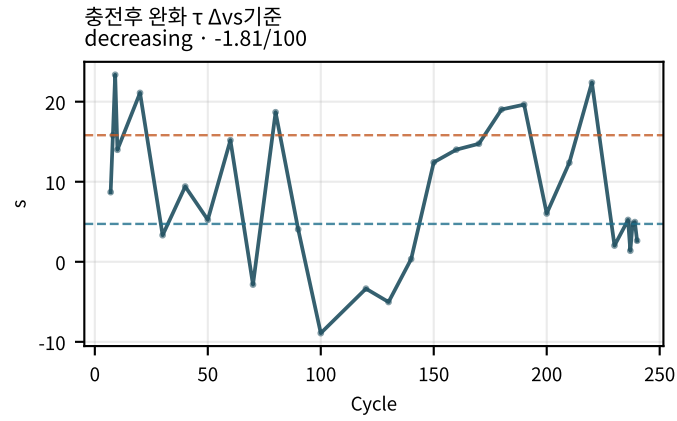
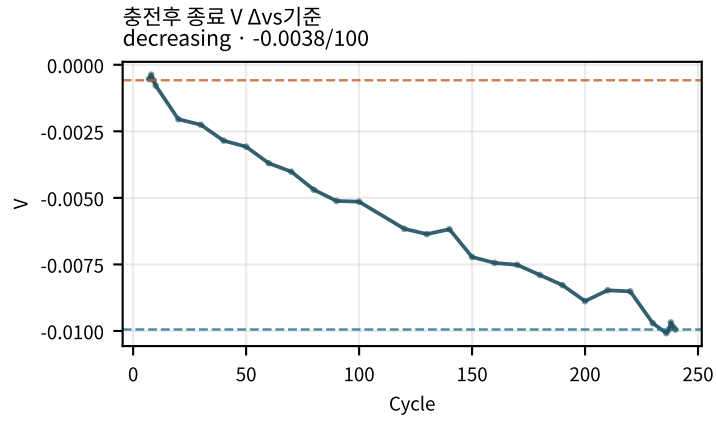
M02Ch103 · 용량 · 효율 · CV



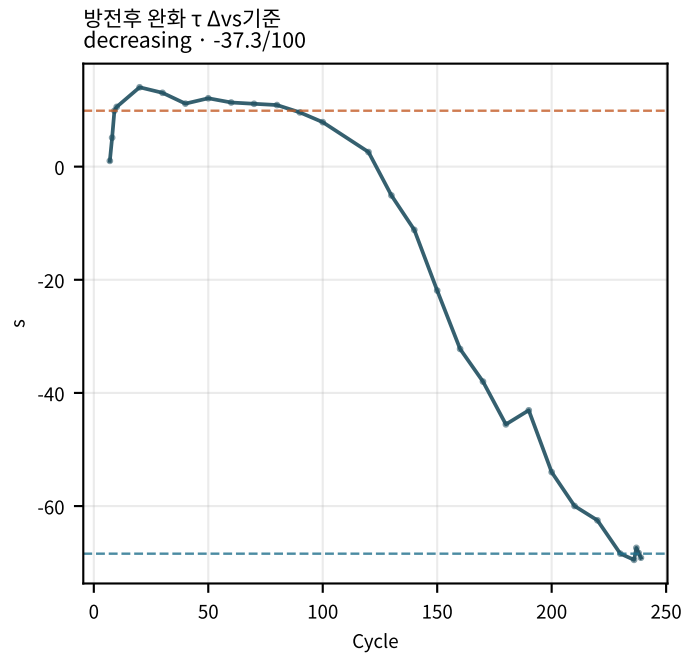
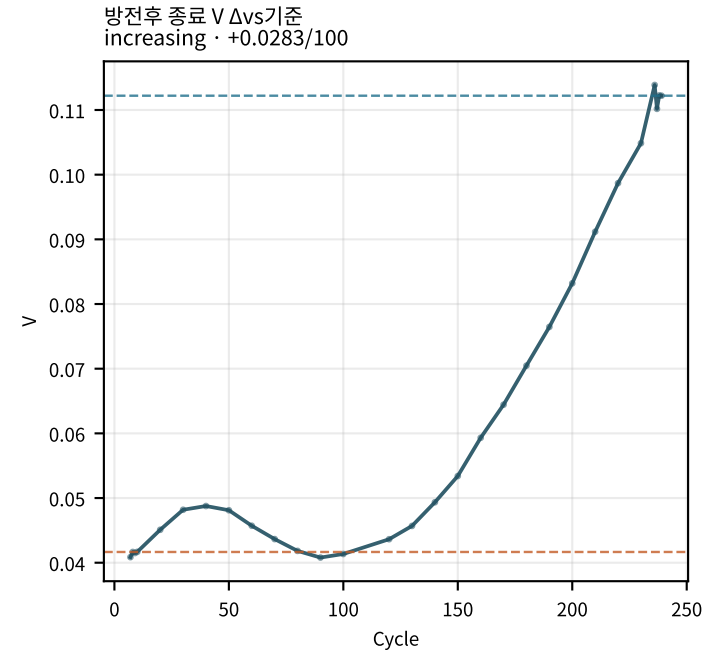
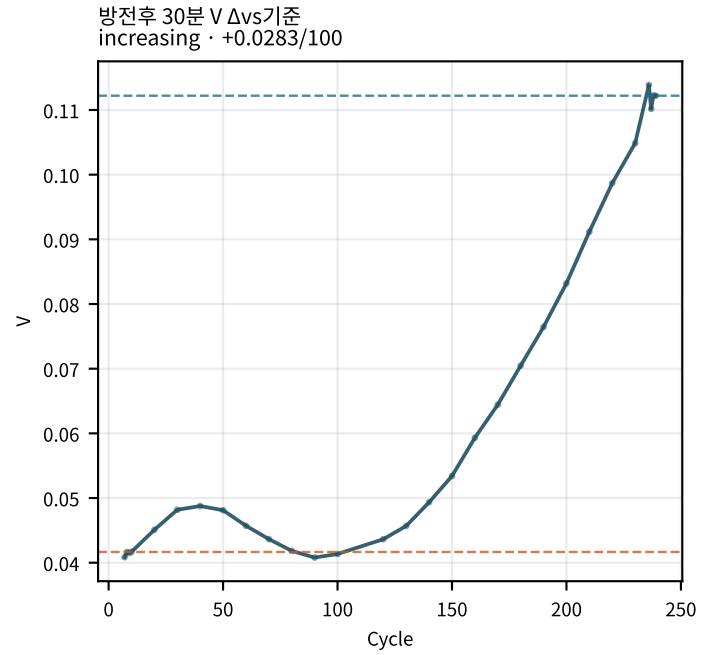
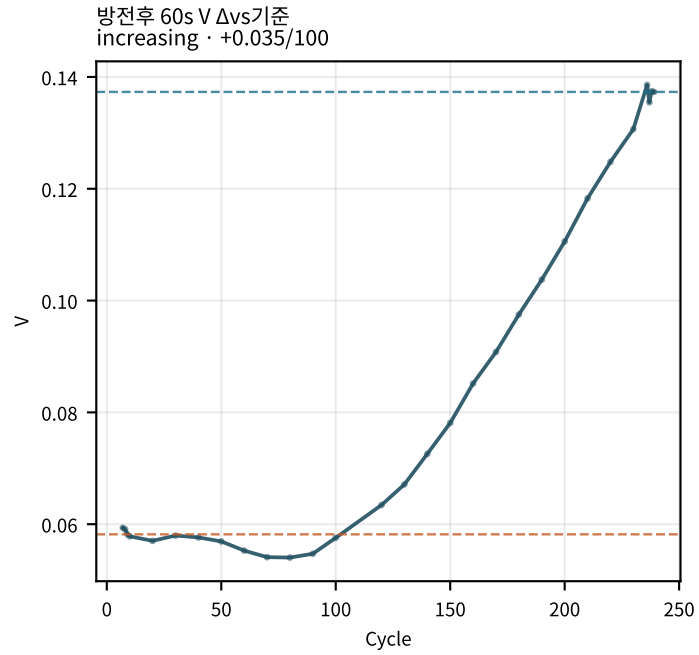
M02Ch103 · 휴지 전압 (충전/방전 후)



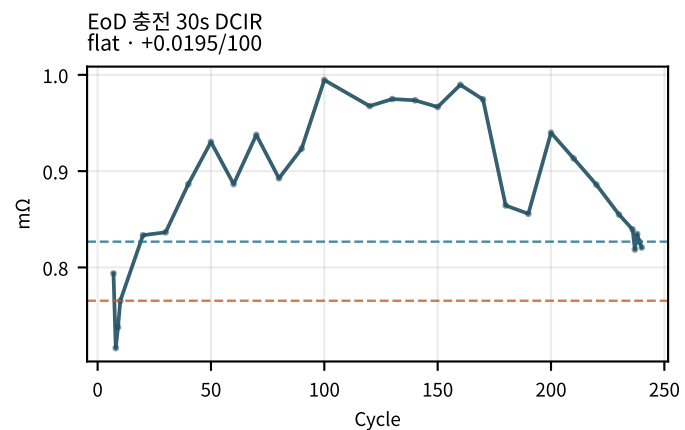
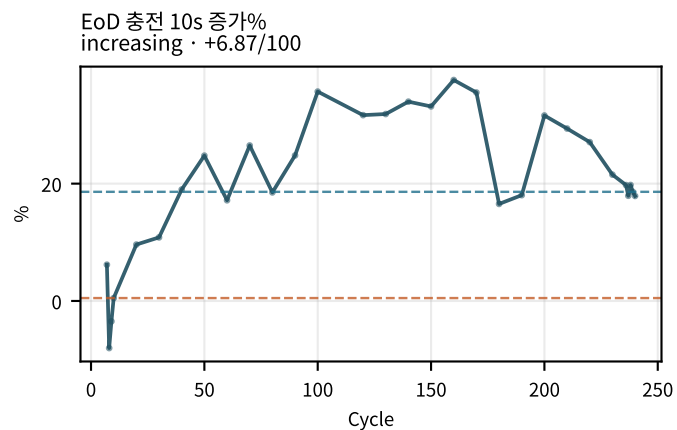
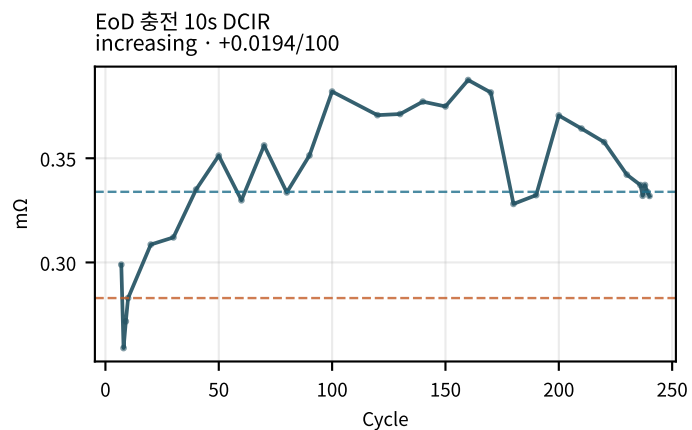
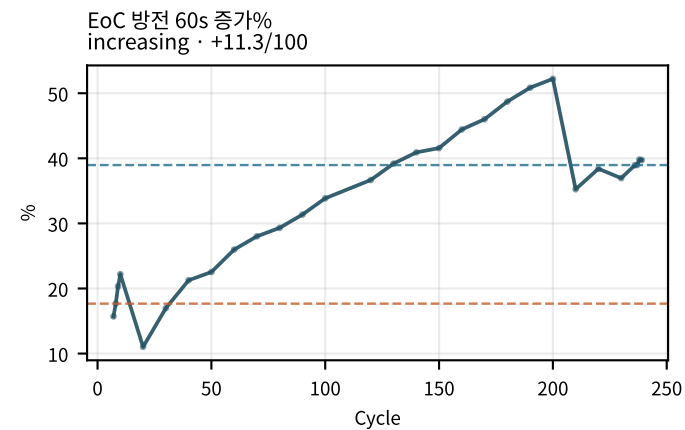
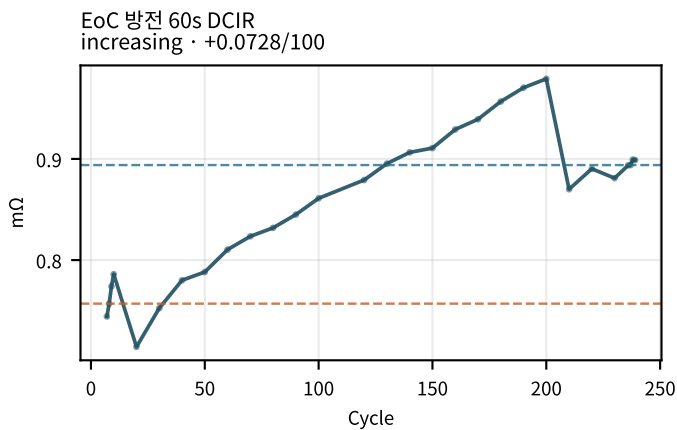
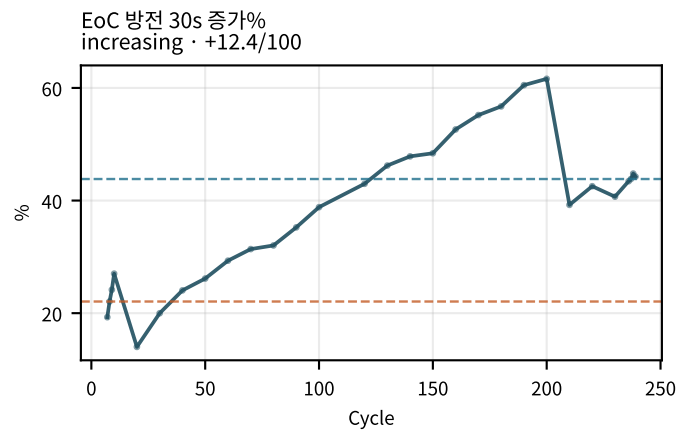
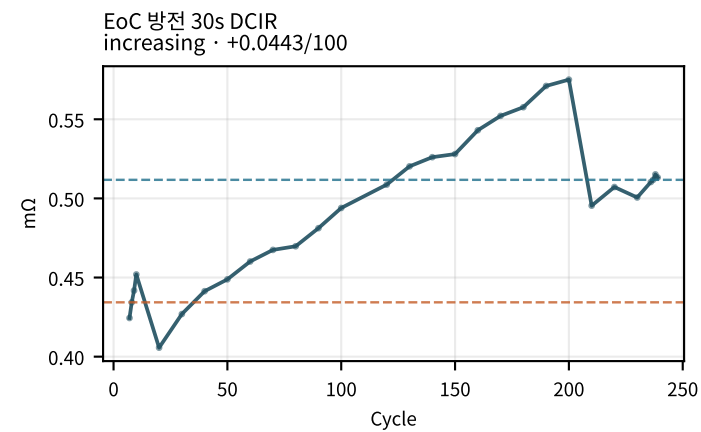
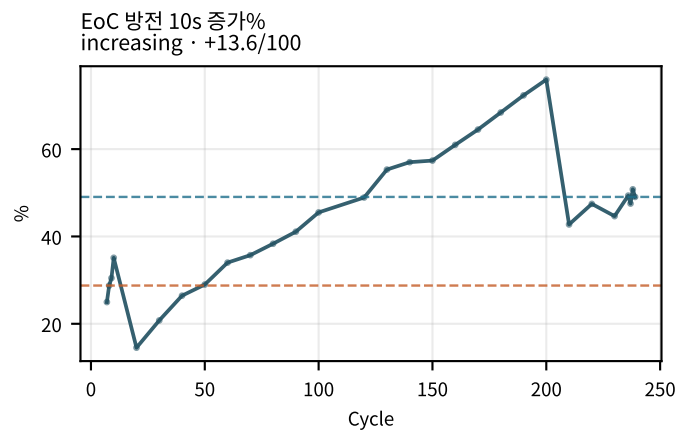
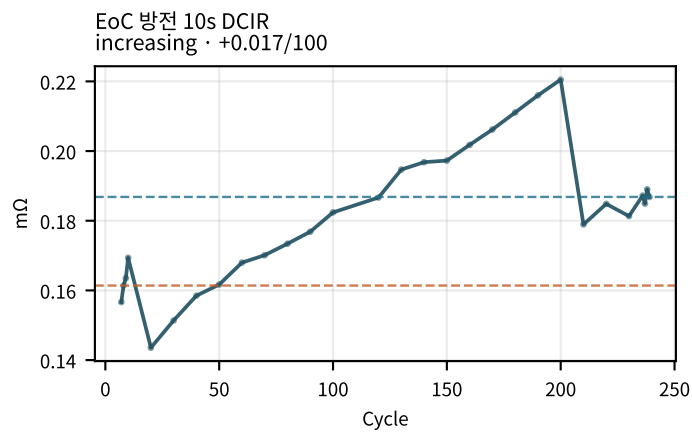
M02Ch103 · 휴지 전압 (충전/방전 후)



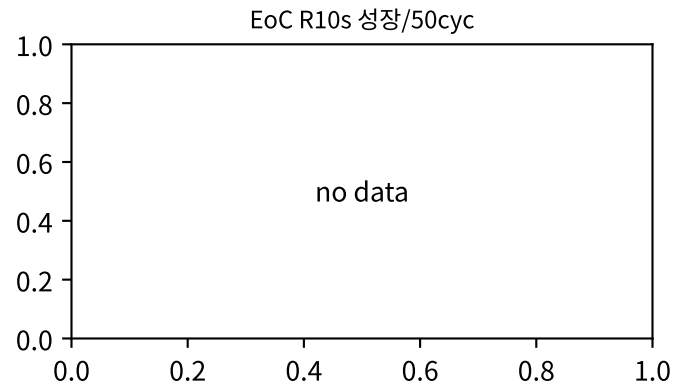
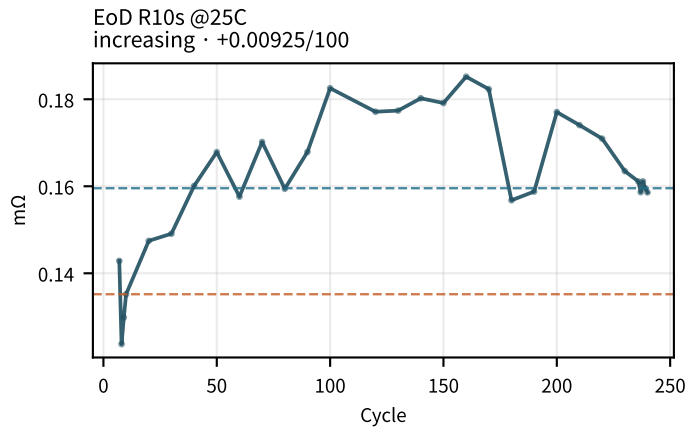
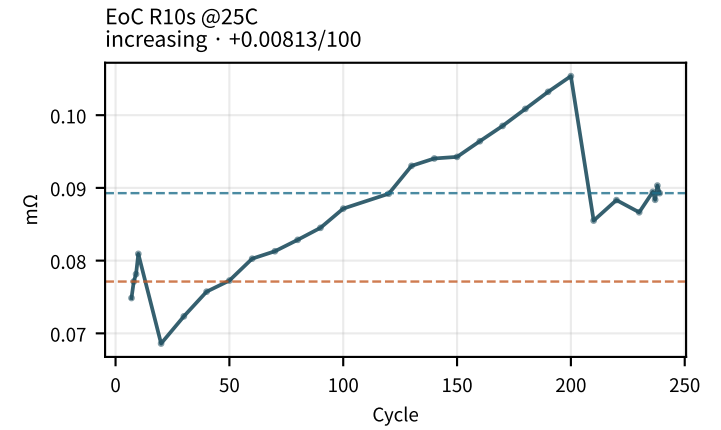
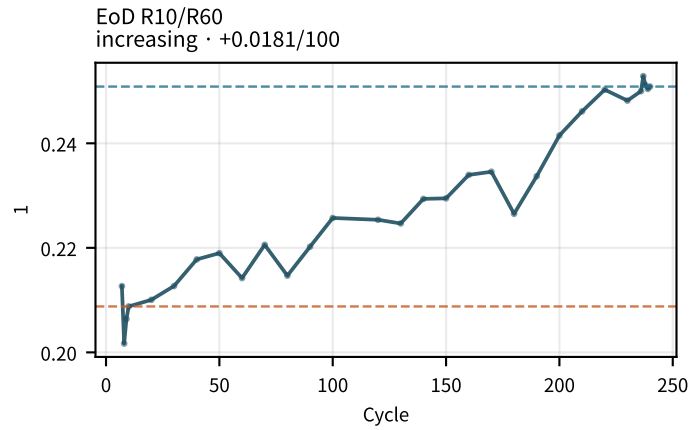
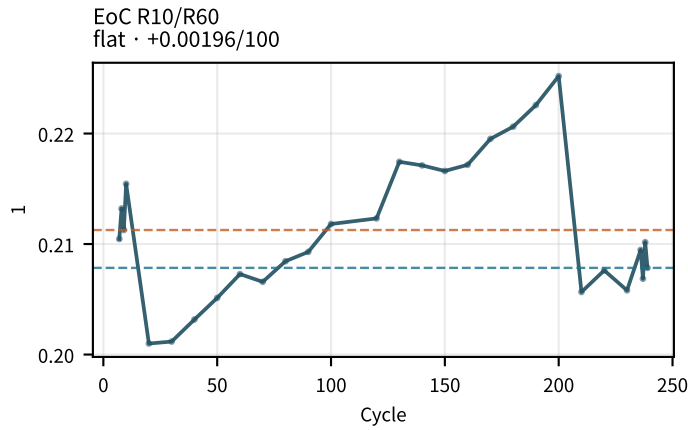
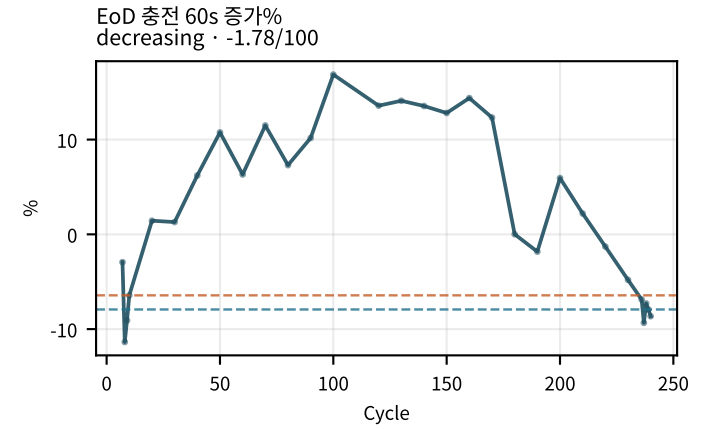
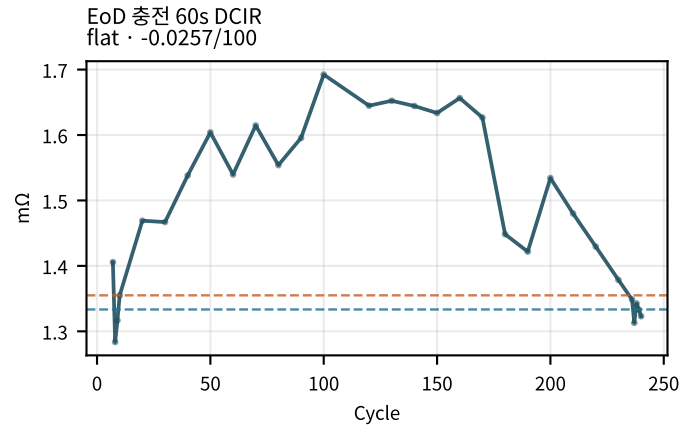
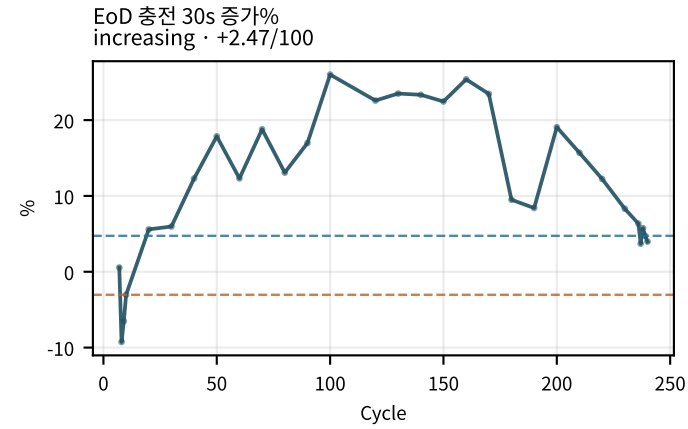
M02Ch103 · 휴지 전압 (충전/방전 후)



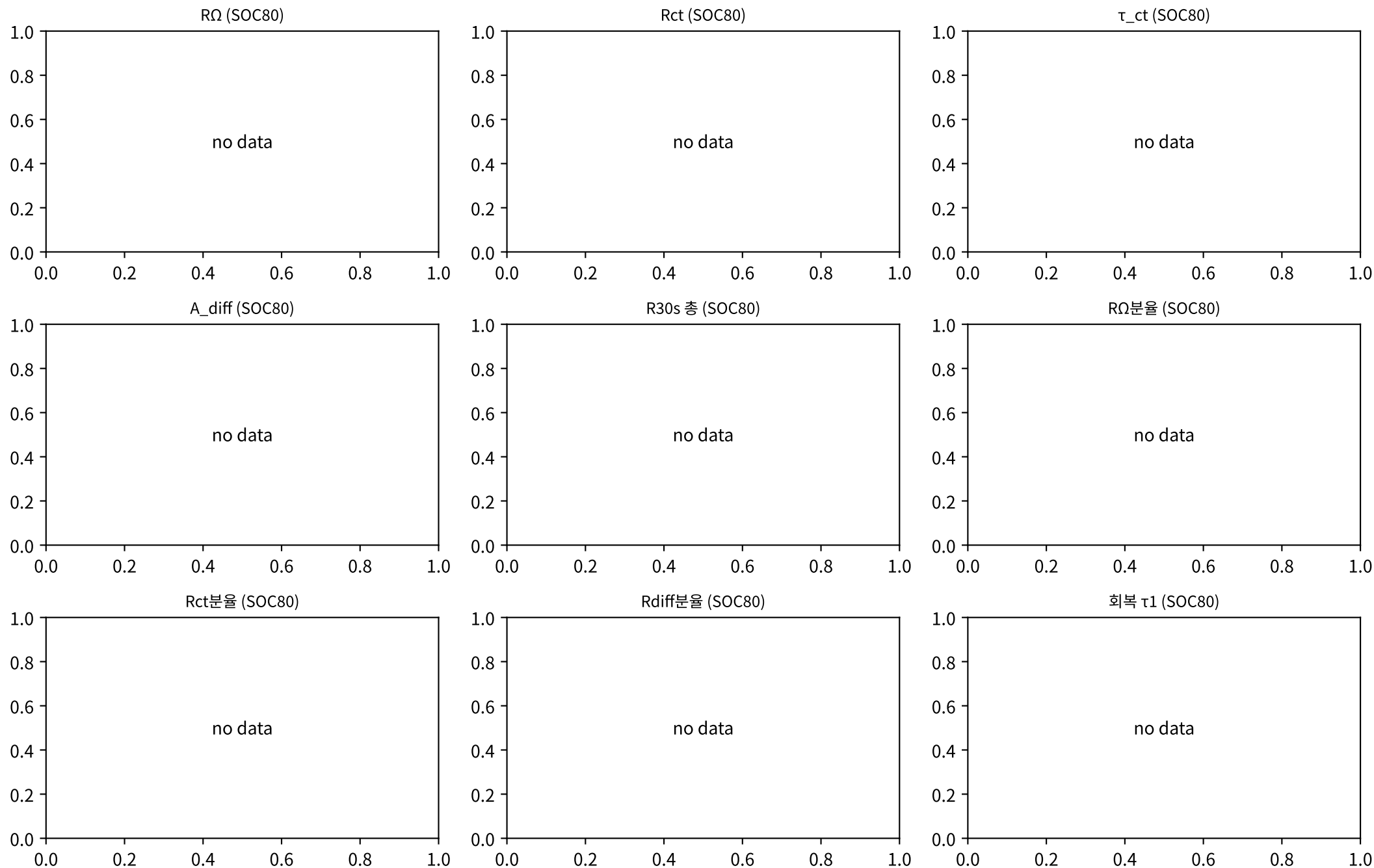
M02Ch103 · 시작 저항 (EoC/EoD)



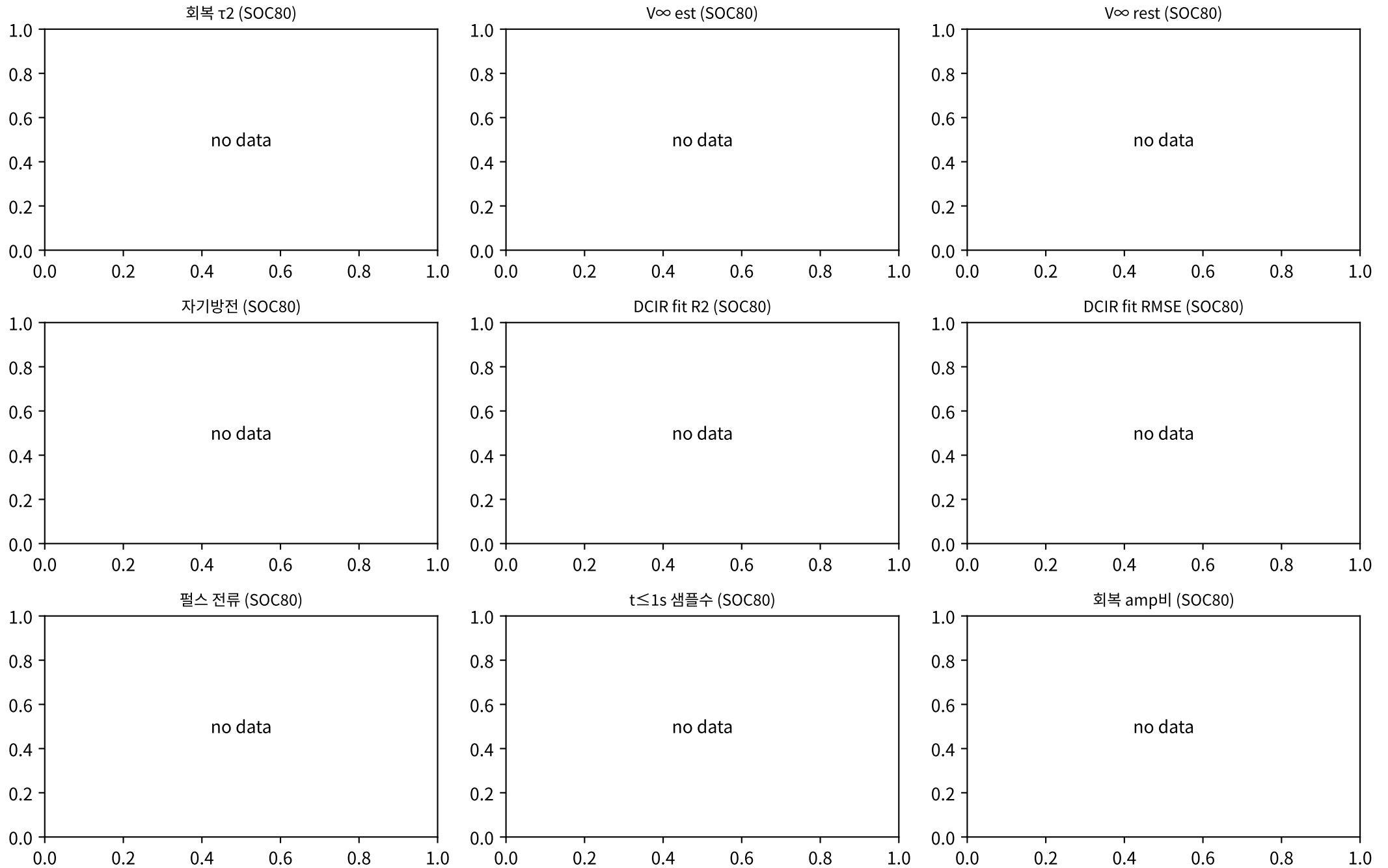
M02Ch103 · 시작 저항 (EoC/EoD)



M02Ch103 · DCIR 분해 · 성장

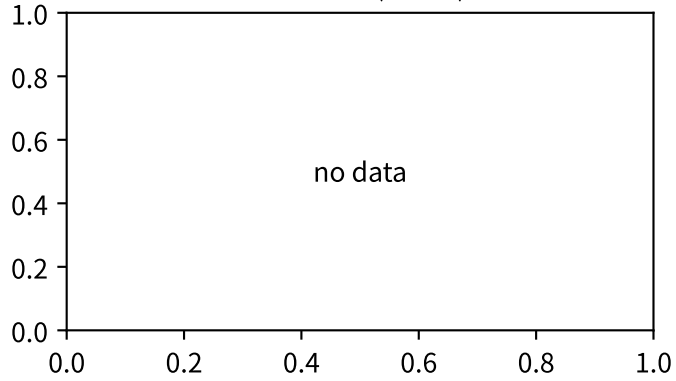


M02Ch103 · DCIR 분해 · 성장

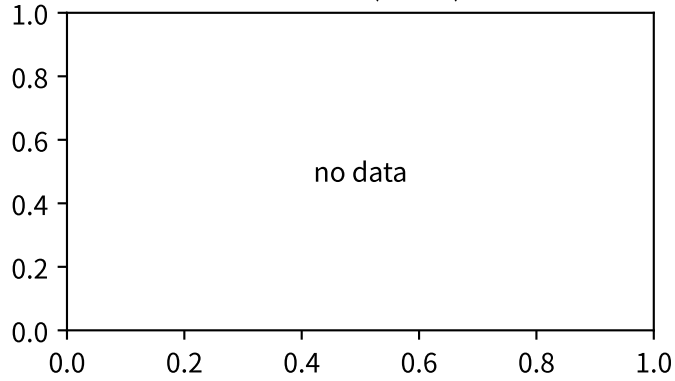


M02Ch103 · DCIR 분해 · 성장

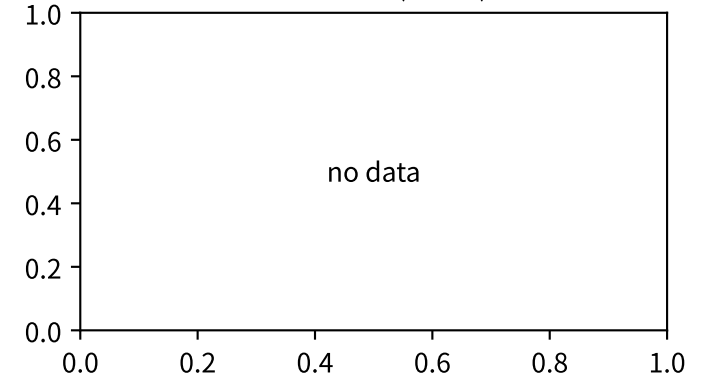
완화 완성도 (SOC80)



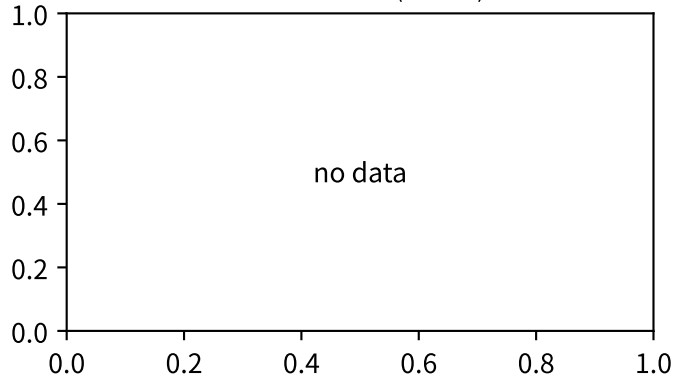
회복 fit R2 (SOC80)



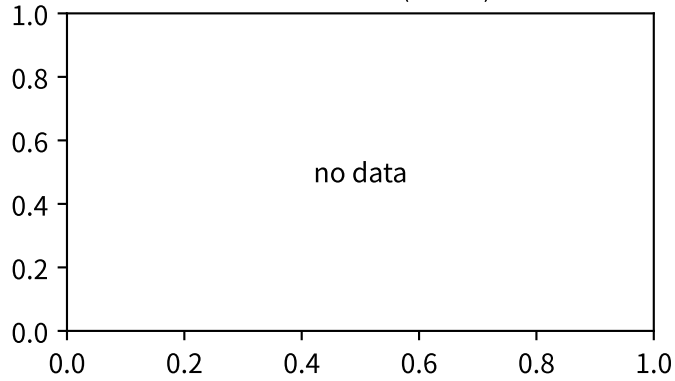
DCIR fit 유효 (SOC80)



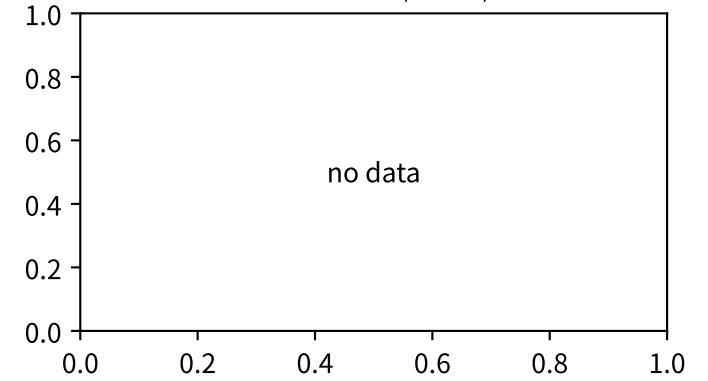
DCIR fit 조건수 (SOC80)



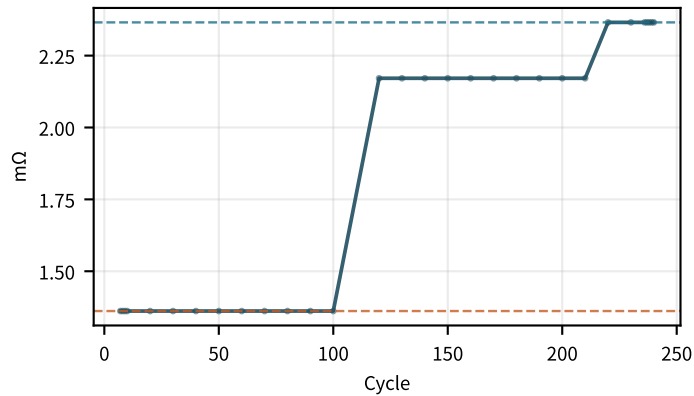
자기방전 fit 유효 (SOC80)



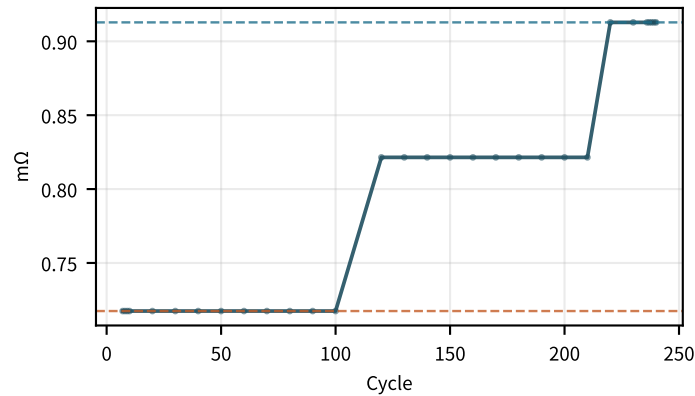
펄스 포인트수 (SOC80)



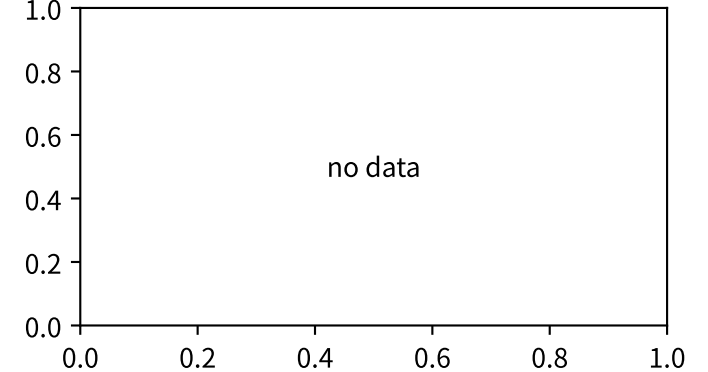
R Ω (SOC50)
increasing · +0.502/100



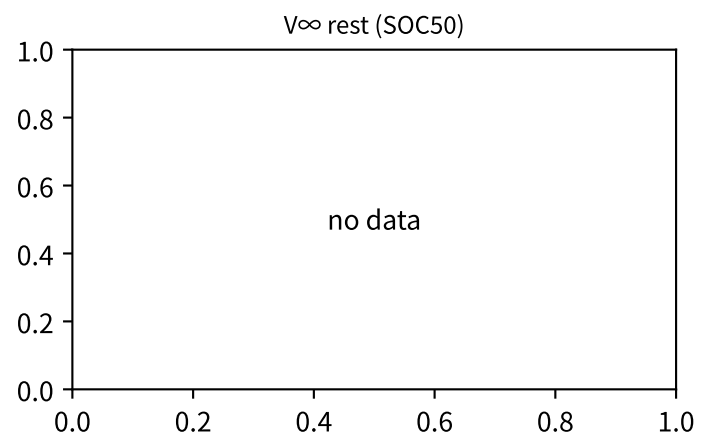
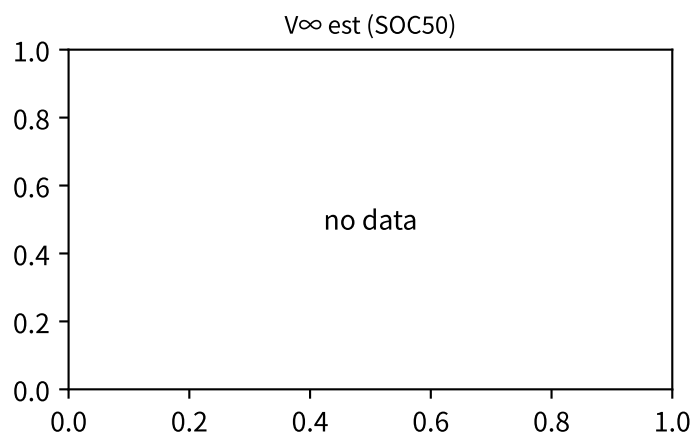
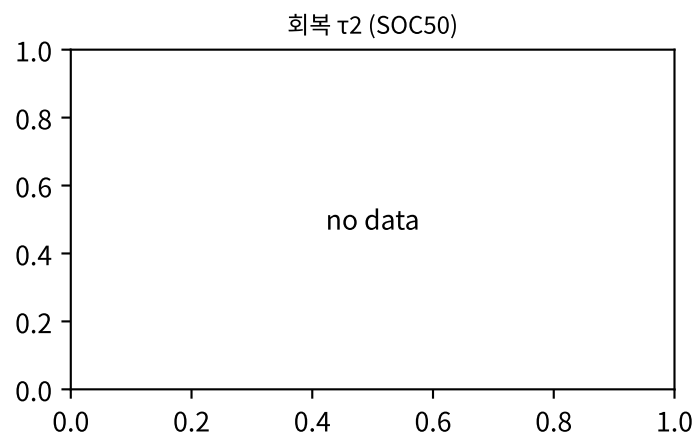
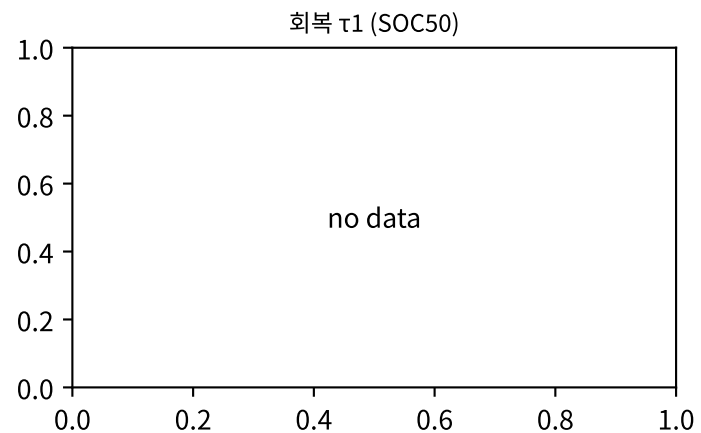
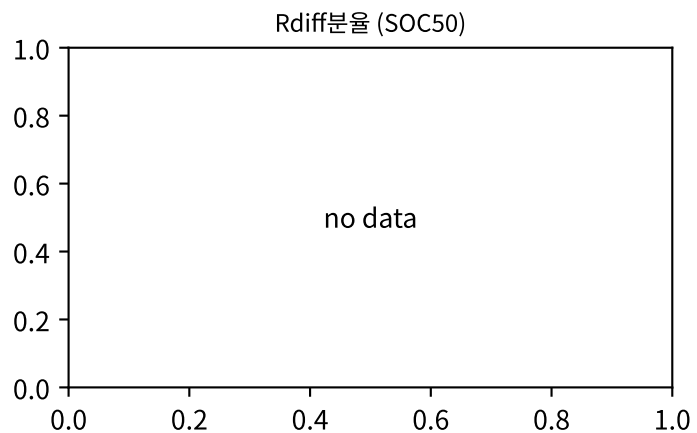
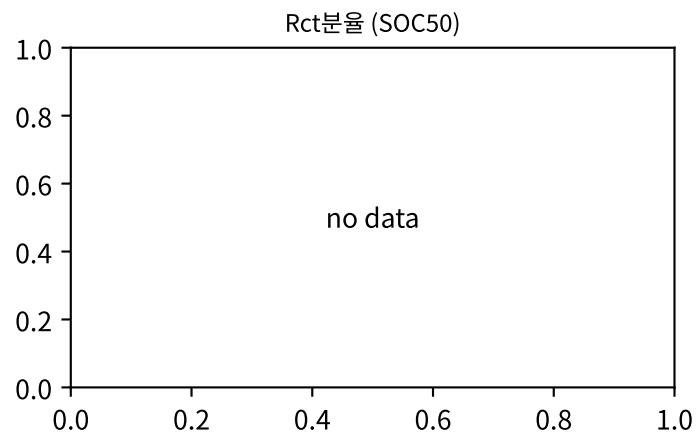
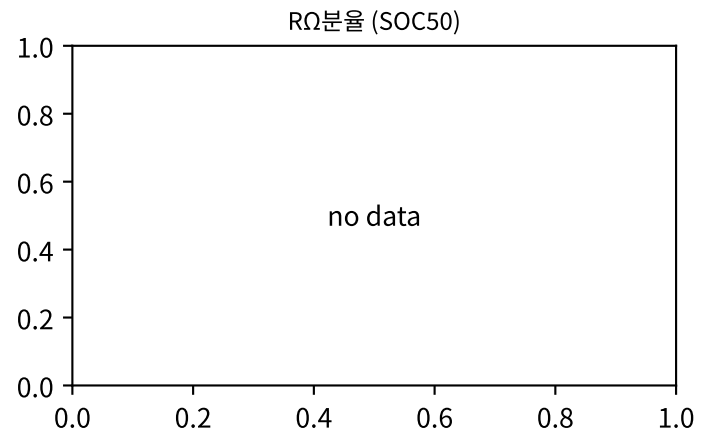
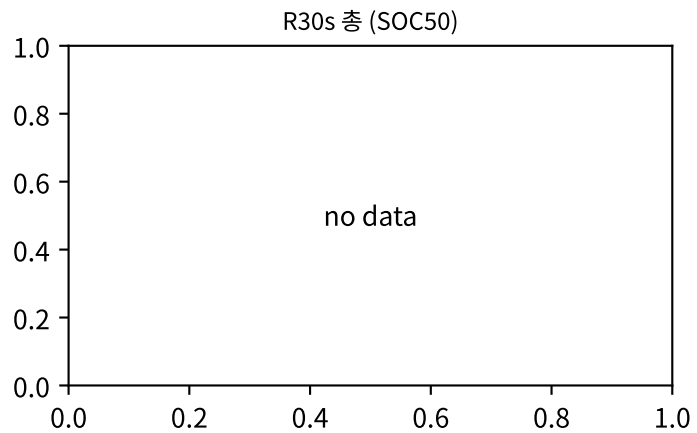
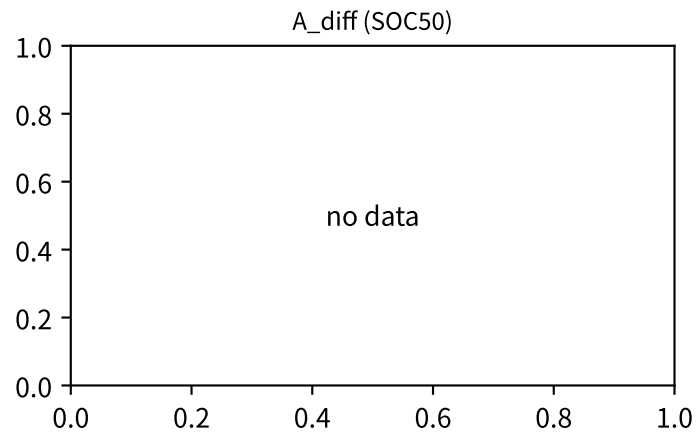
R $_{ct}$ (SOC50)
increasing · +0.0883/100



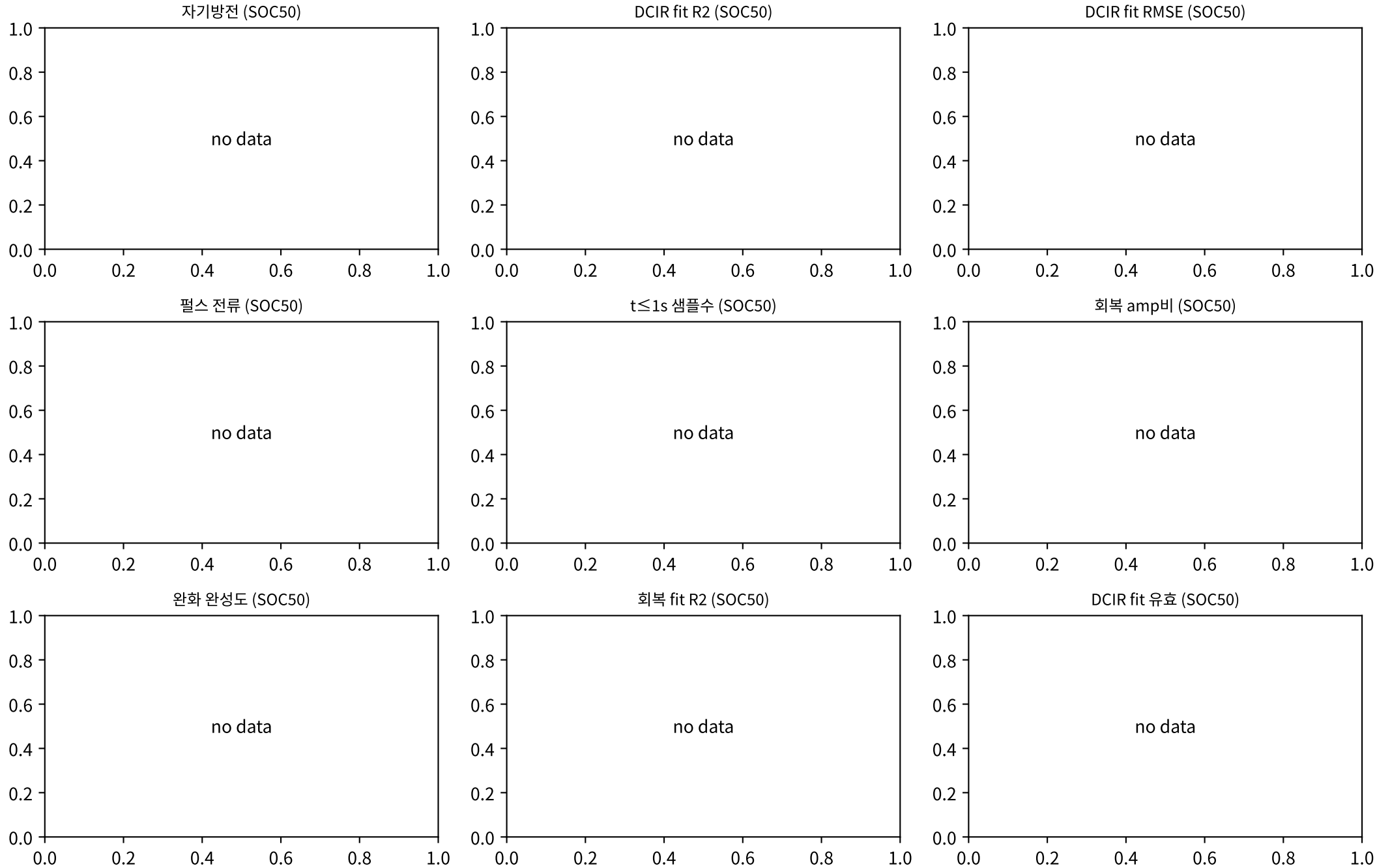
τ_{ct} (SOC50)



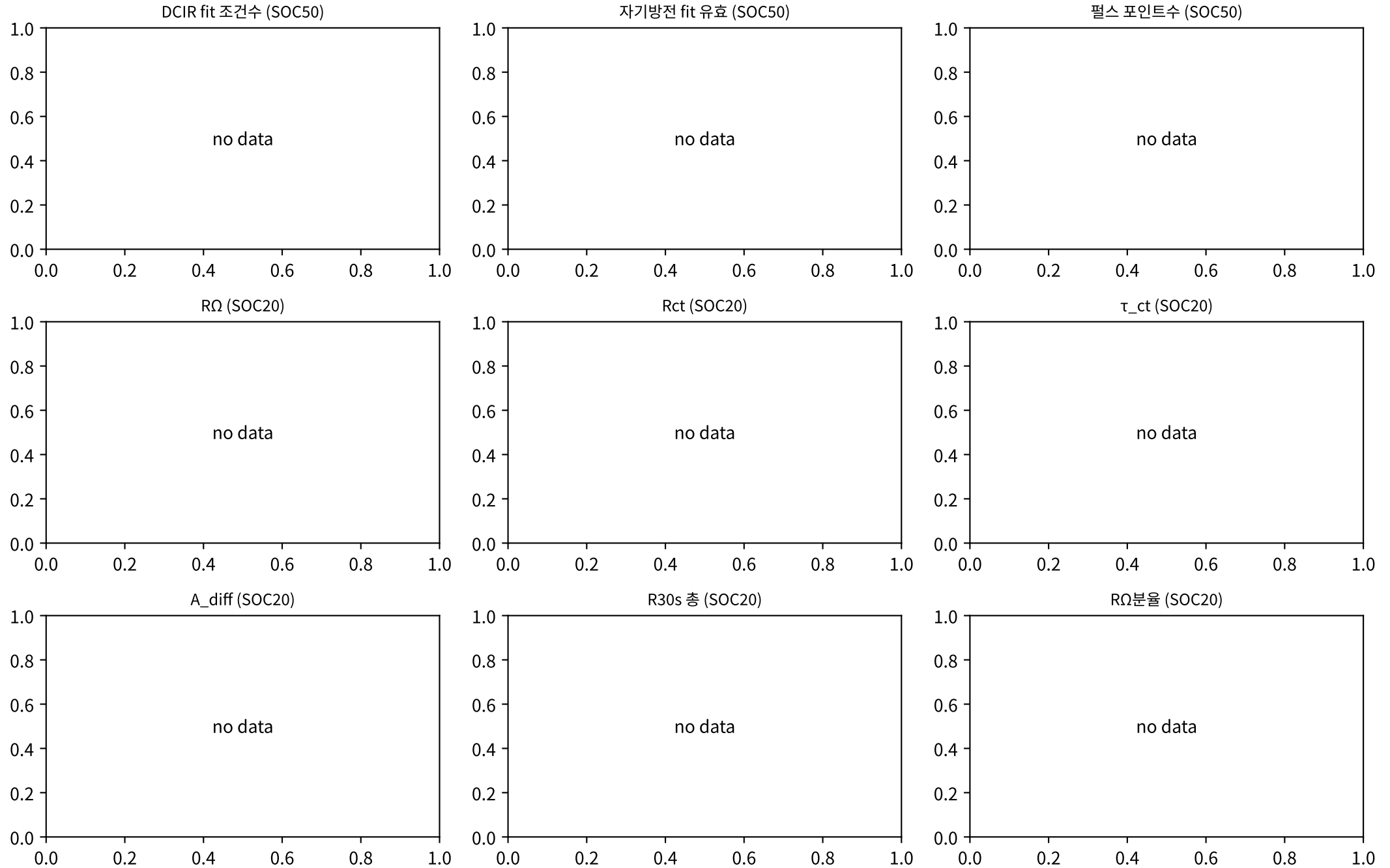
M02Ch103 · DCIR 분해 · 성장



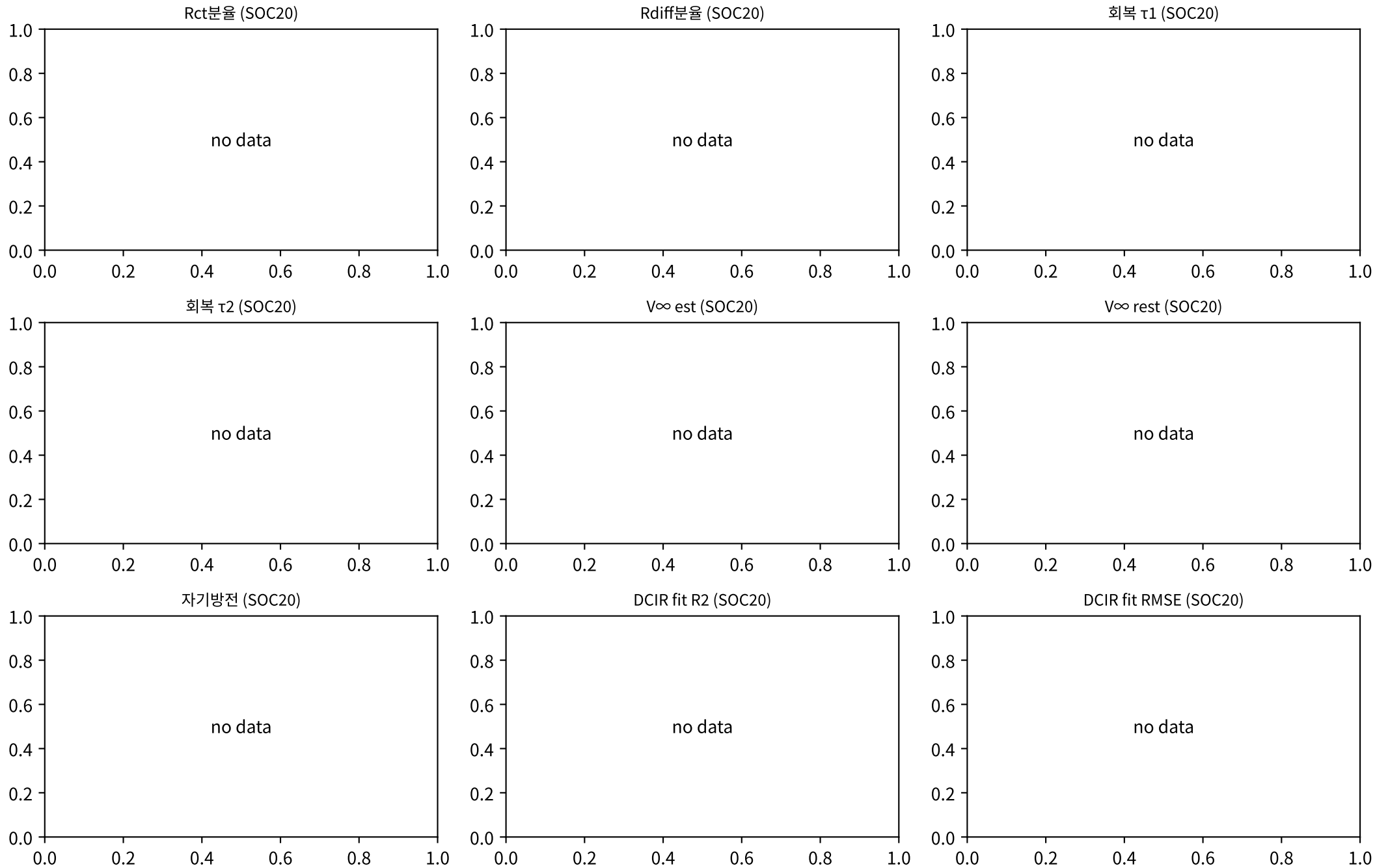
M02Ch103 · DCIR 분해 · 성장



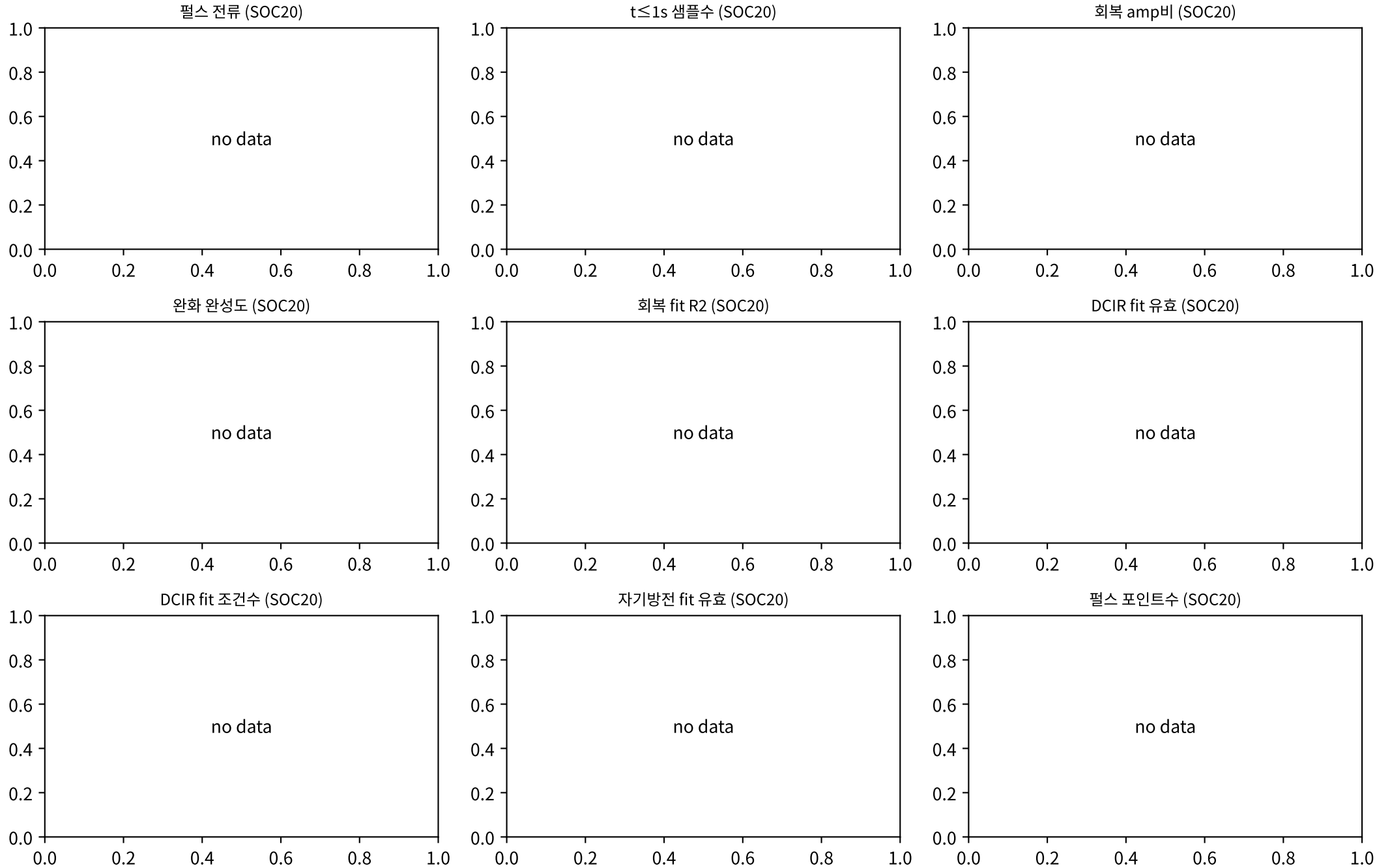
M02Ch103 · DCIR 분해 · 성장



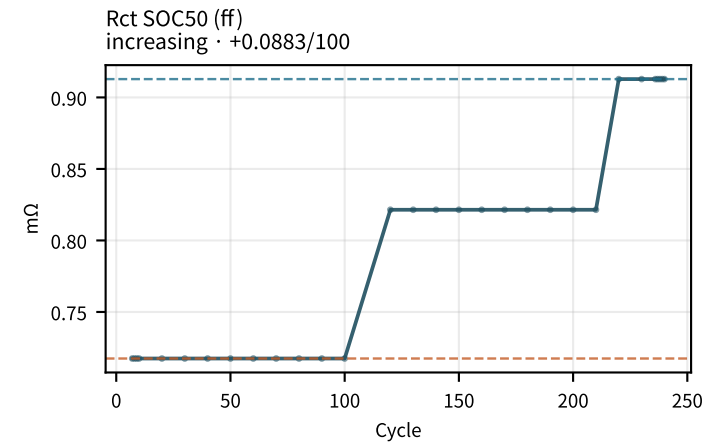
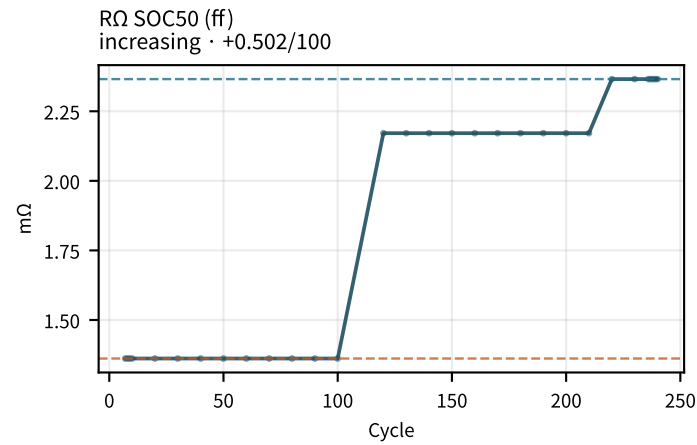
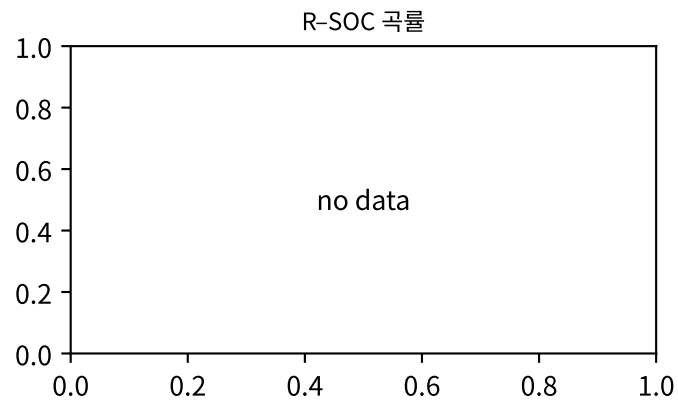
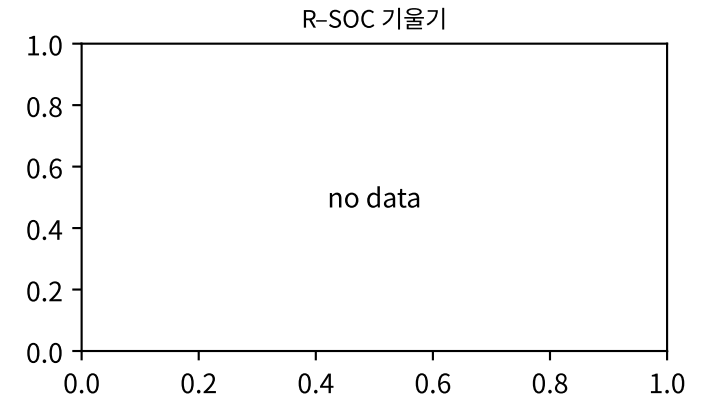
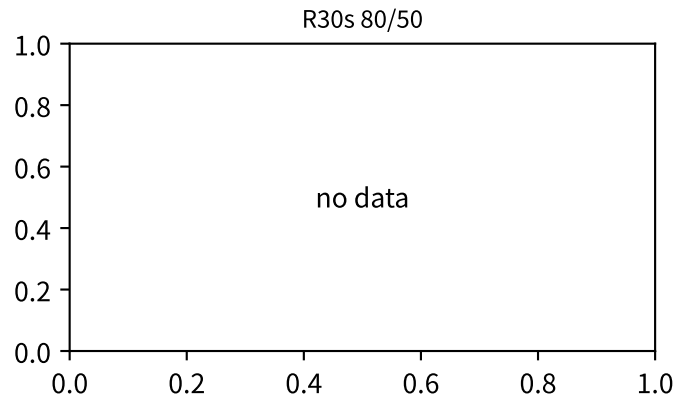
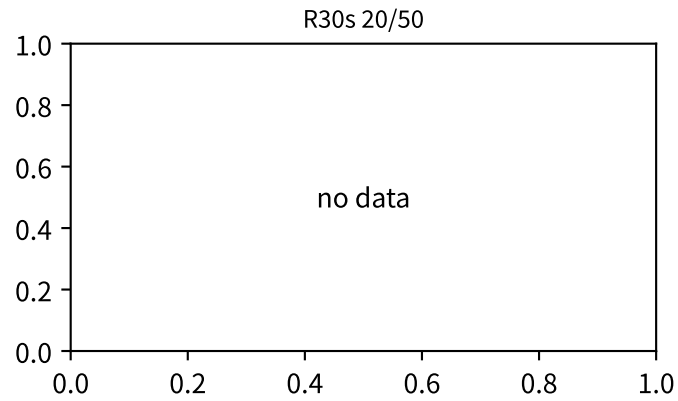
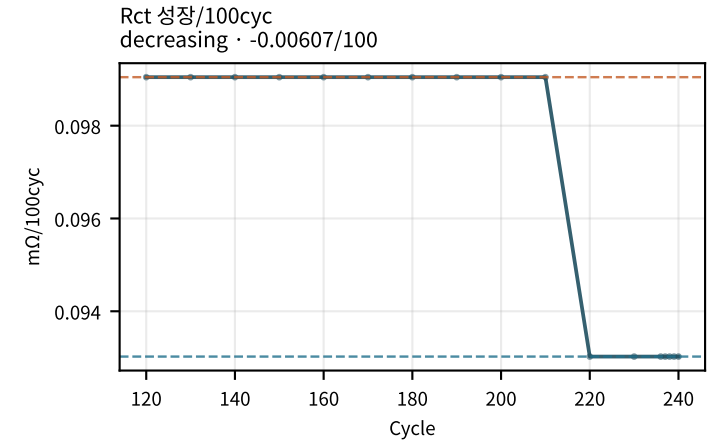
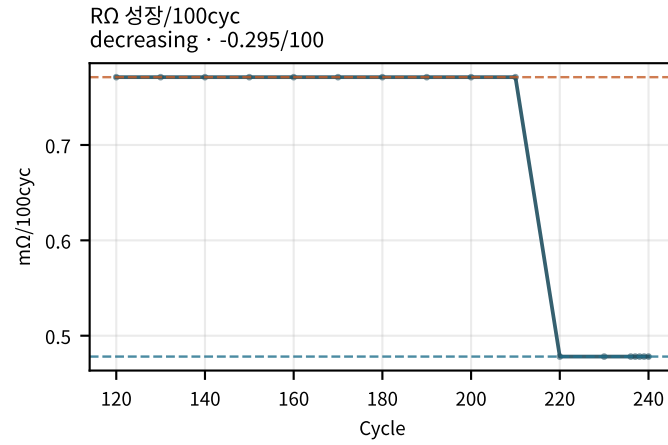
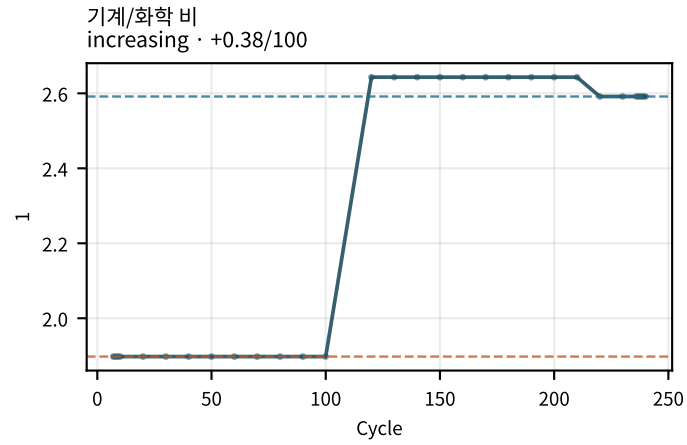
M02Ch103 · DCIR 분해 · 성장



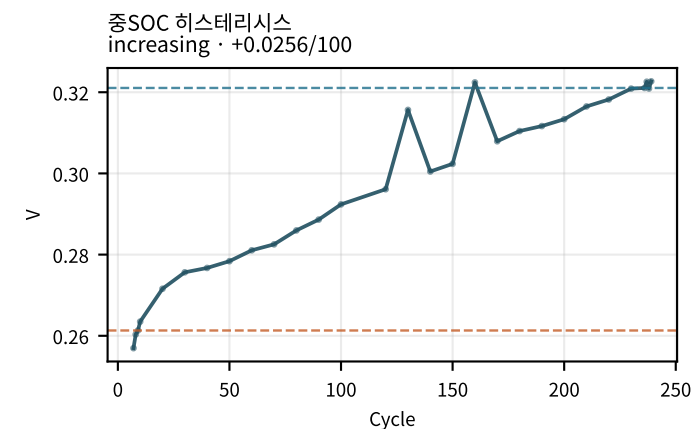
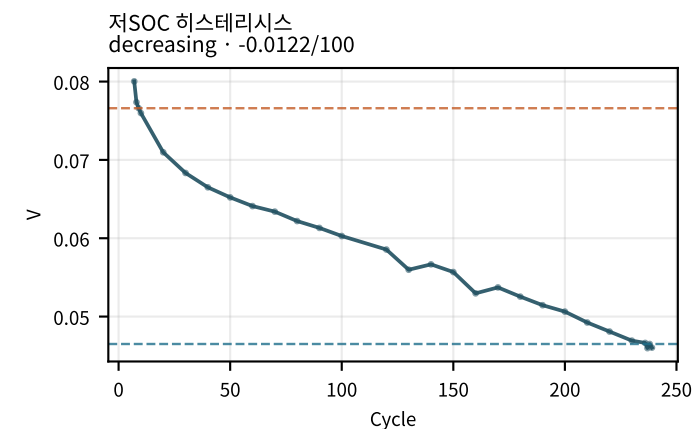
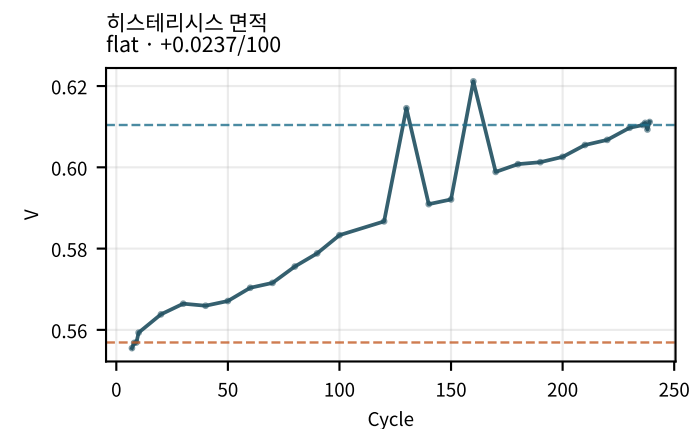
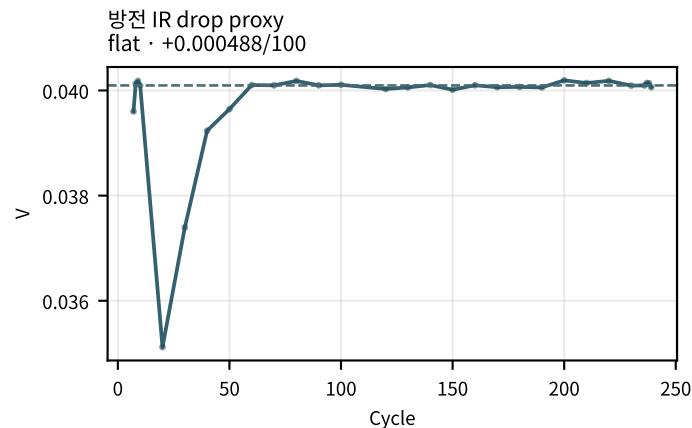
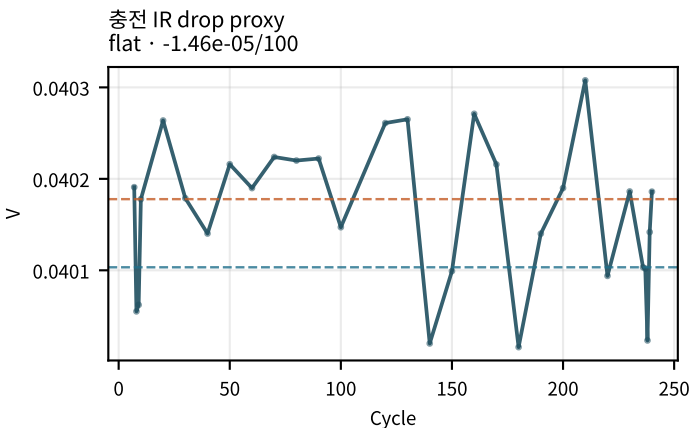
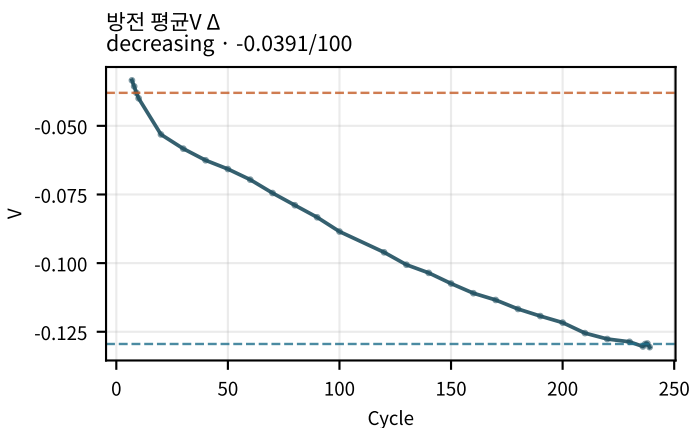
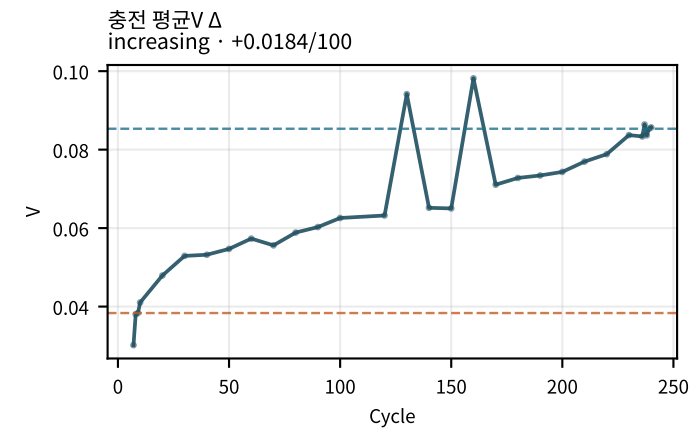
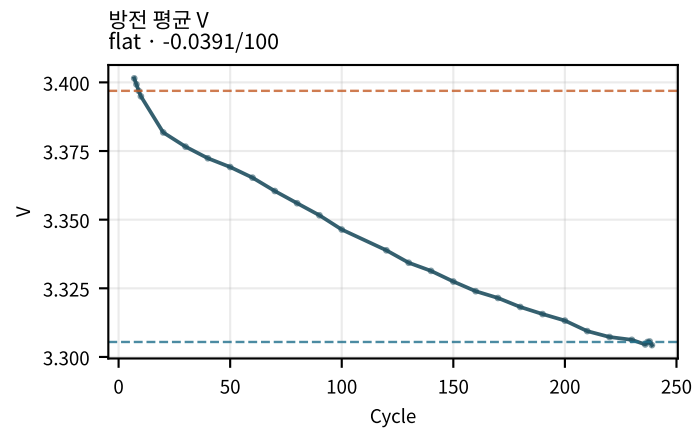
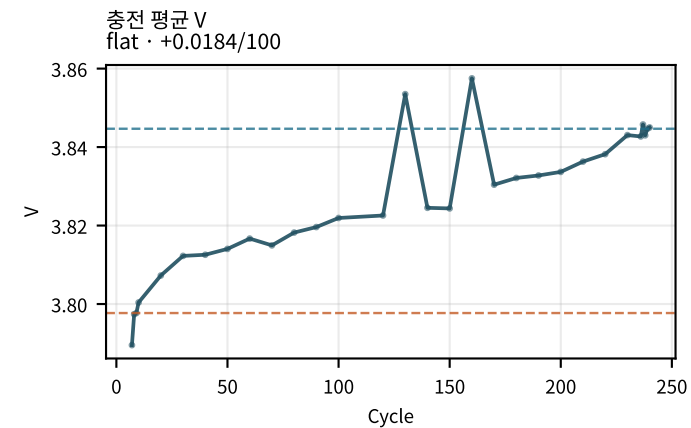
M02Ch103 · DCIR 분해 · 성장



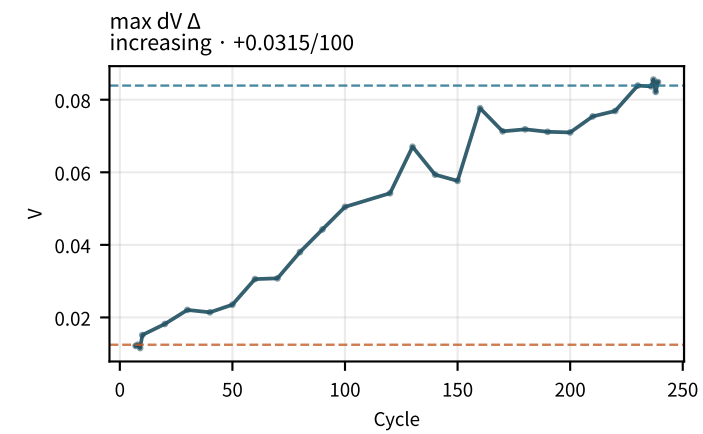
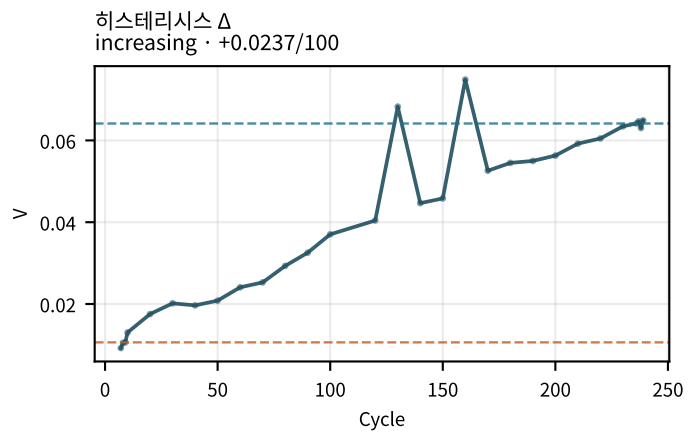
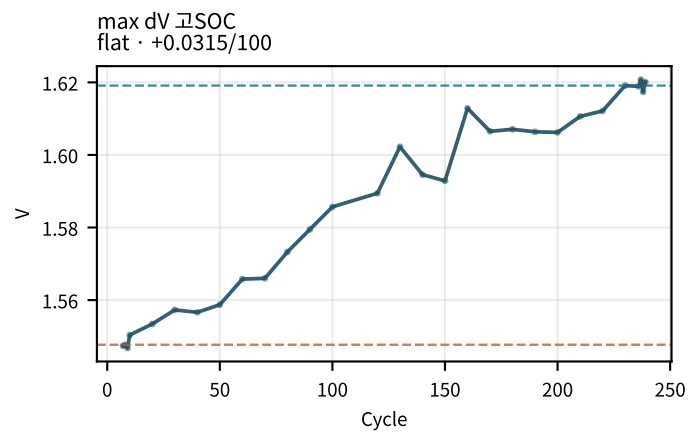
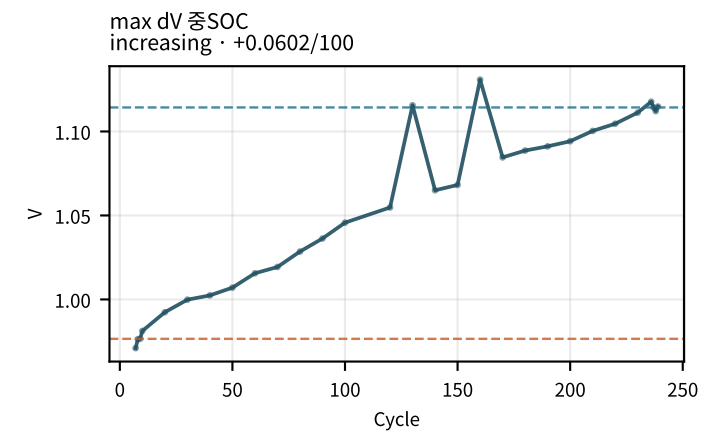
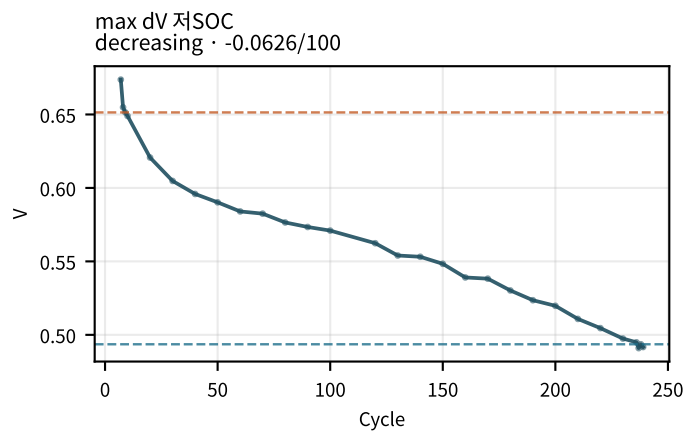
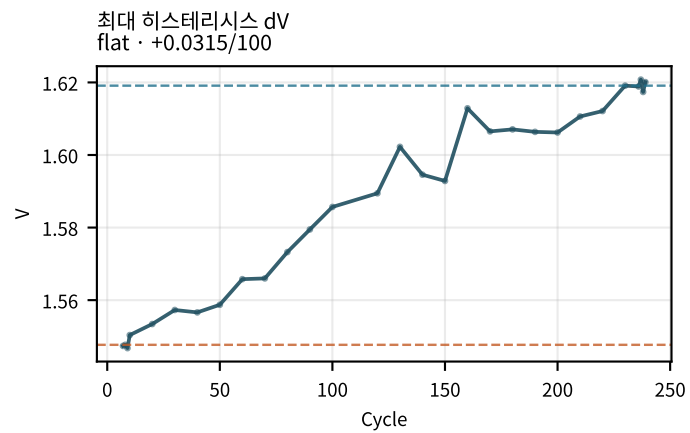
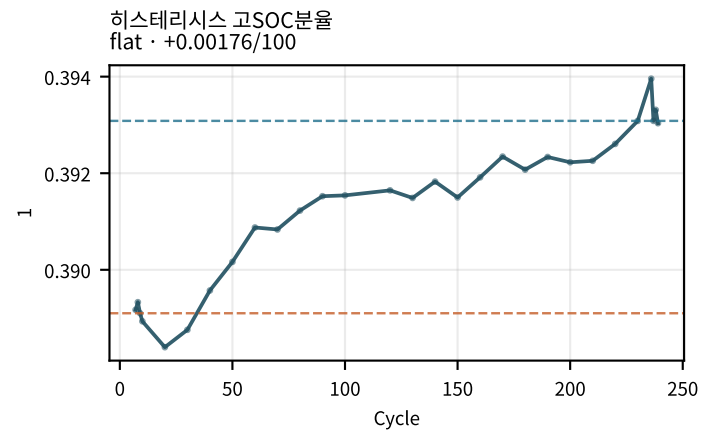
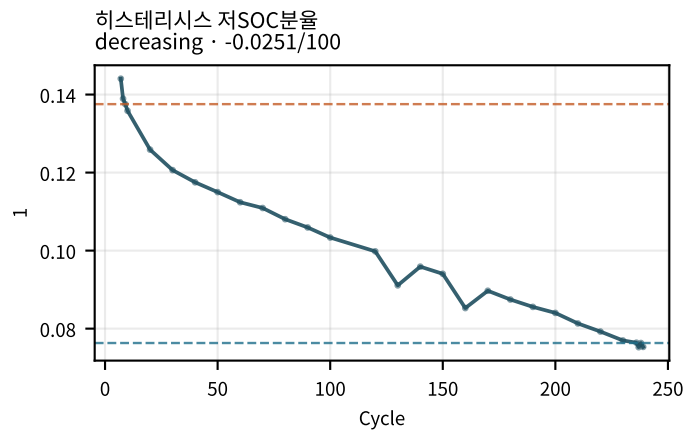
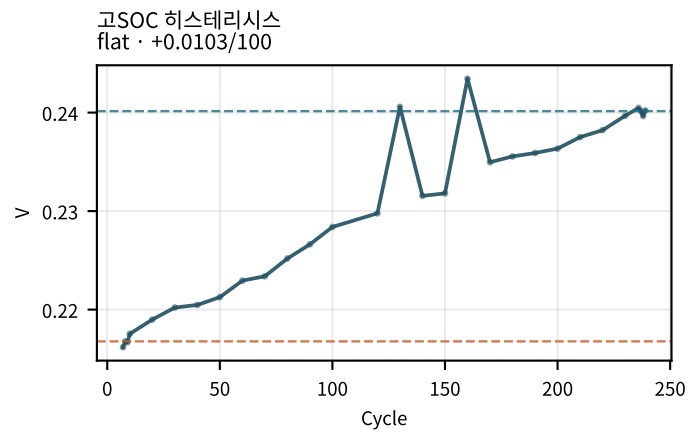
M02Ch103 · DCIR 분해 · 성장



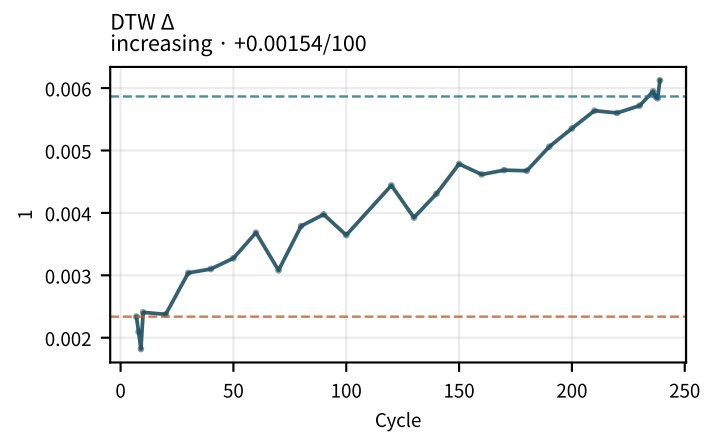
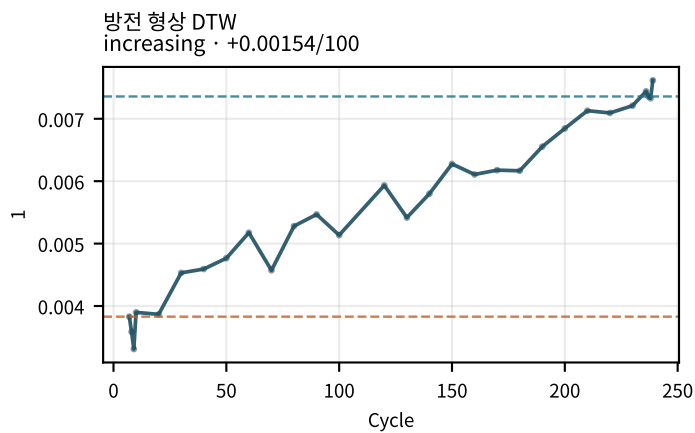
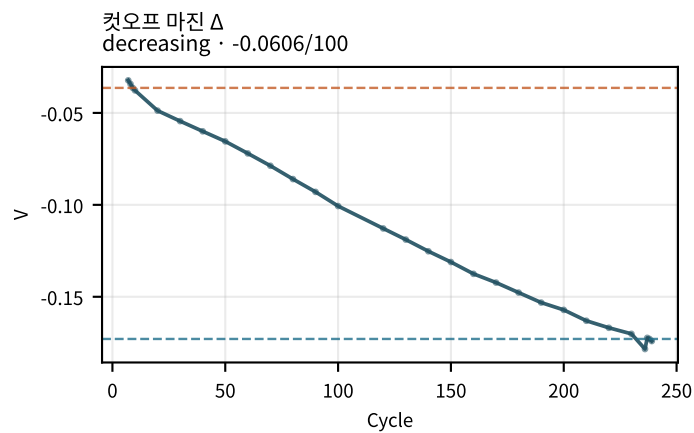
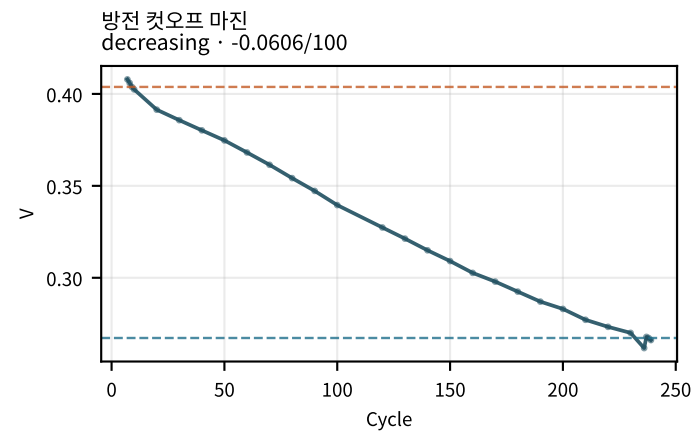
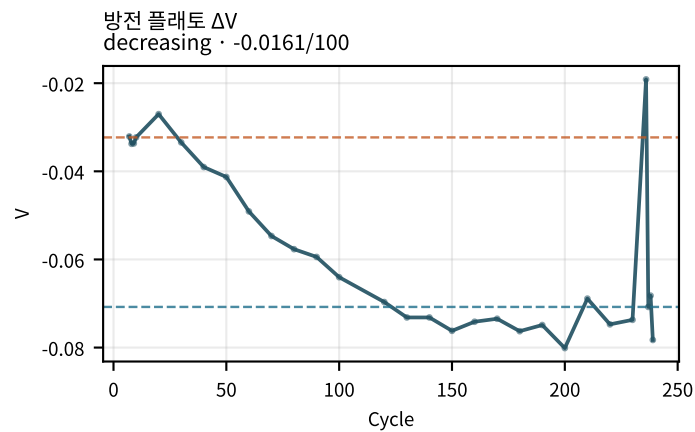
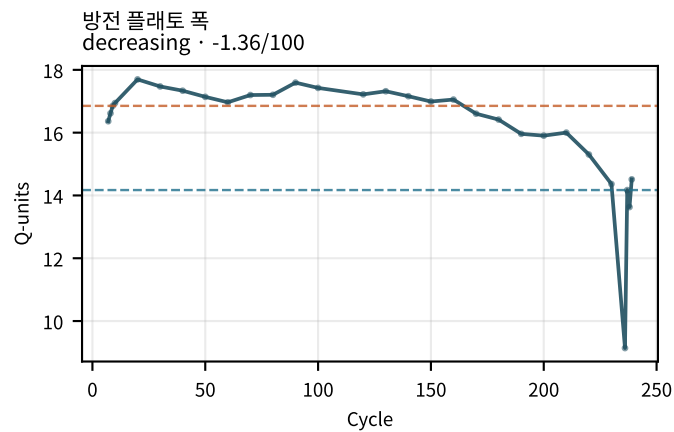
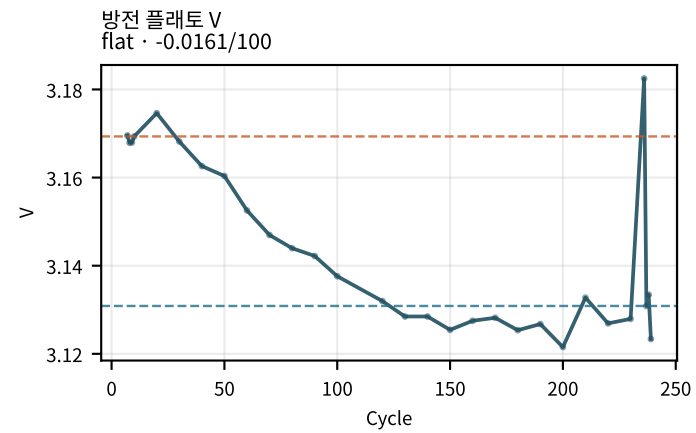
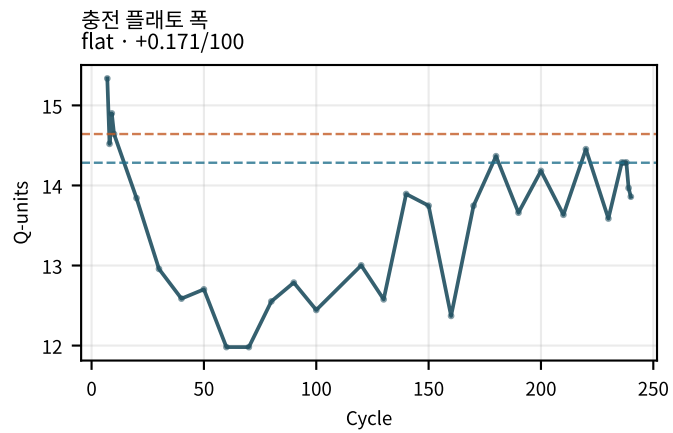
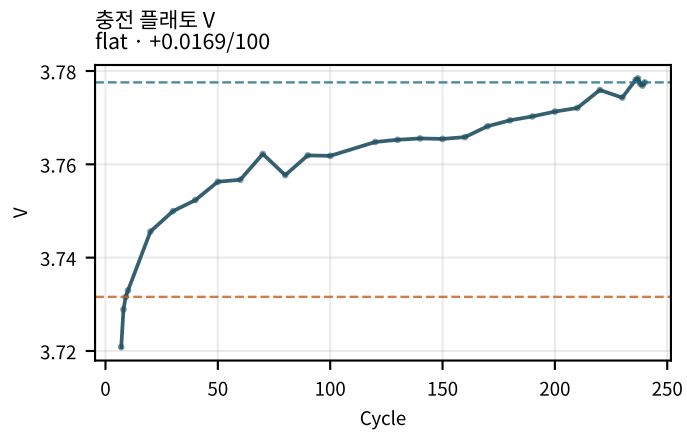
M02Ch103 · 곡선 형상 · 히스테리시스



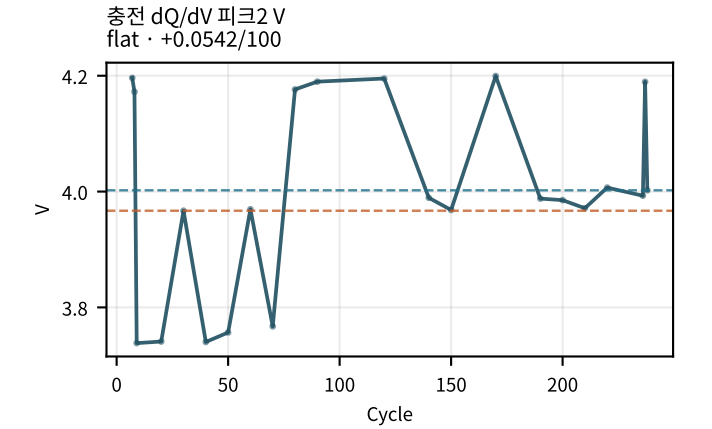
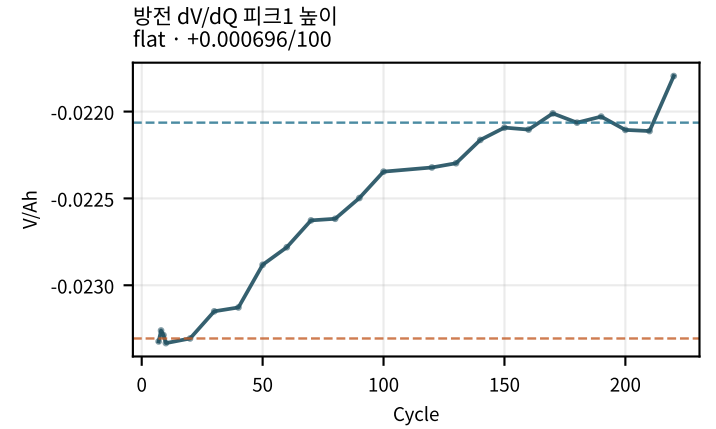
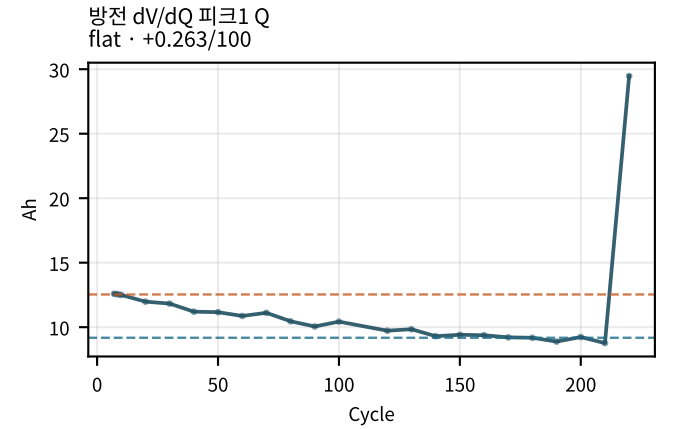
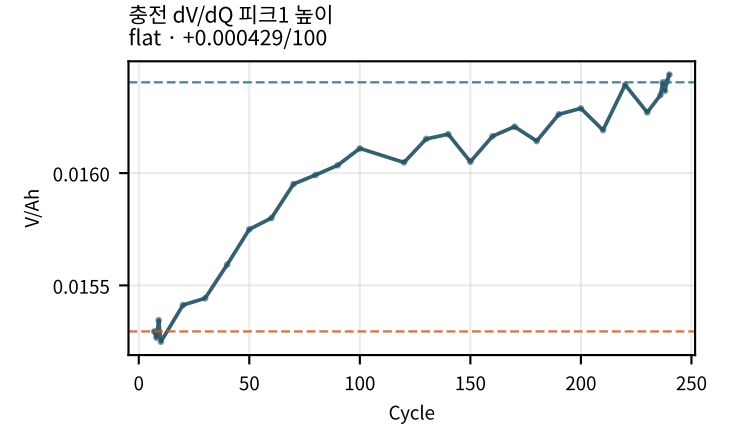
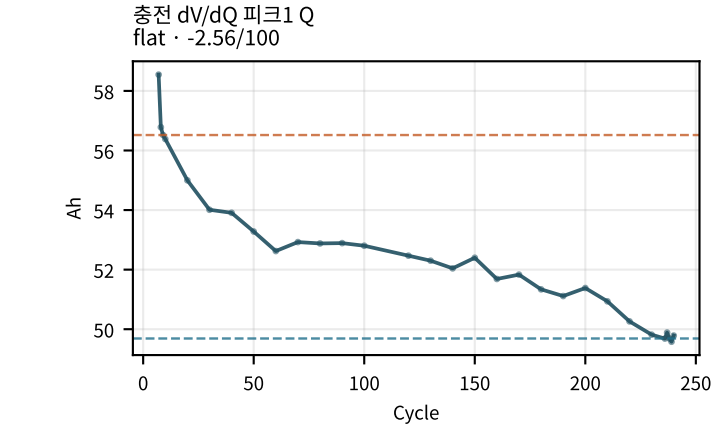
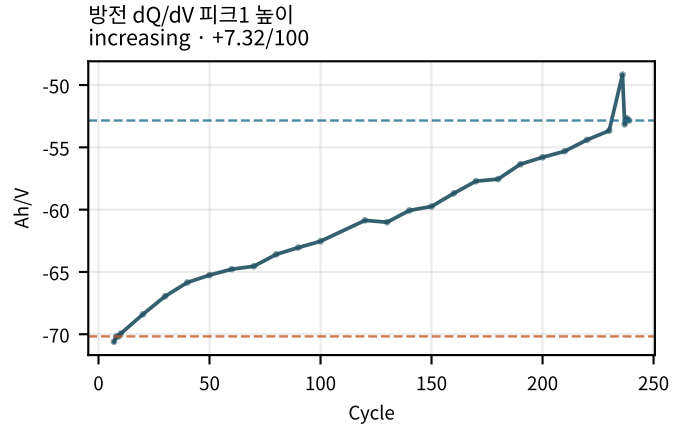
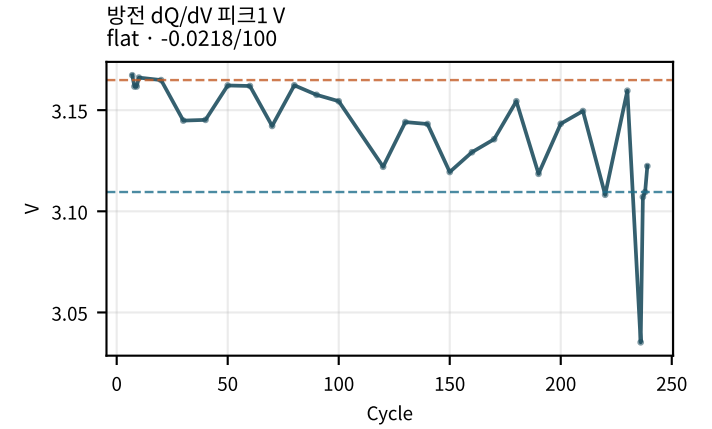
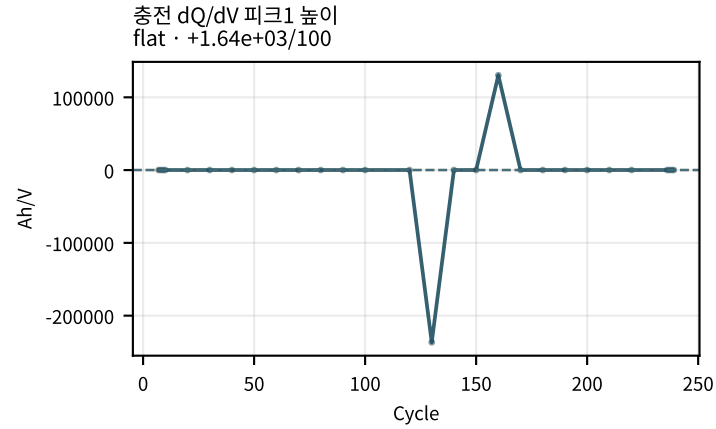
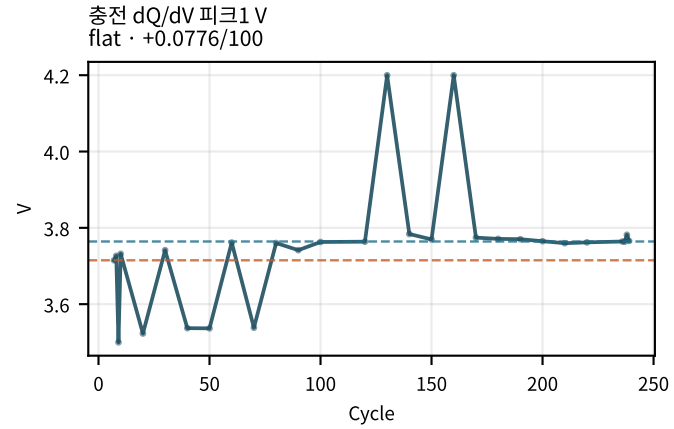
M02Ch103 · 곡선 형상 · 히스테리시스



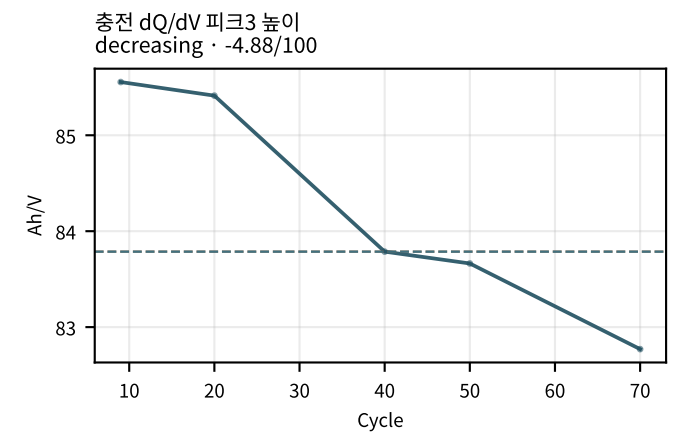
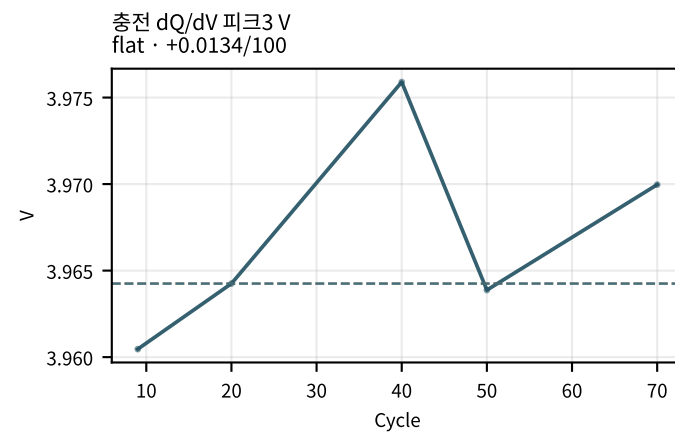
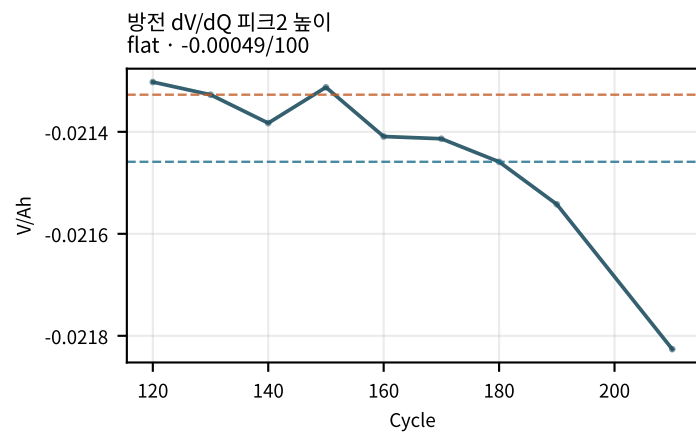
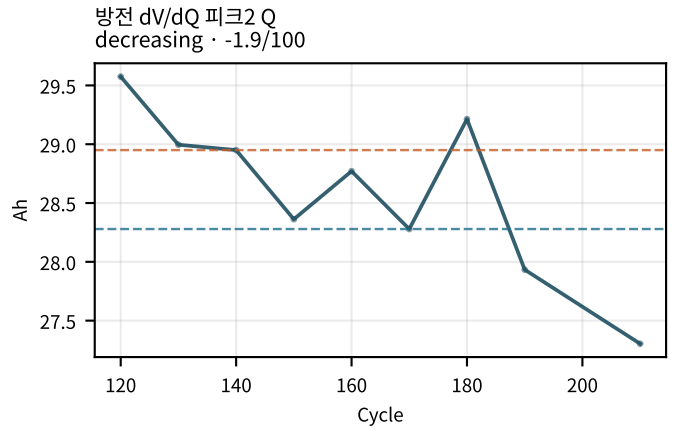
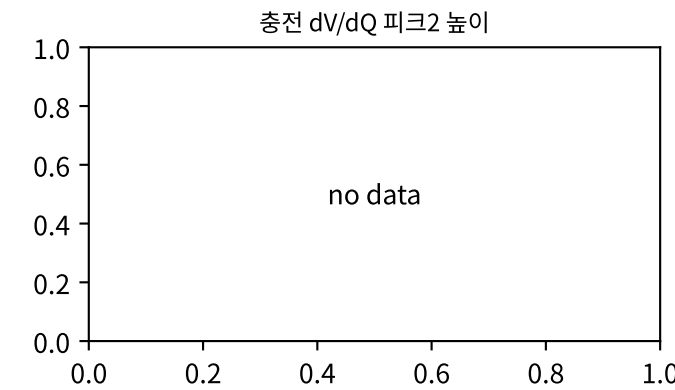
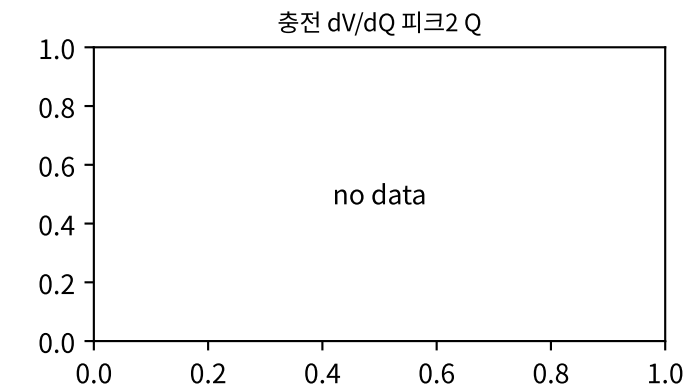
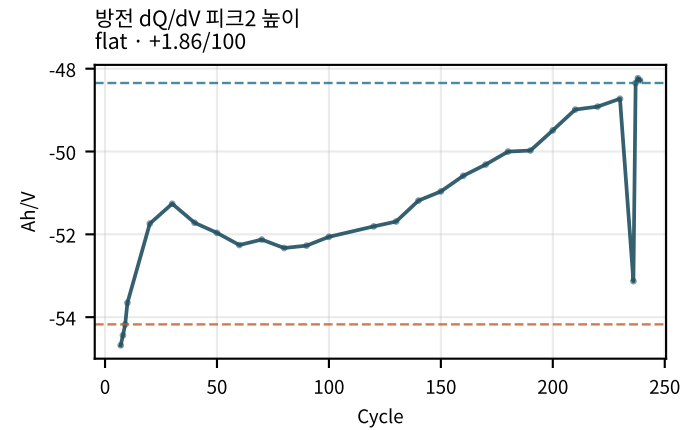
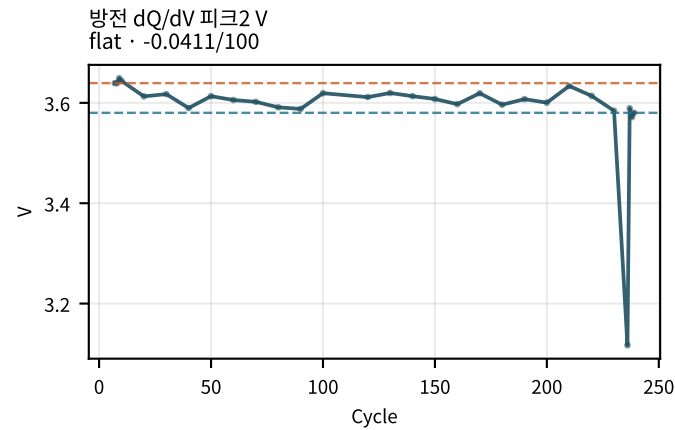
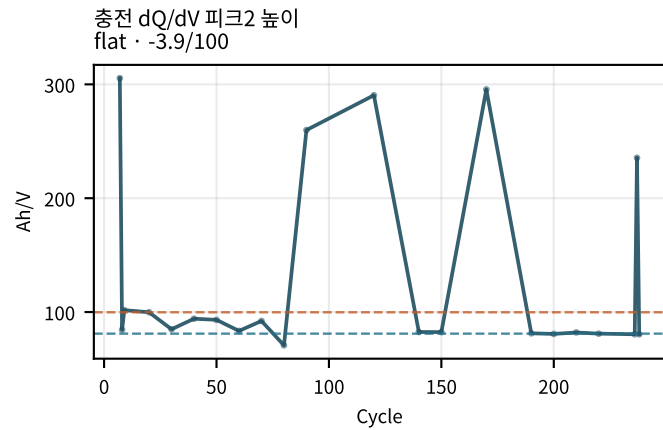
M02Ch103 · 곡선 형상 · 히스테리시스



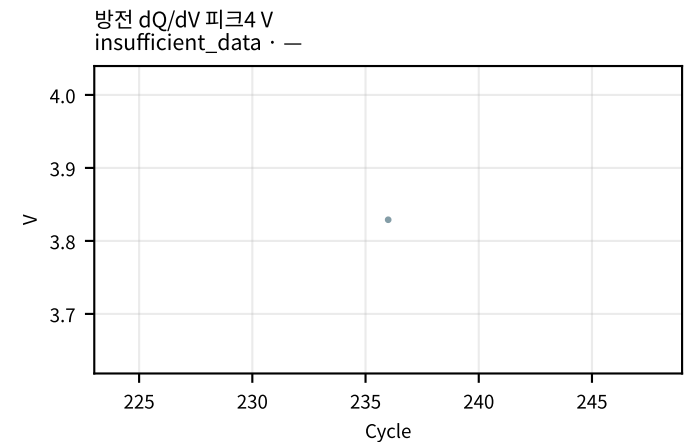
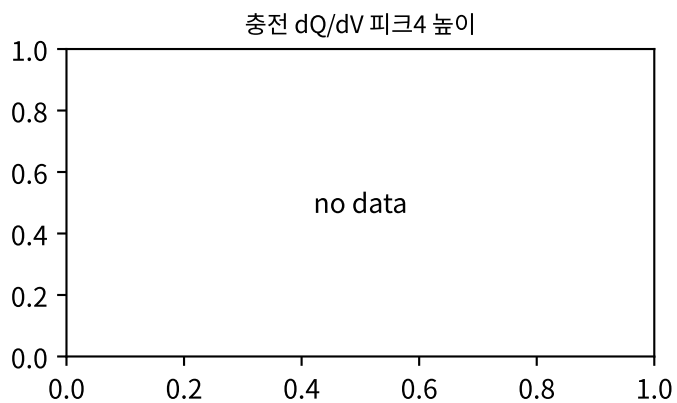
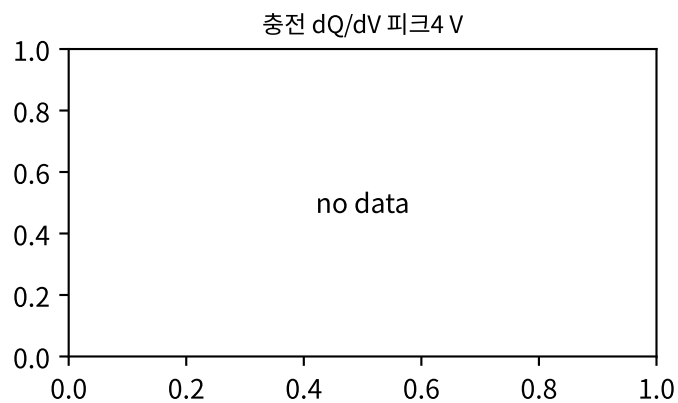
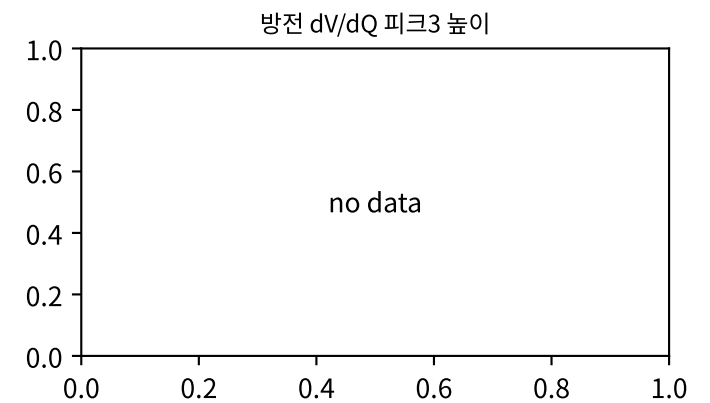
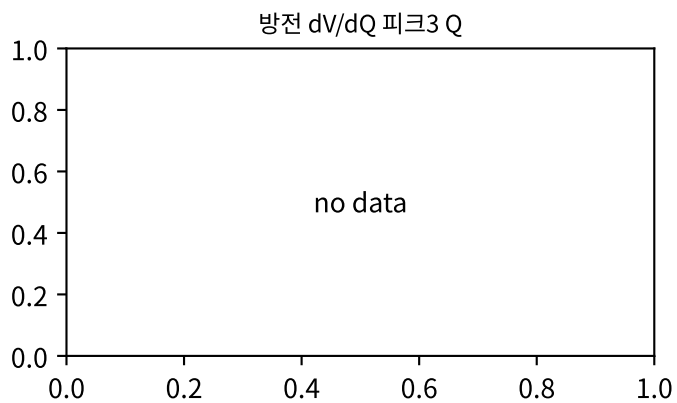
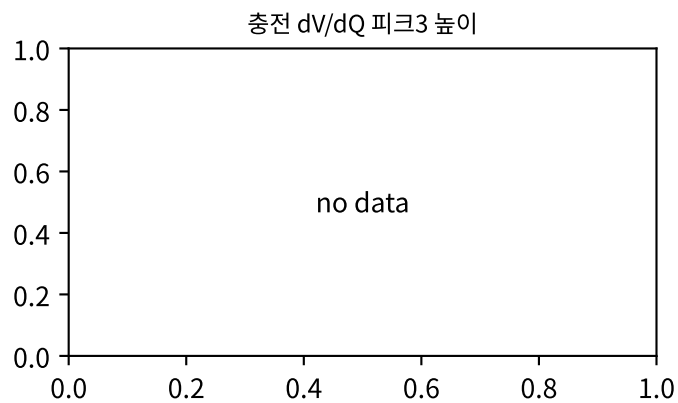
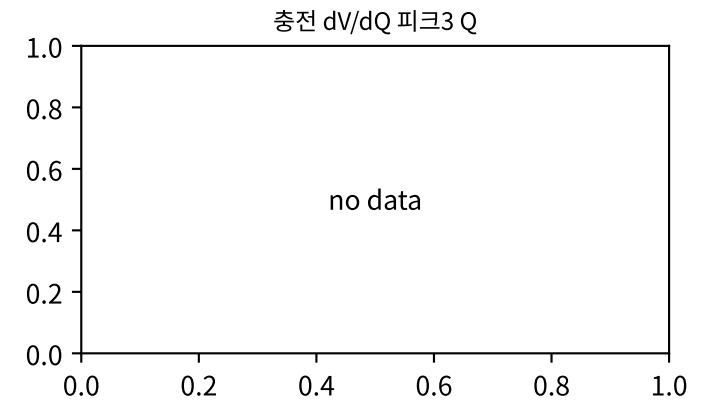
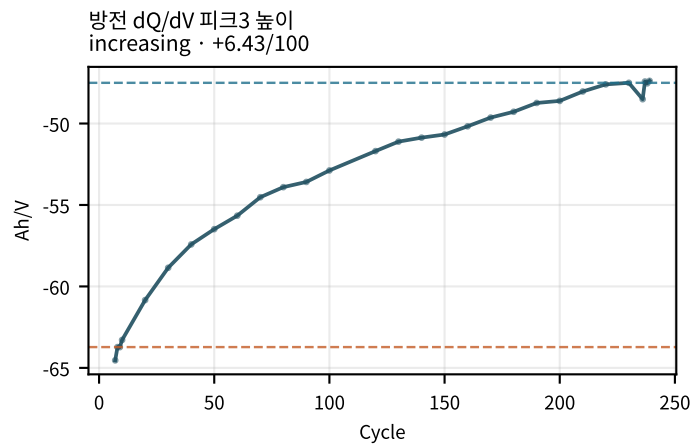
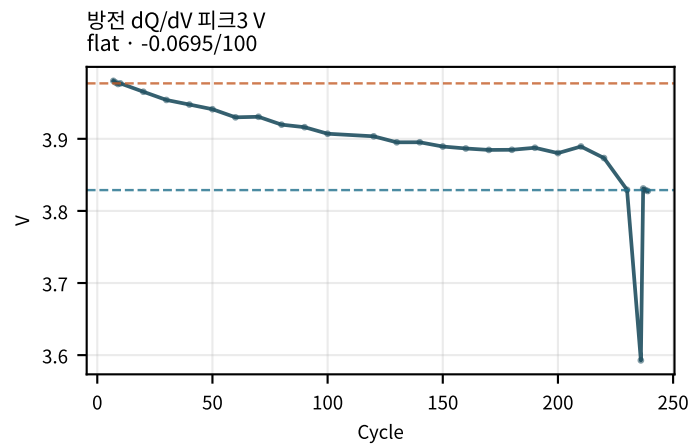
M02Ch103 · dQ/dV · dV/dQ 피크



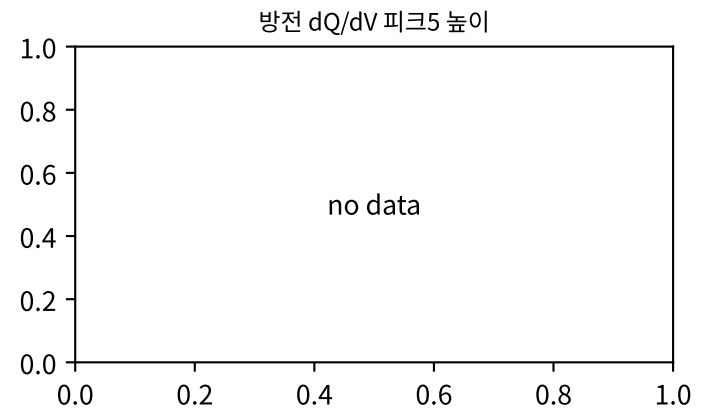
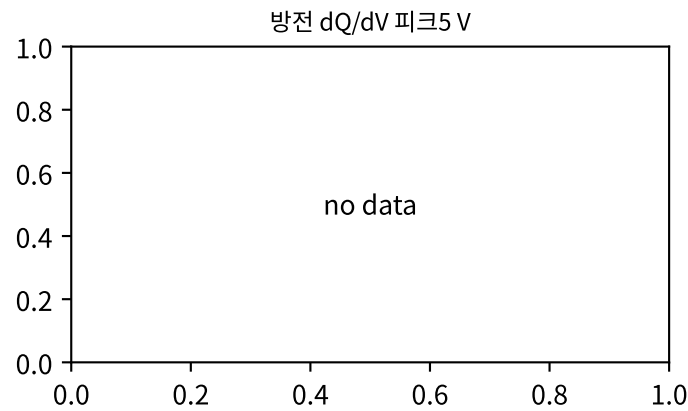
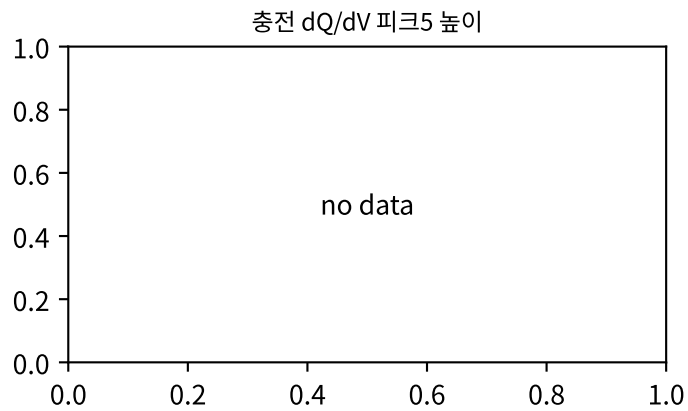
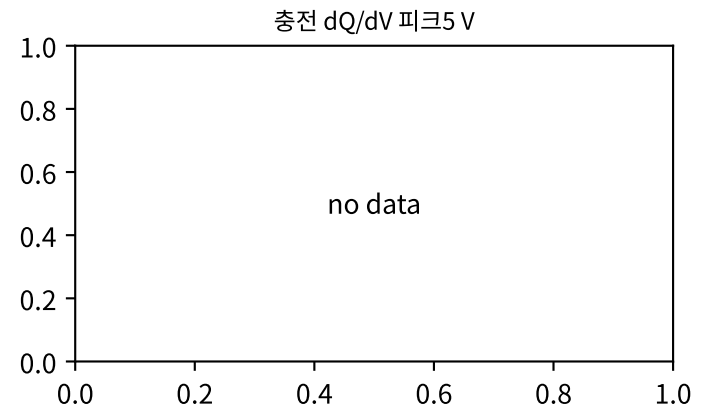
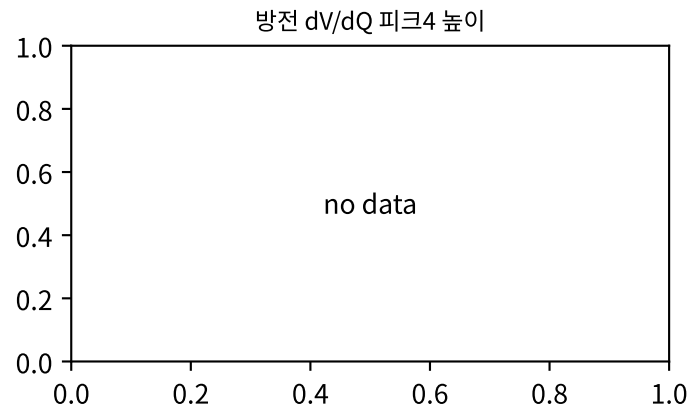
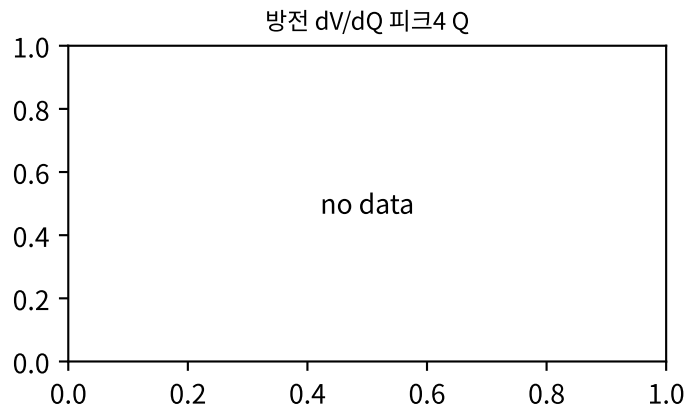
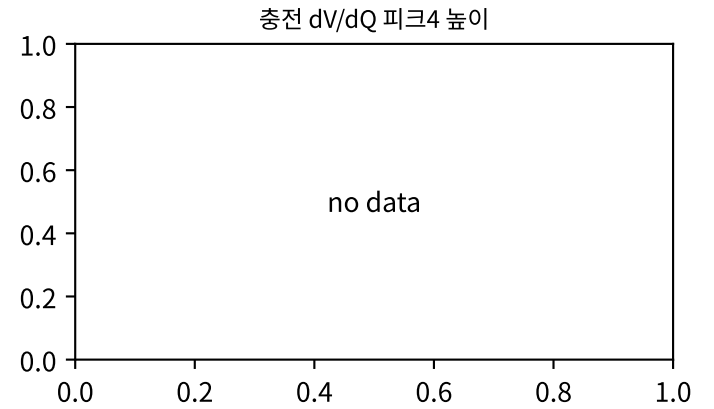
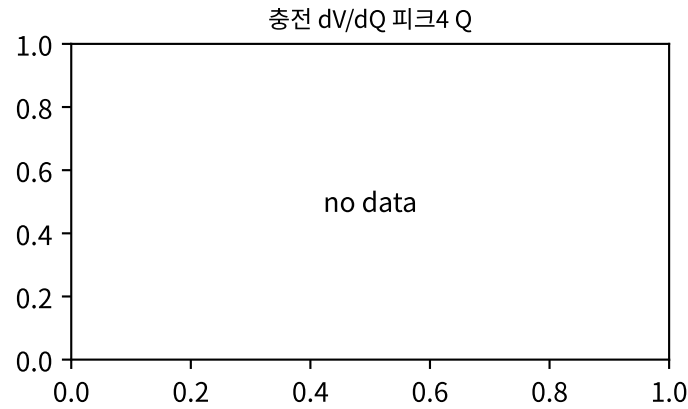
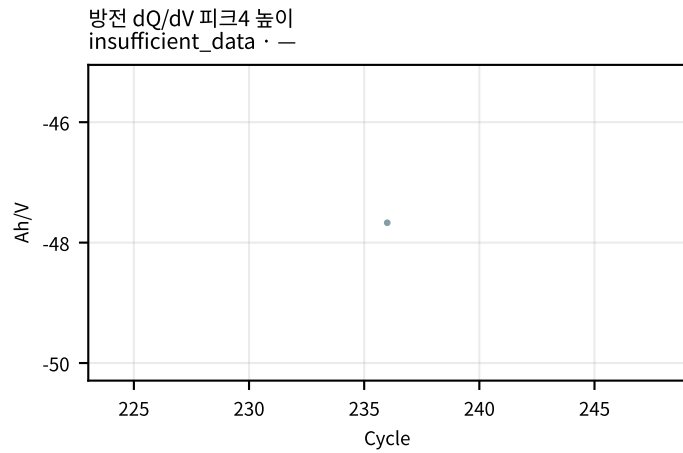
M02Ch103 · dQ/dV · dV/dQ 피크



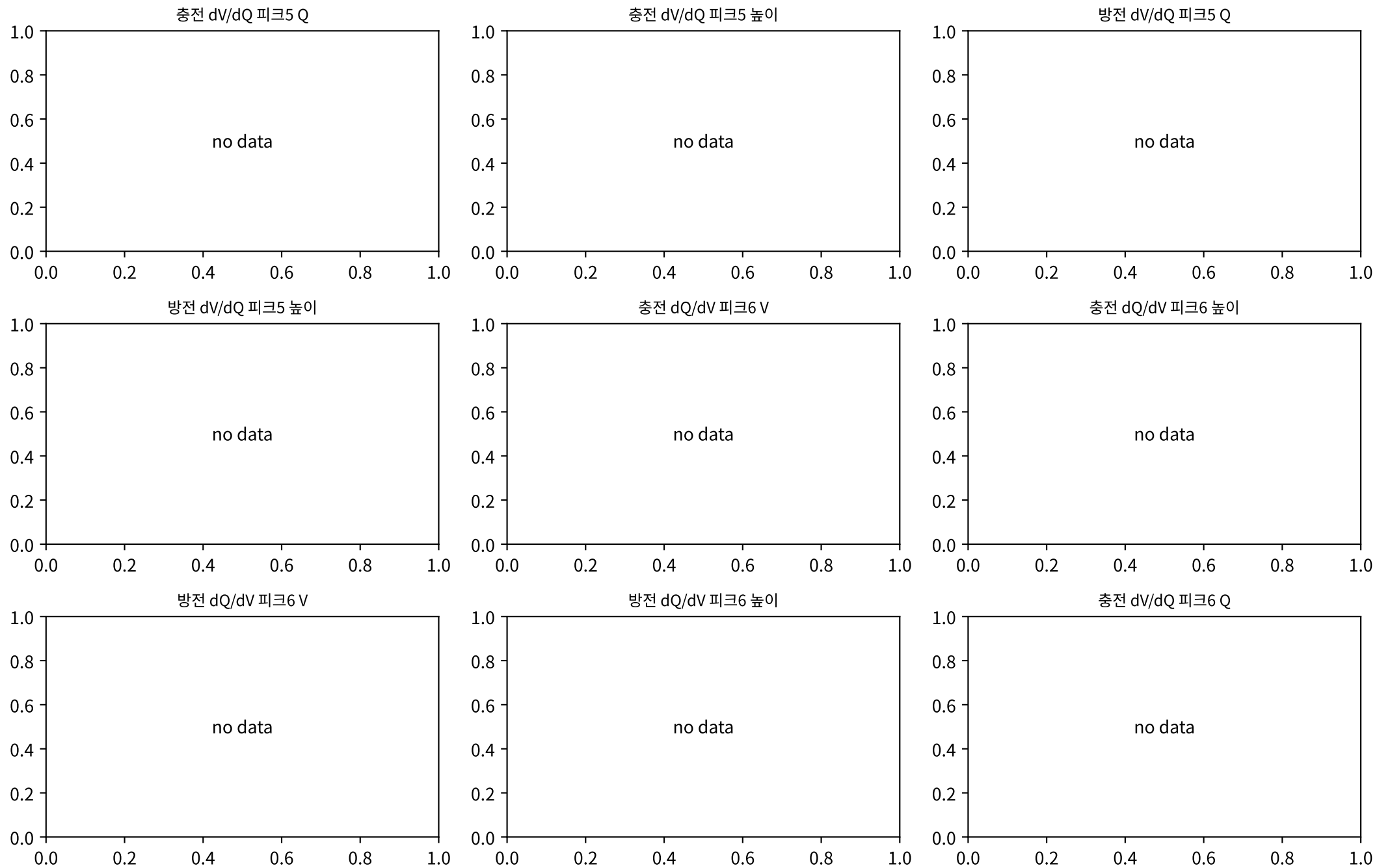
M02Ch103 · dQ/dV · dV/dQ 피크



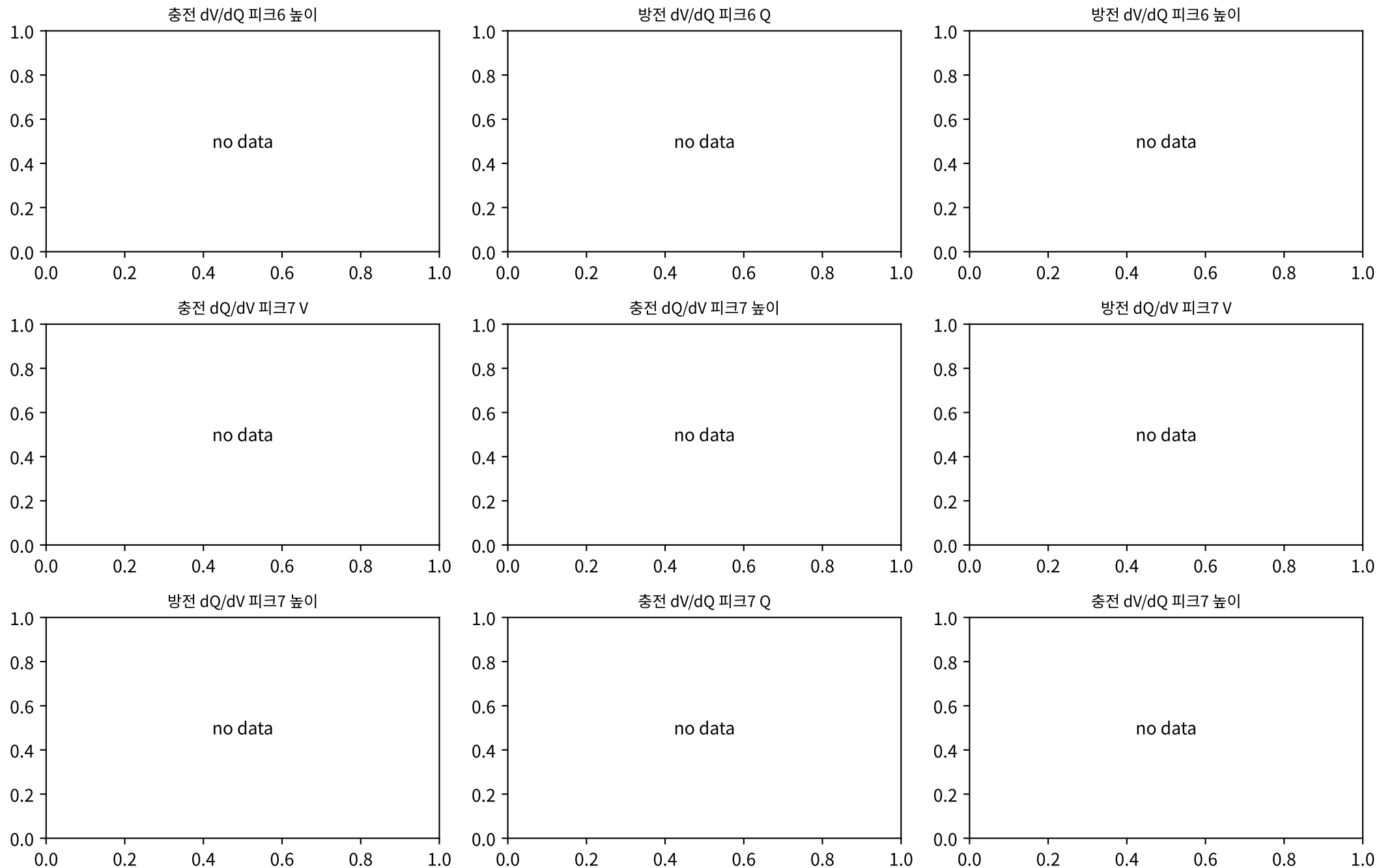
M02Ch103 · dQ/dV · dV/dQ 피크



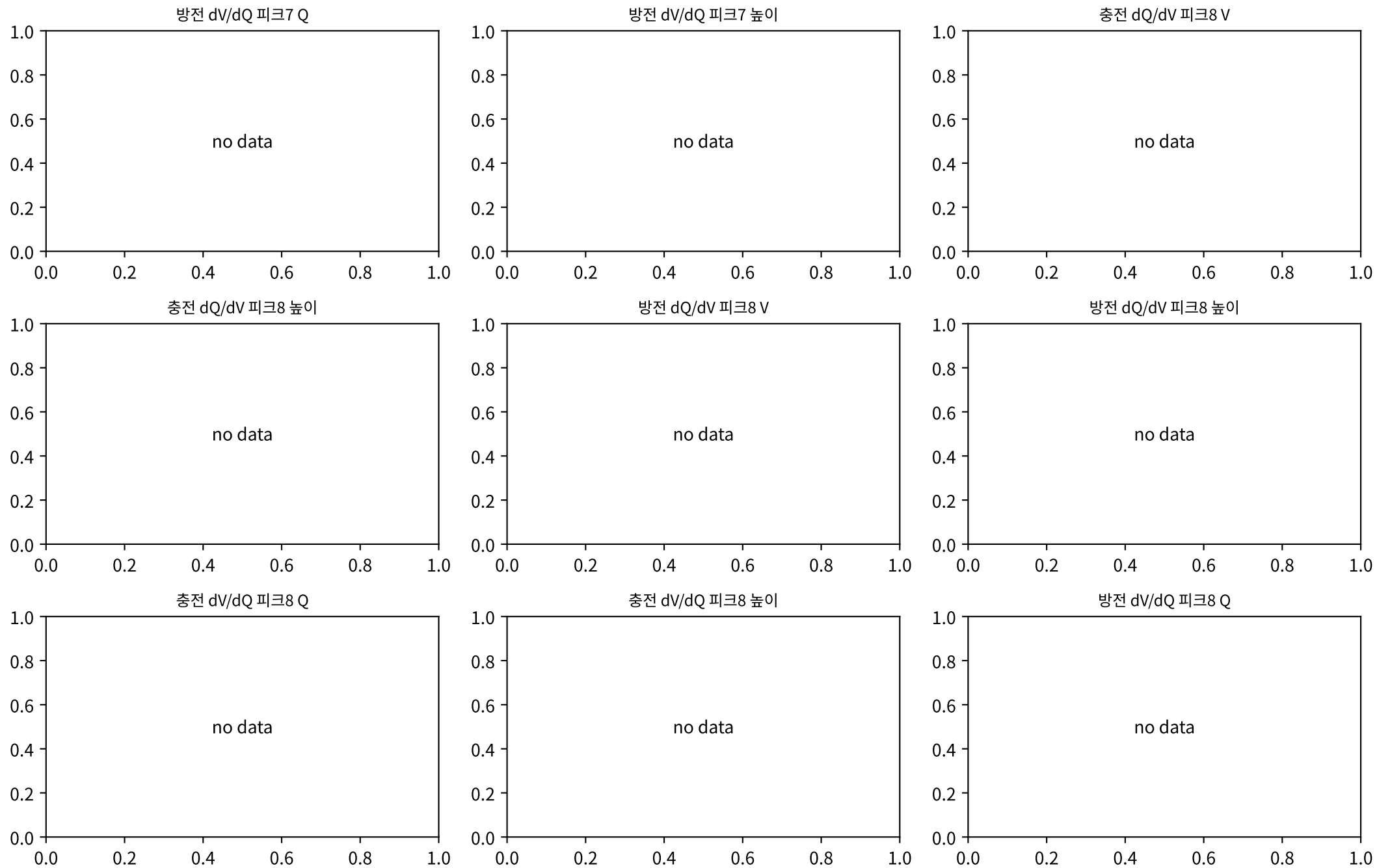
M02Ch103 · dQ/dV · dV/dQ 피크



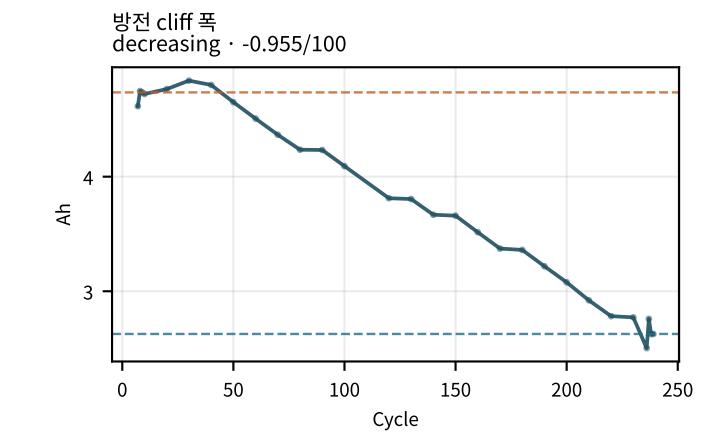
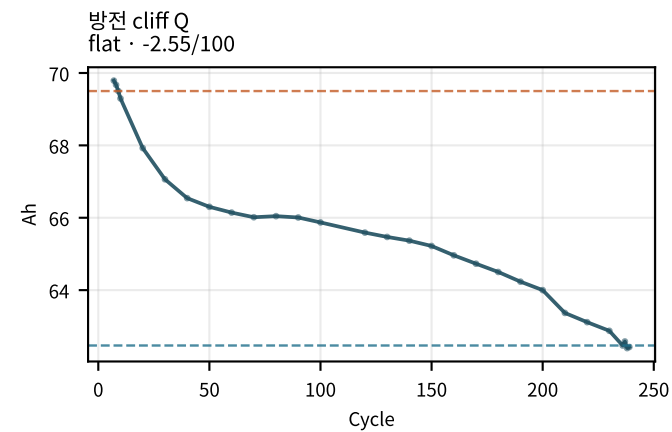
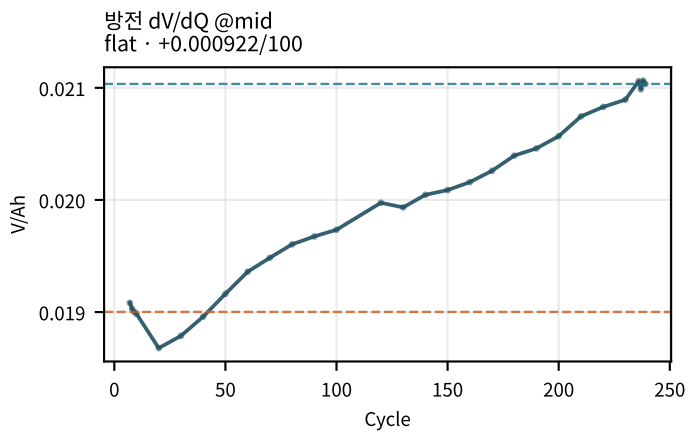
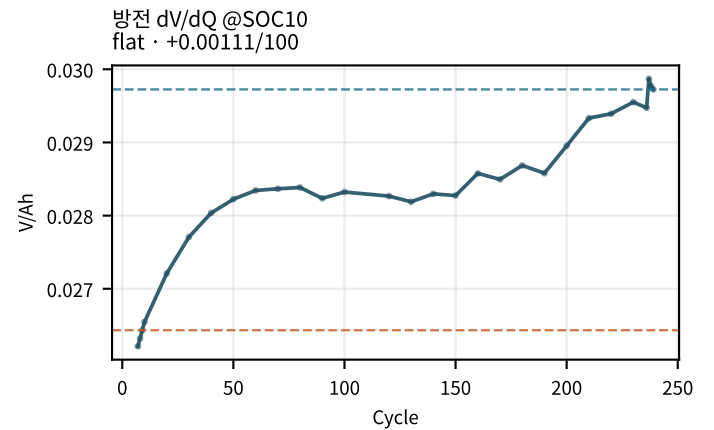
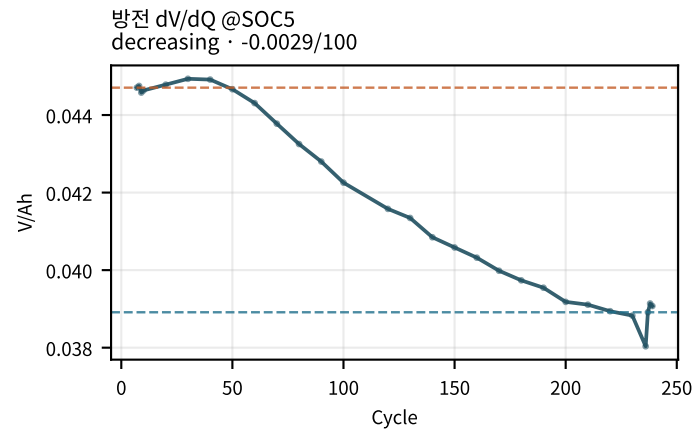
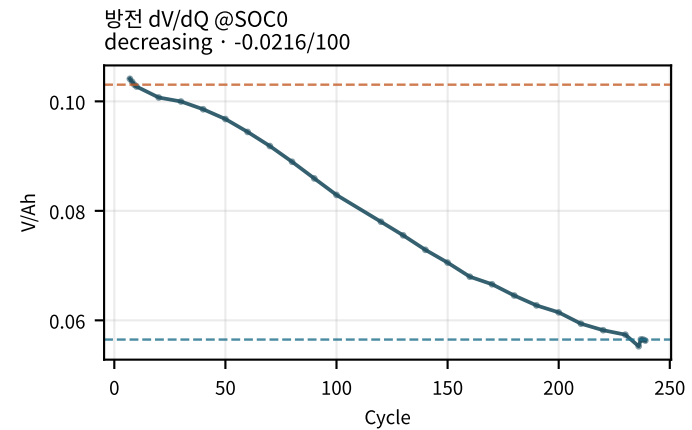
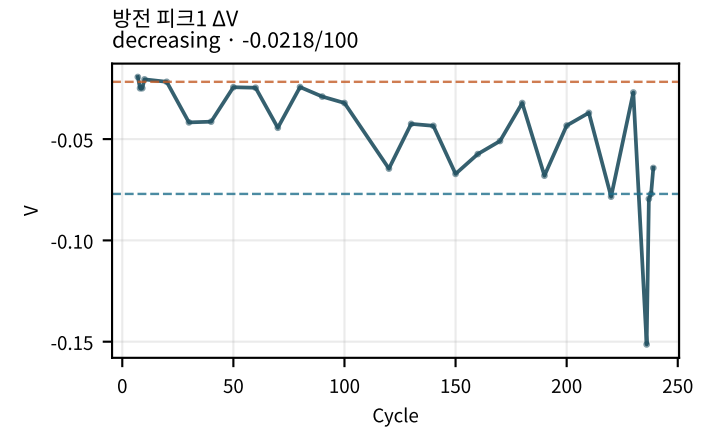
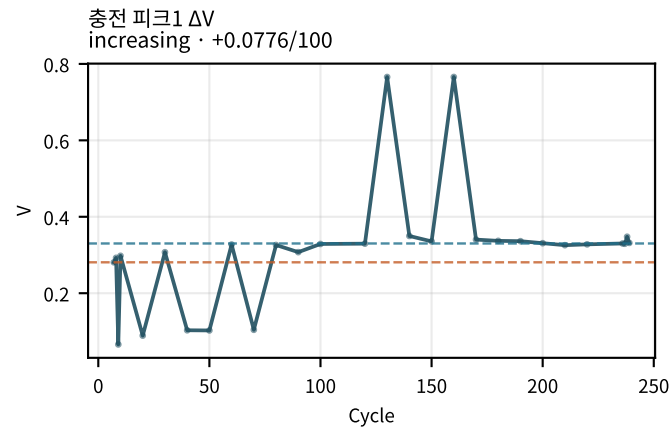
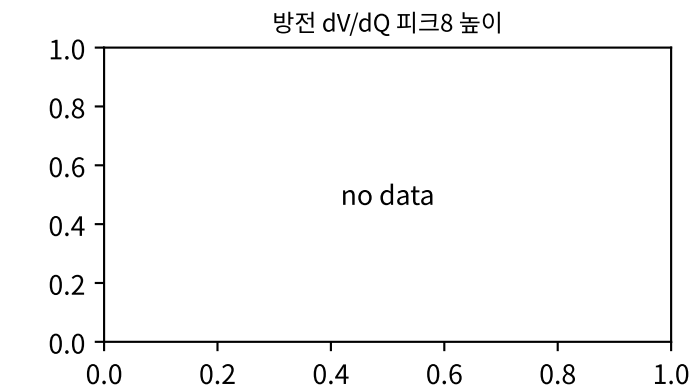
M02Ch103 · dQ/dV · dV/dQ 피크



M02Ch103 · dQ/dV · dV/dQ 피크

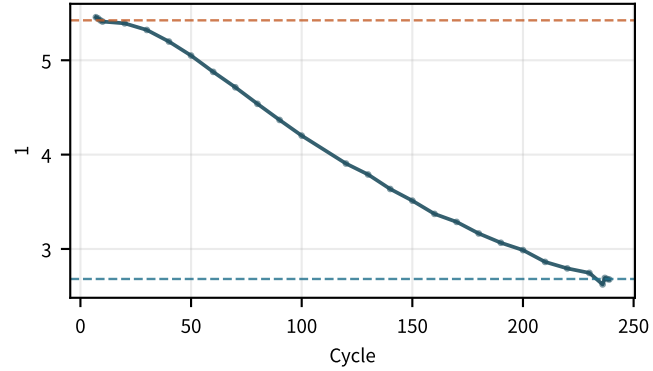


M02Ch103 · dQ/dV · dV/dQ 피크

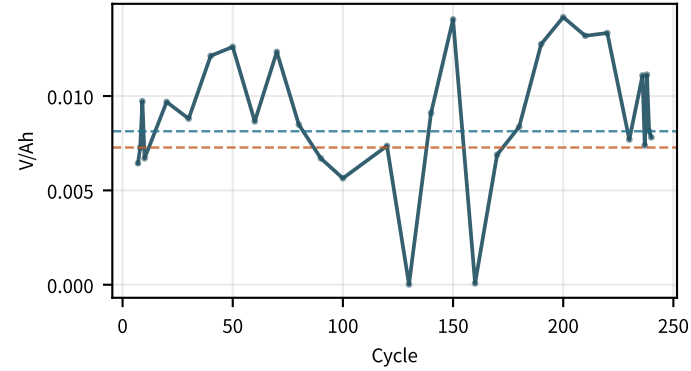


M02Ch103 · dQ/dV · dV/dQ 피크

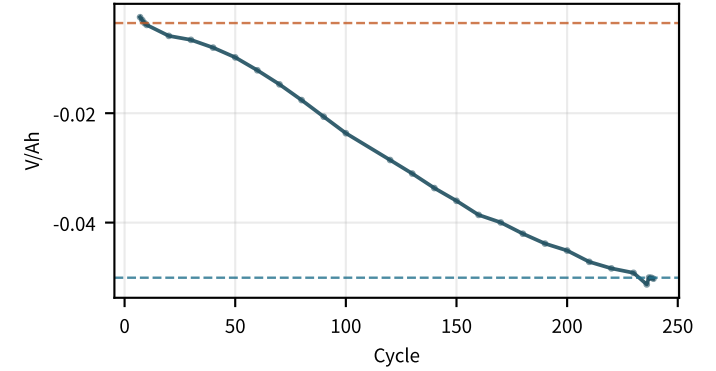
cliff/mid 비
decreasing · -1.27/100



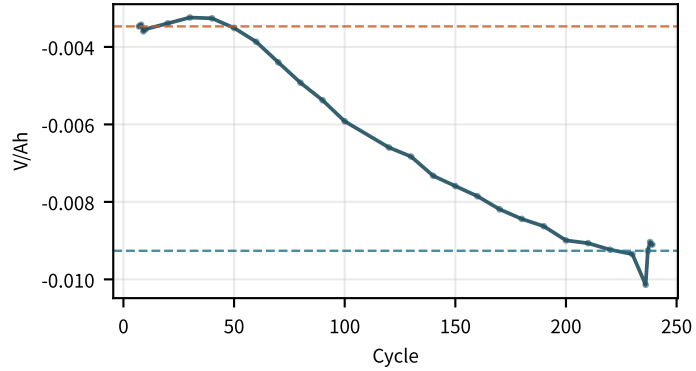
충전 dV/dQ @100
increasing · +0.000534/100



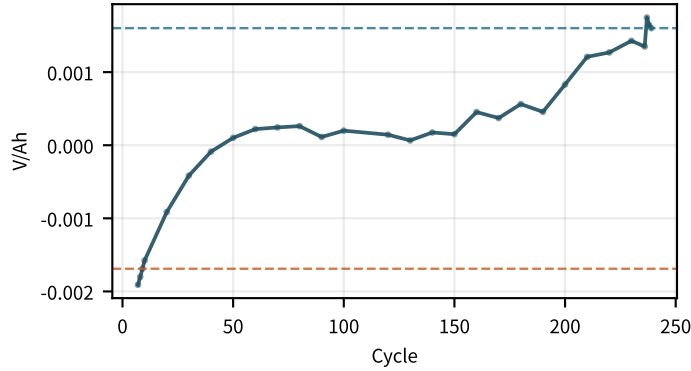
dV/dQ SOC0 Δ
decreasing · -0.0216/100



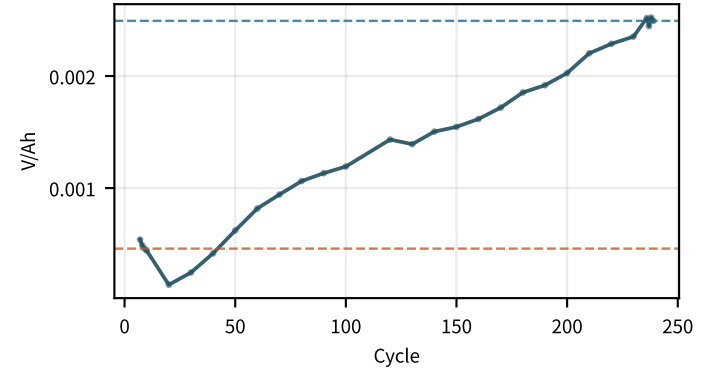
dV/dQ SOC5 Δ
decreasing · -0.0029/100



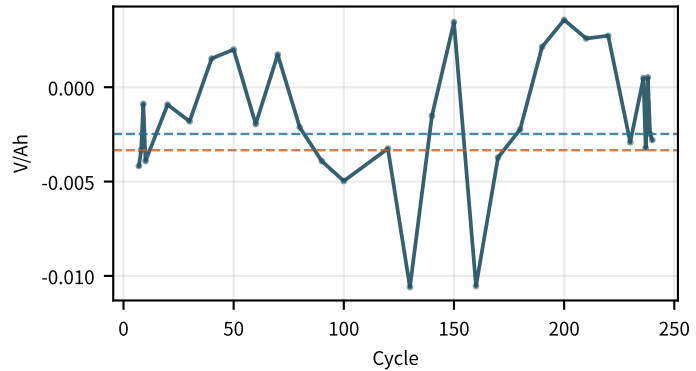
dV/dQ SOC10 Δ
increasing · +0.00111/100



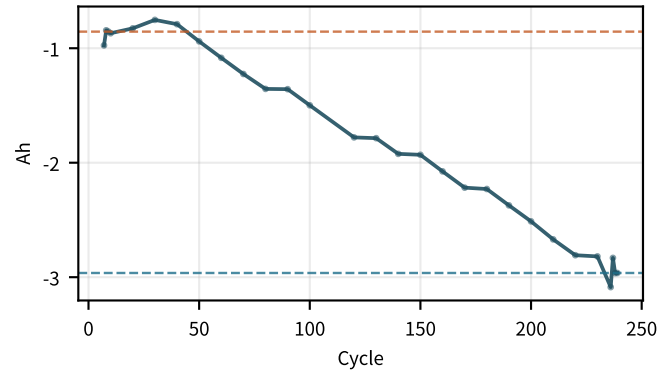
dV/dQ mid Δ
increasing · +0.000922/100



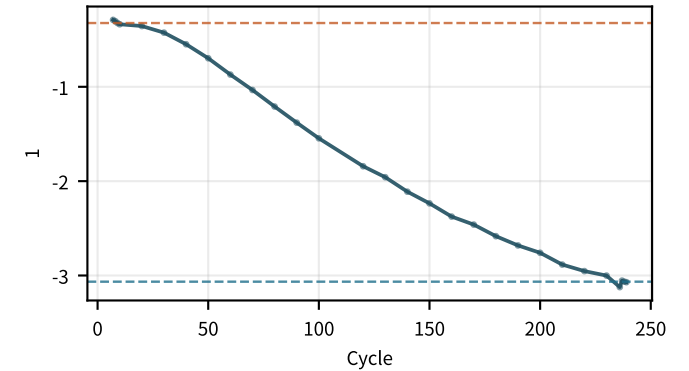
dV/dQ 100 Δ
increasing · +0.000534/100



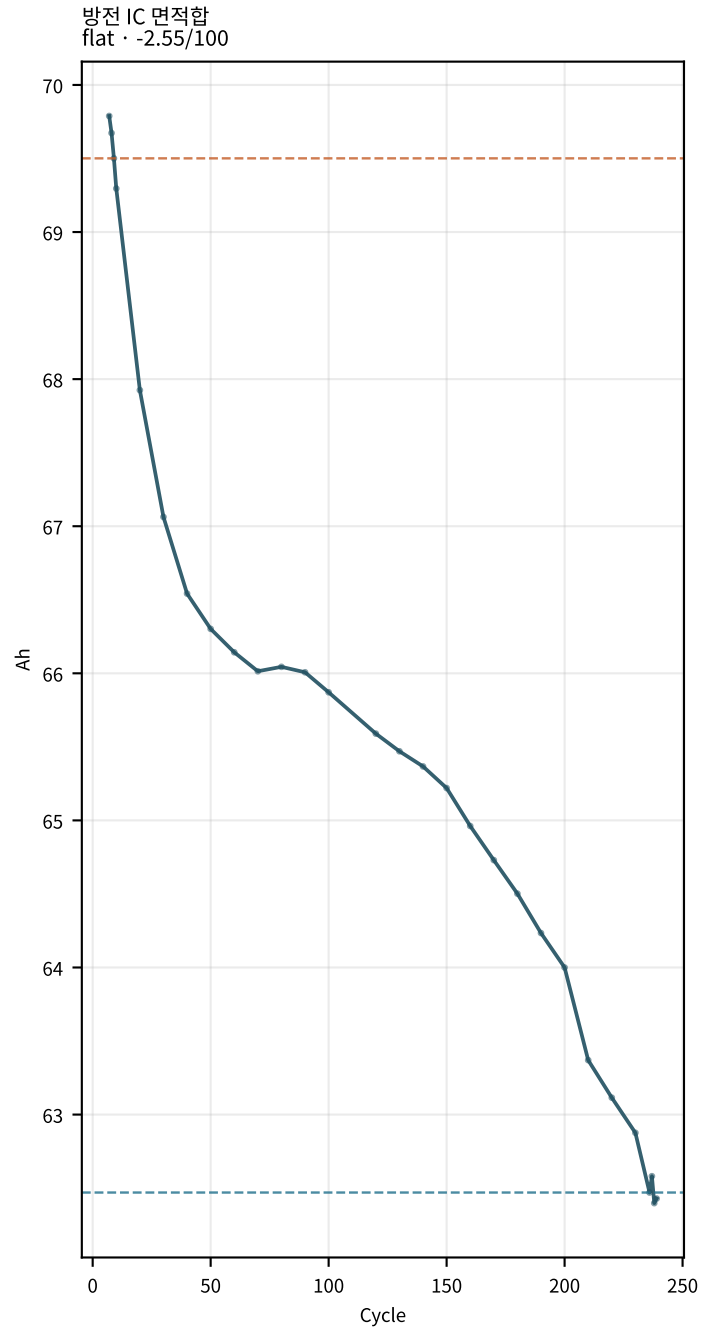
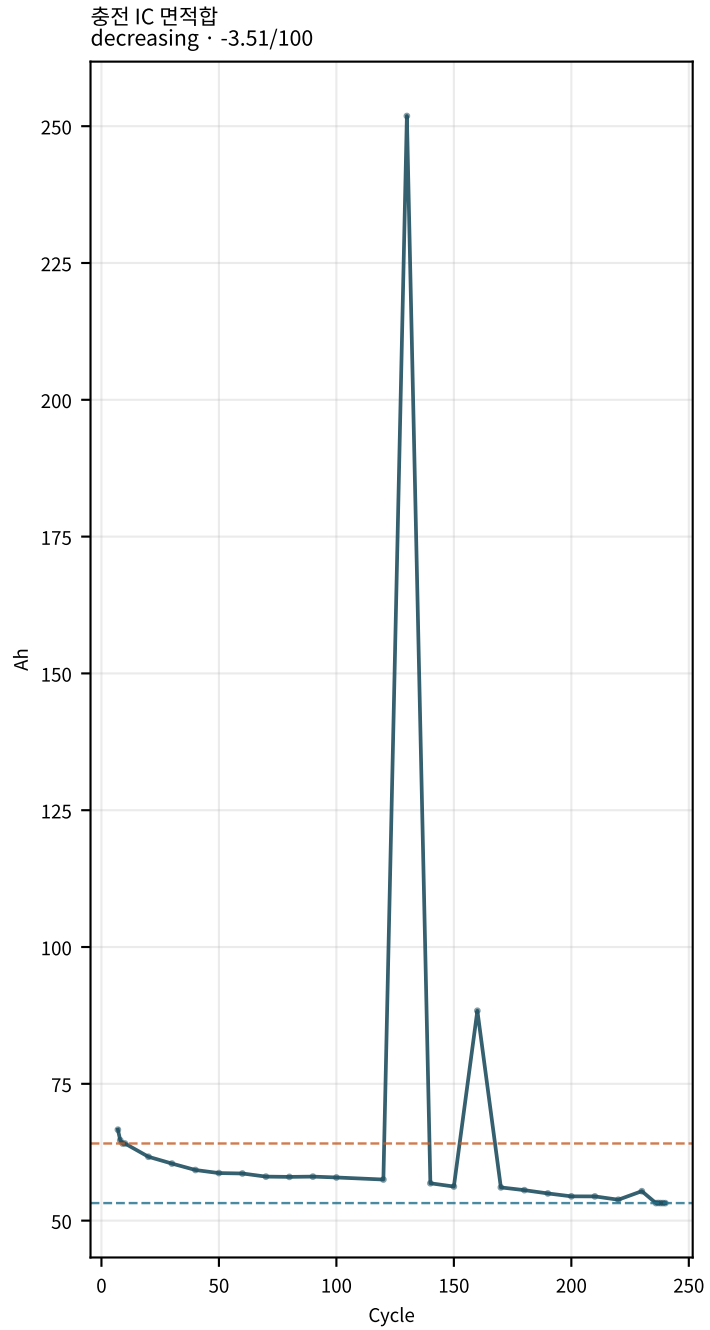
cliff 폭 Δ
decreasing · -0.955/100



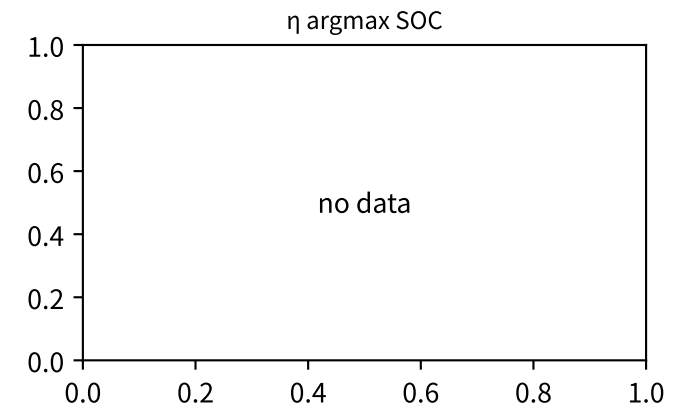
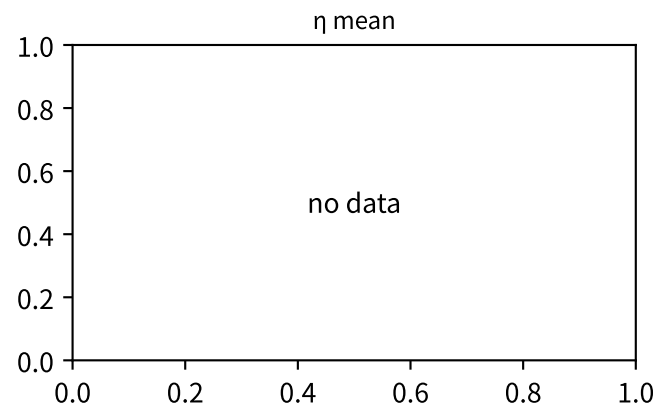
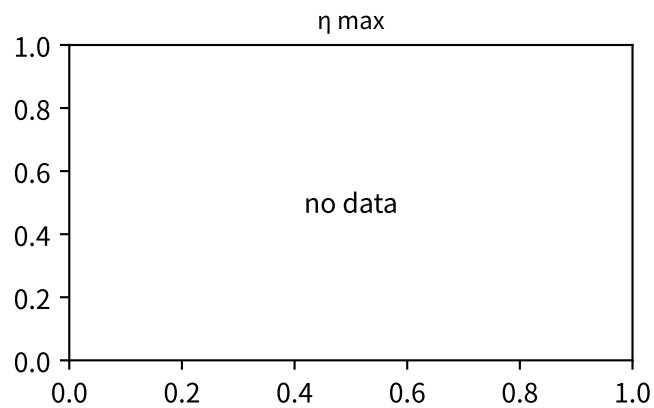
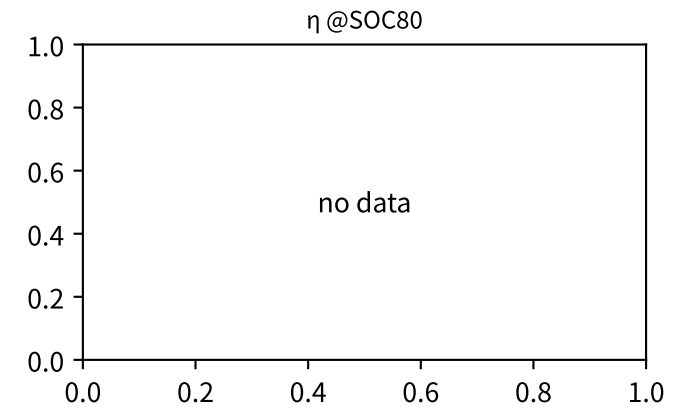
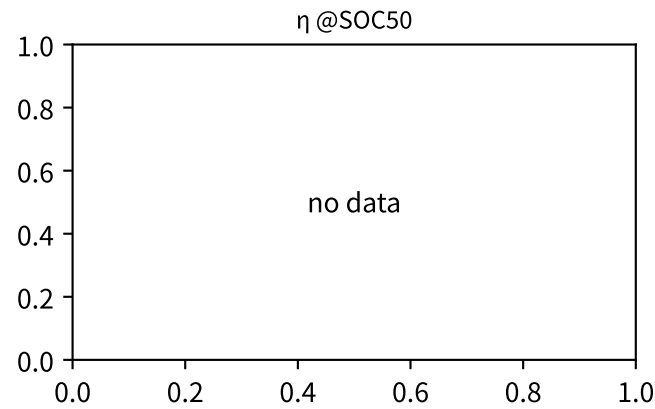
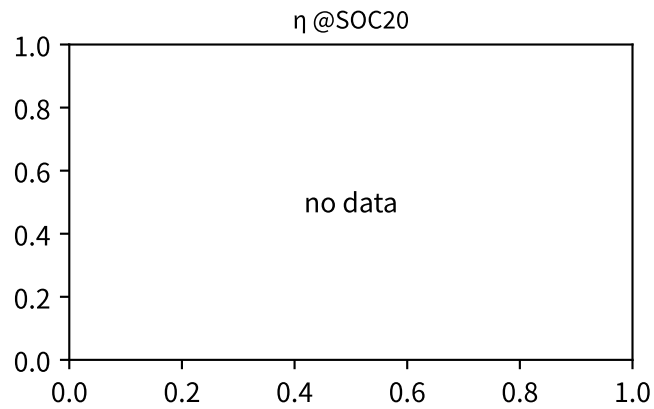
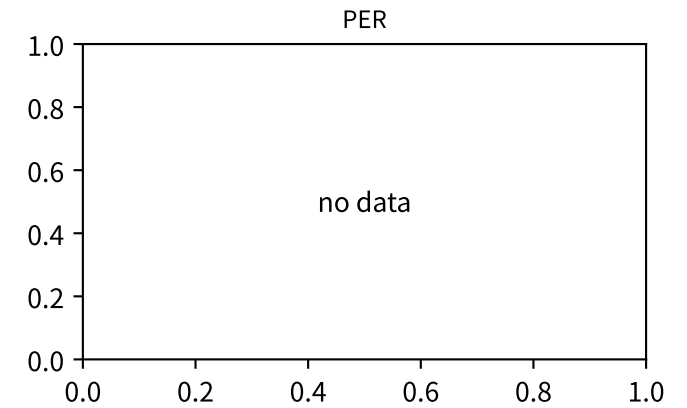
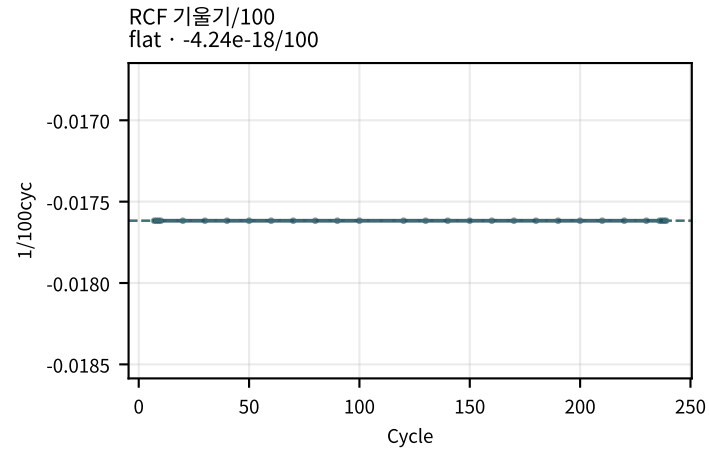
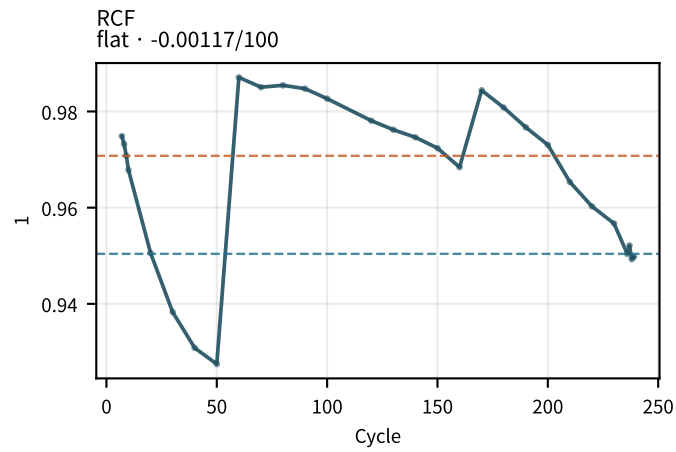
cliff/mid Δ
decreasing · -1.27/100



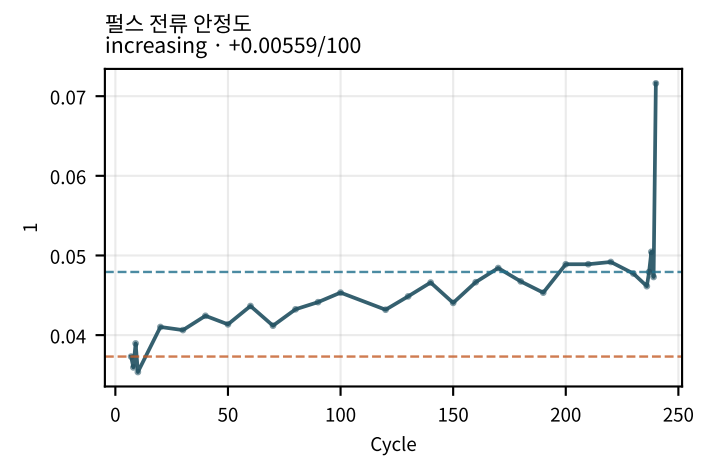
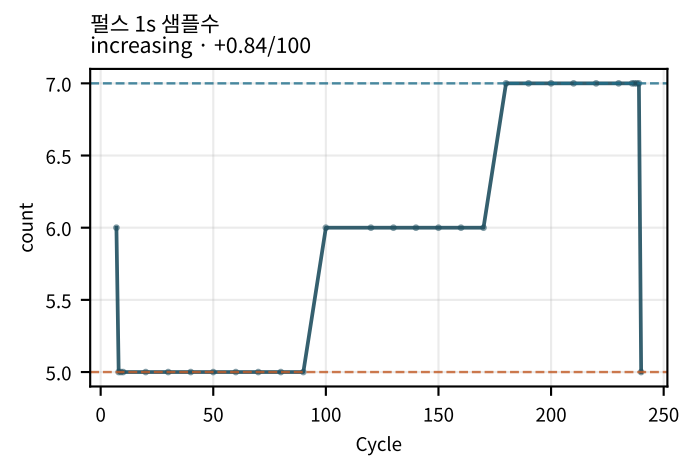
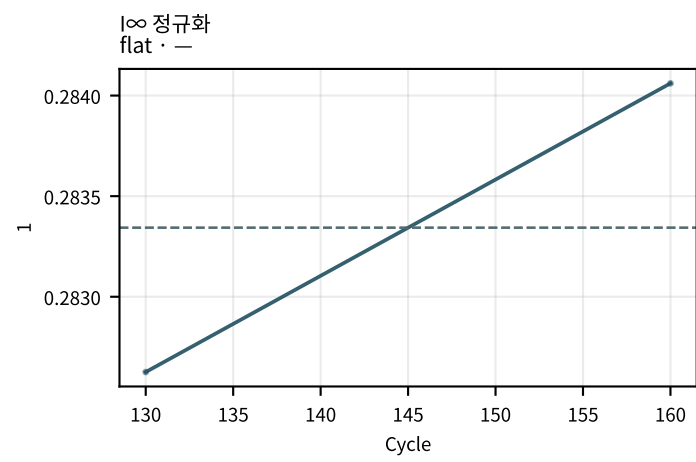
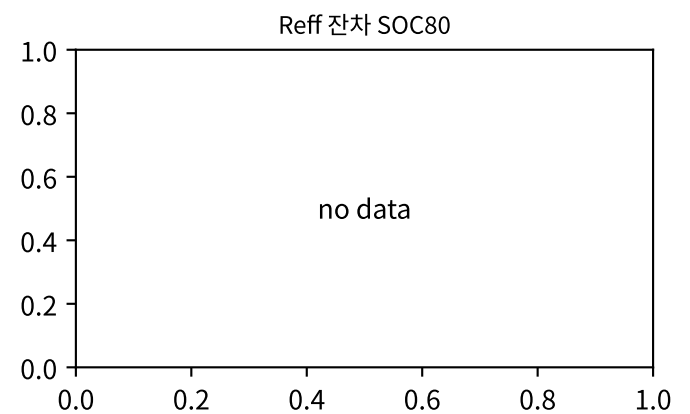
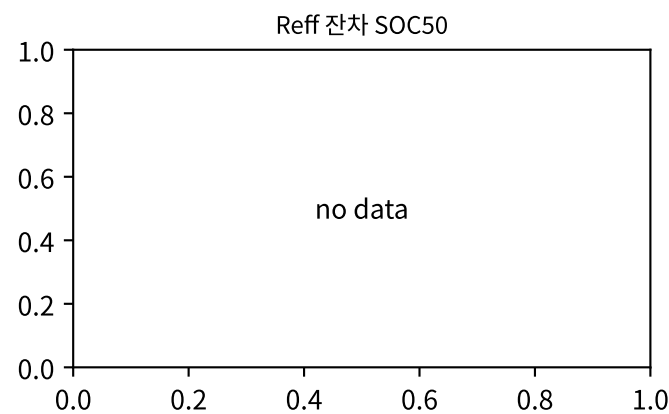
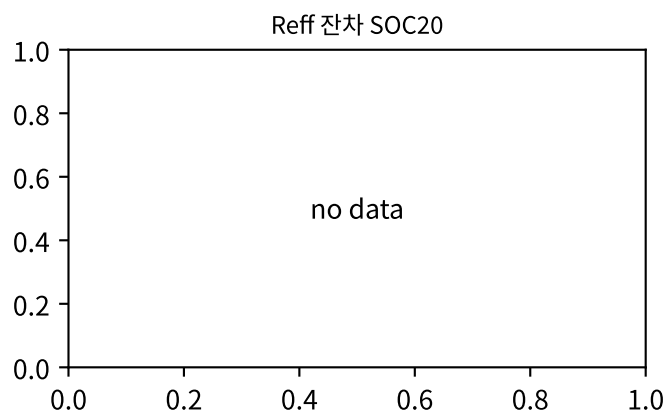
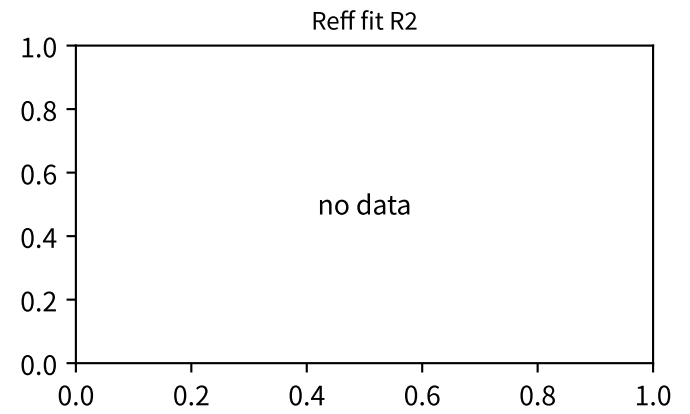
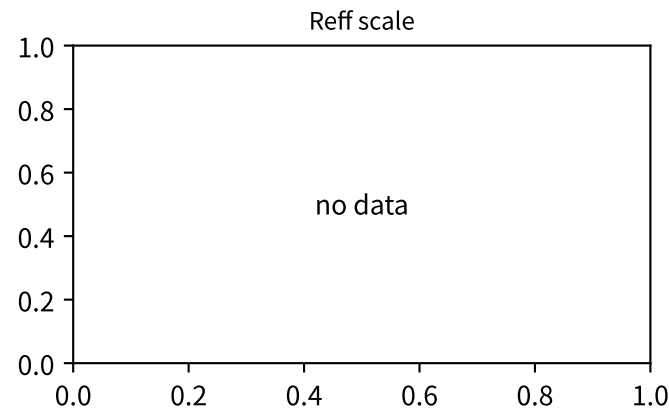
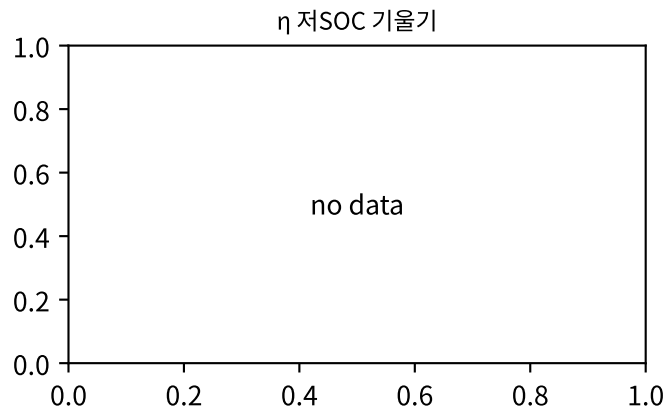
M02Ch103 · dQ/dV · dV/dQ 피크



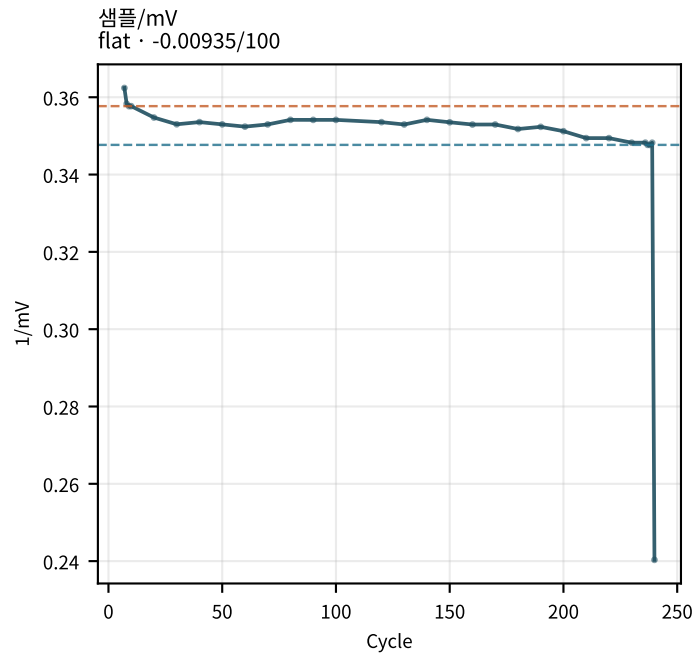
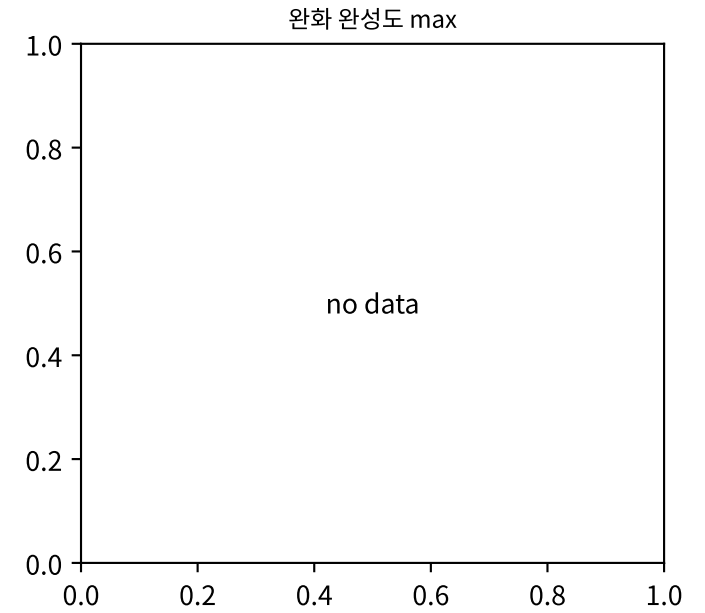
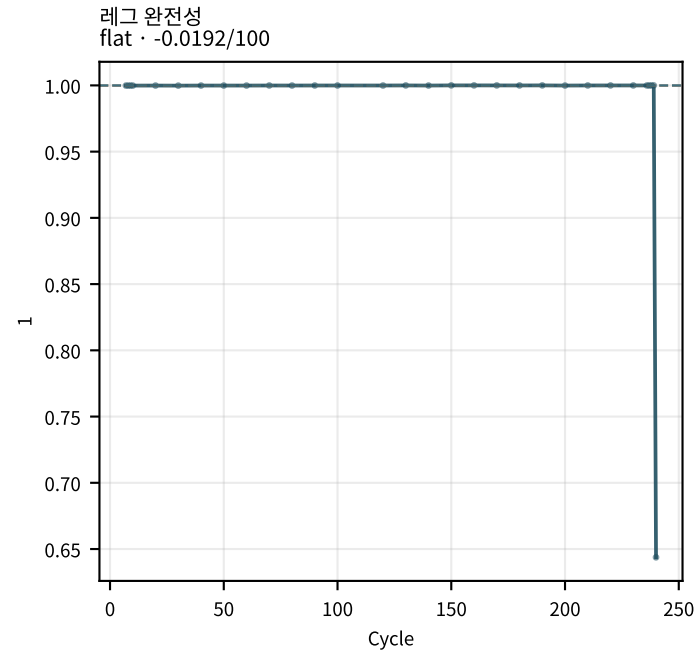
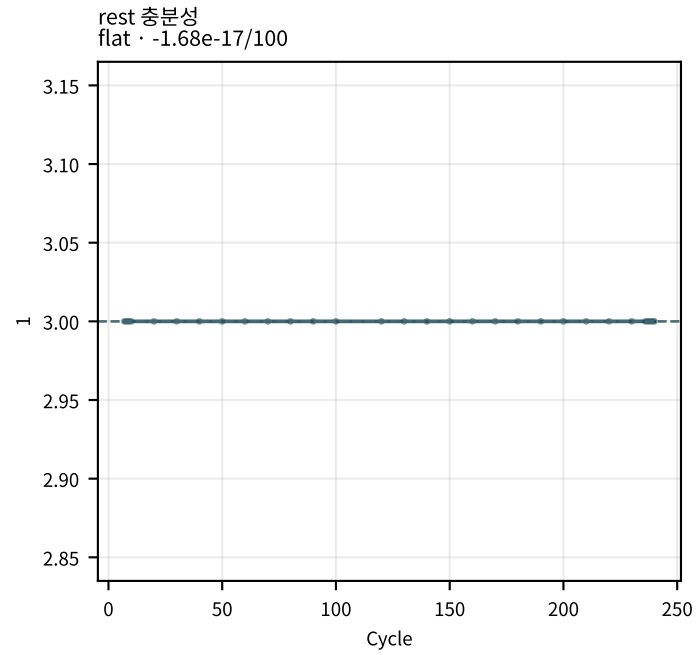
M02Ch103 · 수송 · rate · η



M02Ch103 · 수송 · rate · η

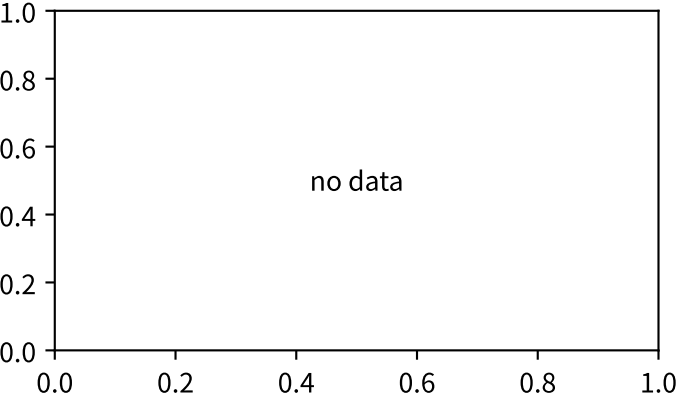


M02Ch103 · 수송 · rate · η

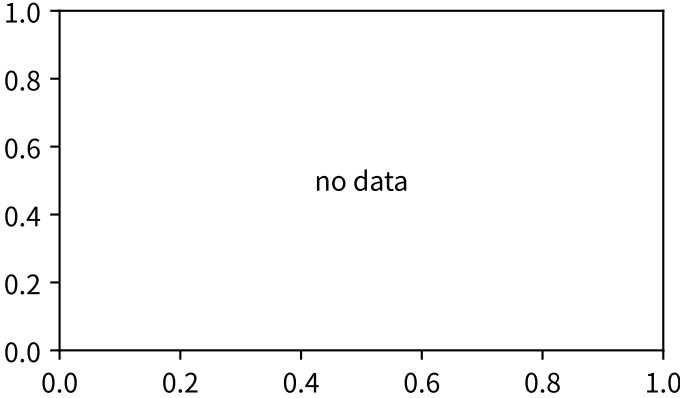


M02Ch103 · OCV · 자기방전

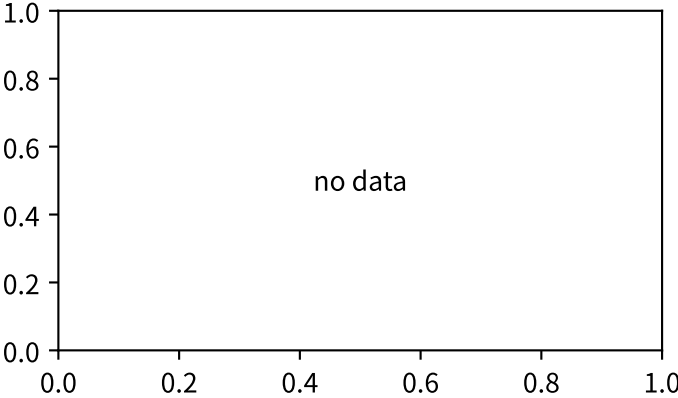
OCV V_{∞} SOC80



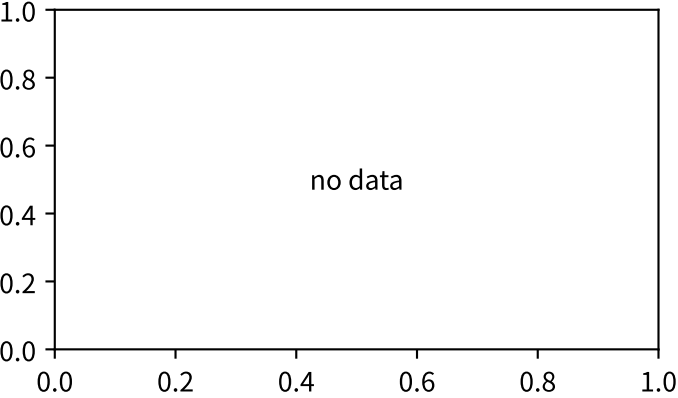
OCV V_{∞} SOC50



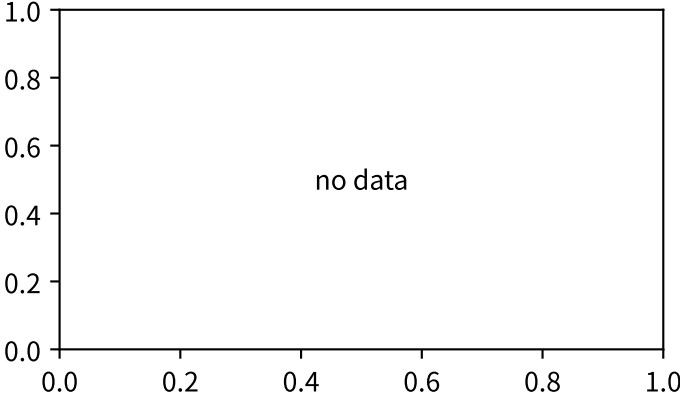
OCV V_{∞} SOC20



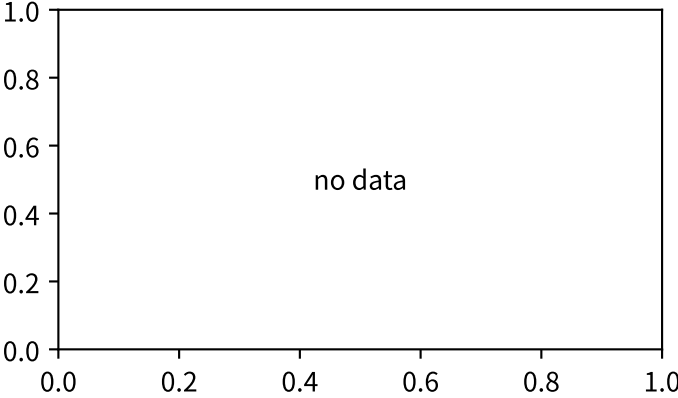
OCV spread 20–80



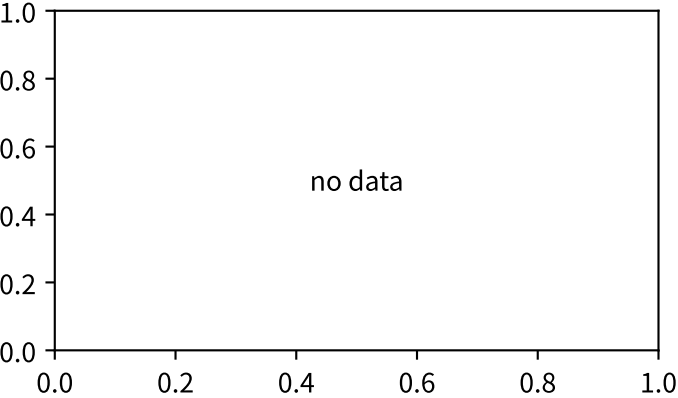
OCV spread 50–80



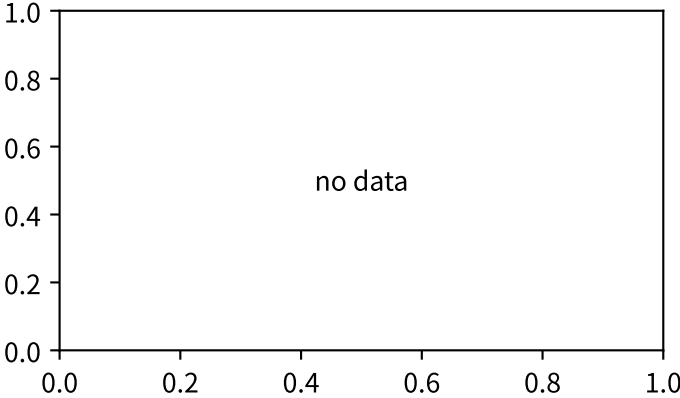
OCV spread 20–50



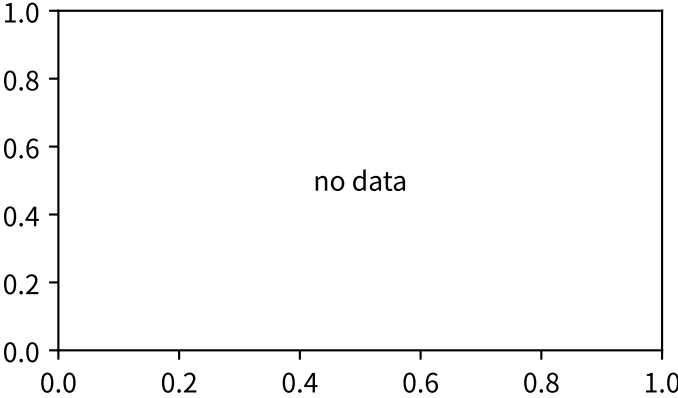
OCV 평행이동



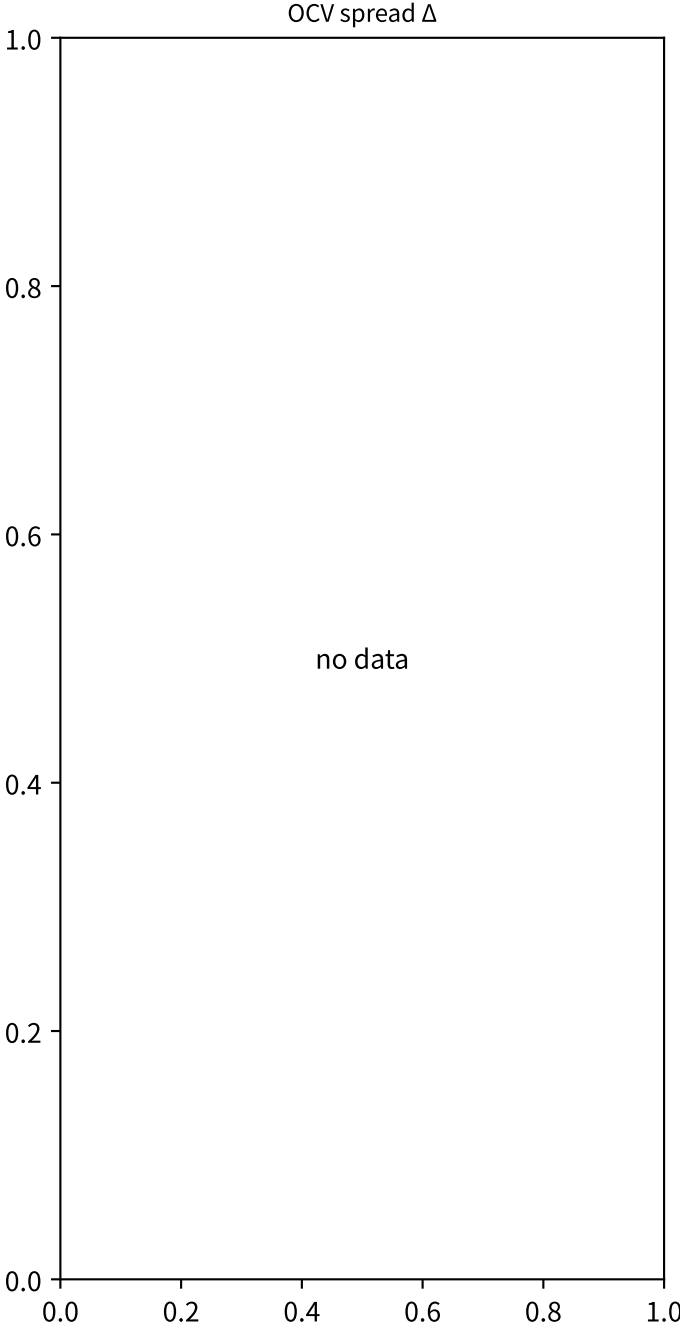
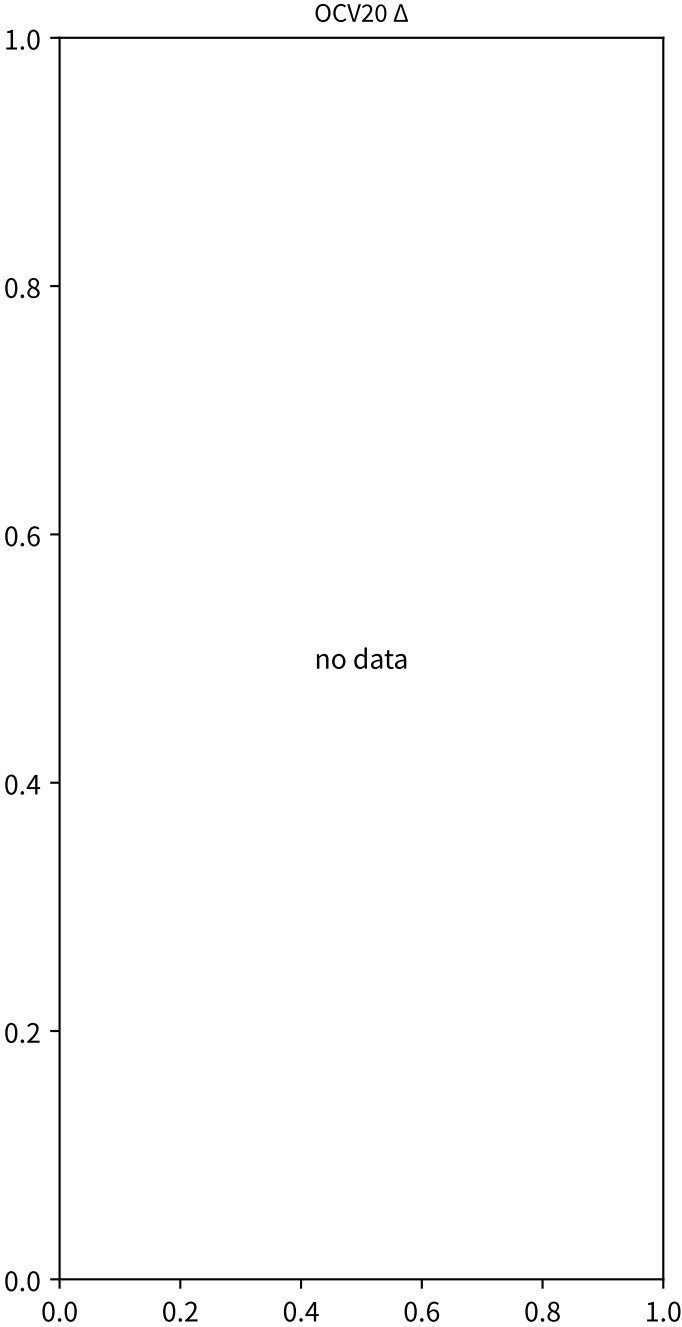
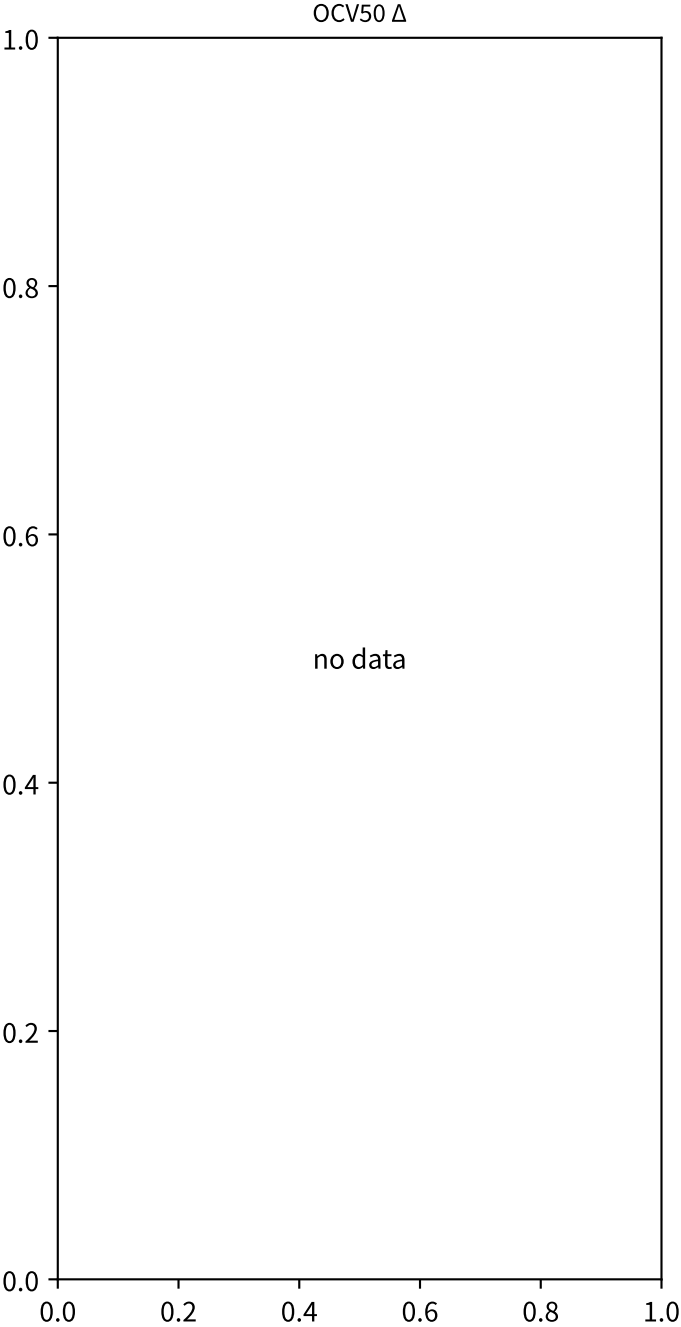
OCV 폭 압축



OCV80 Δ

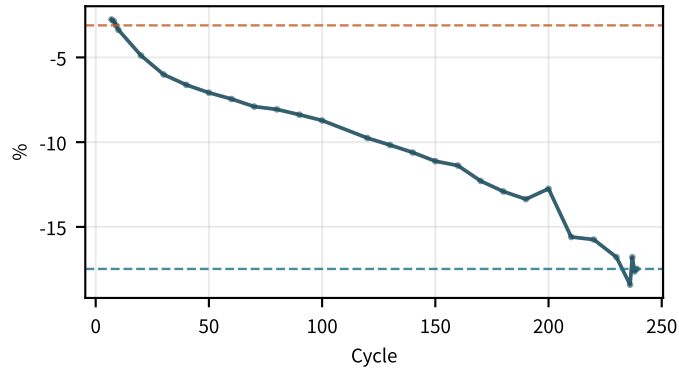


M02Ch103 · OCV · 자기방전

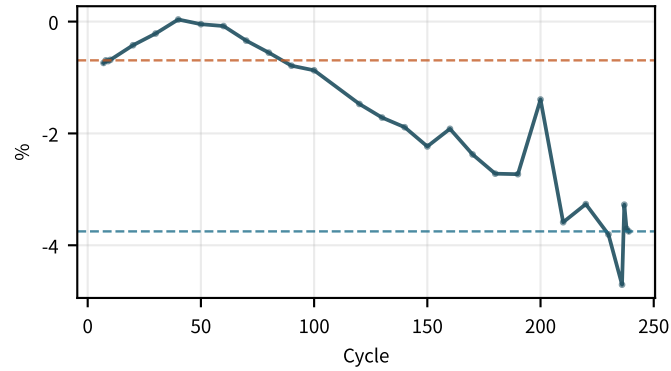


M02Ch103 · 곡선 fit · $\Delta Q(V)$

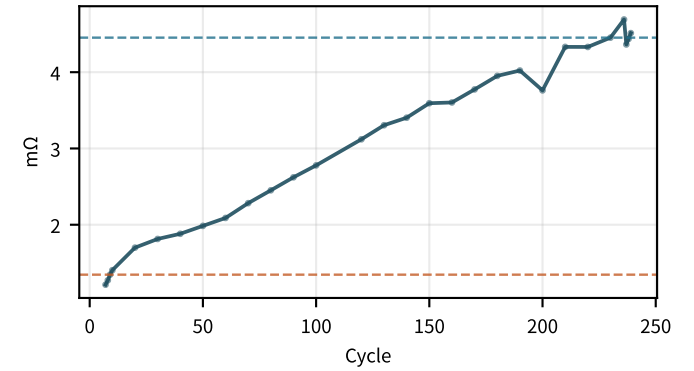
LAM 곡선 proxy
decreasing · $-5.72/100$



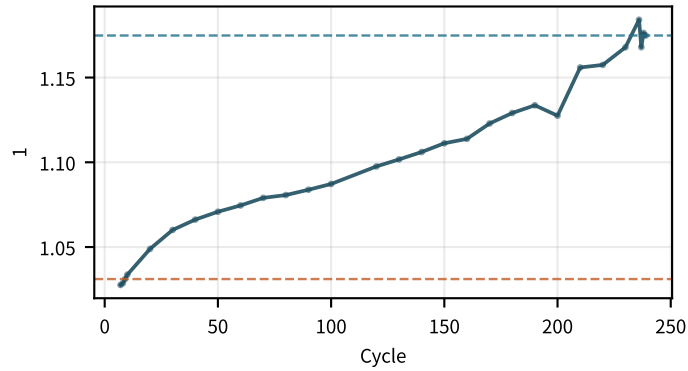
LLI 곡선 proxy
decreasing · $-1.53/100$



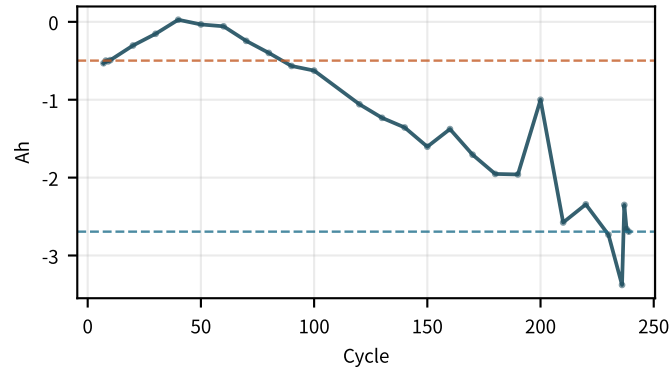
R 곡선 proxy
increasing · $+1.38/100$



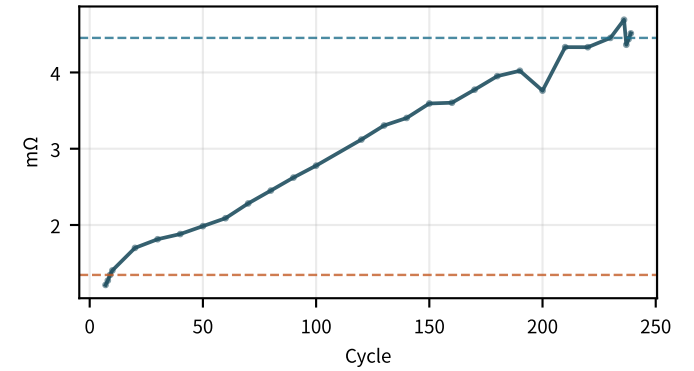
방전 fit scale
flat · $+0.0572/100$



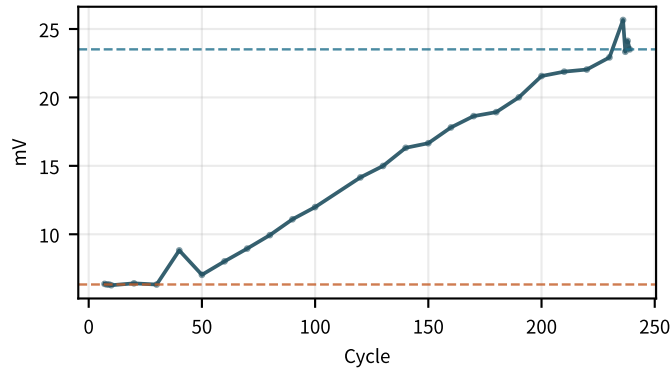
방전 fit offset
decreasing · $-1.1/100$



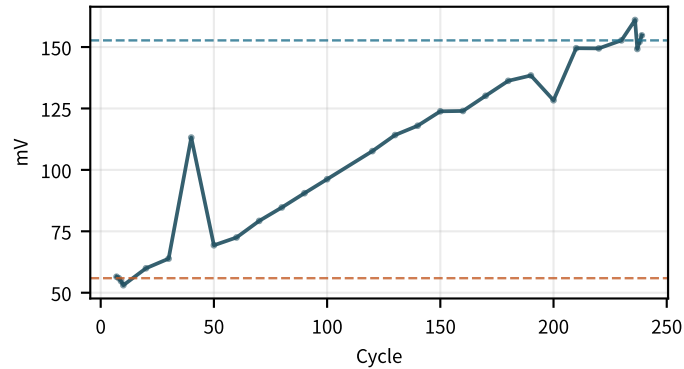
방전 fit dR
increasing · $+1.38/100$



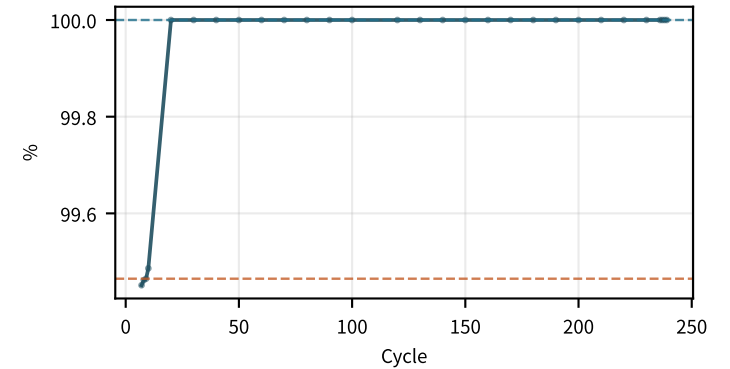
fit 잔차 RMS
increasing · $+8.11/100$



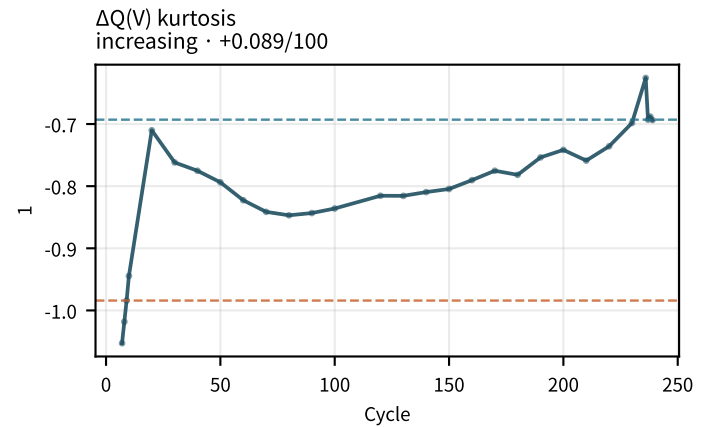
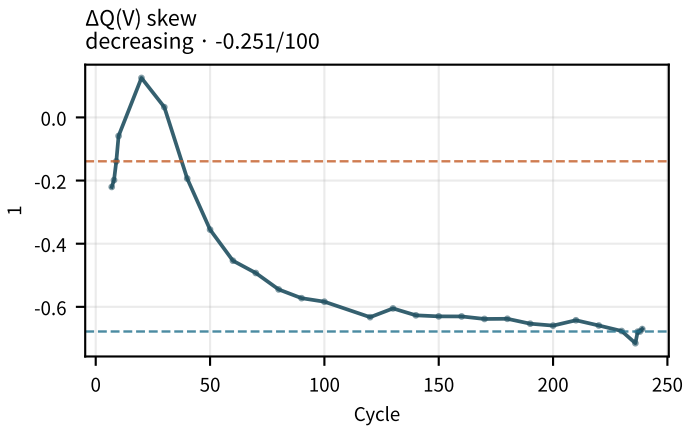
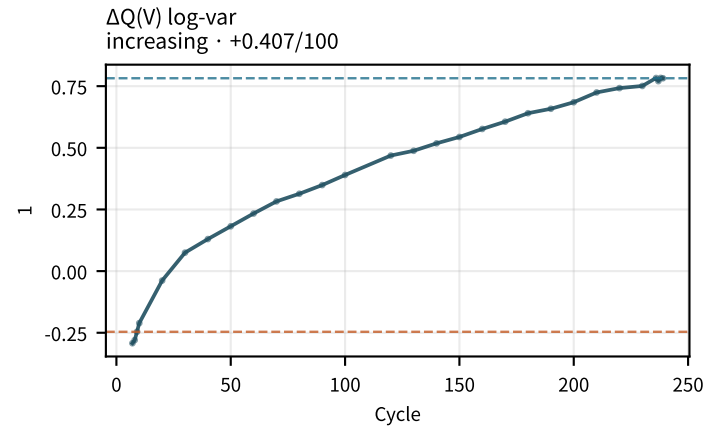
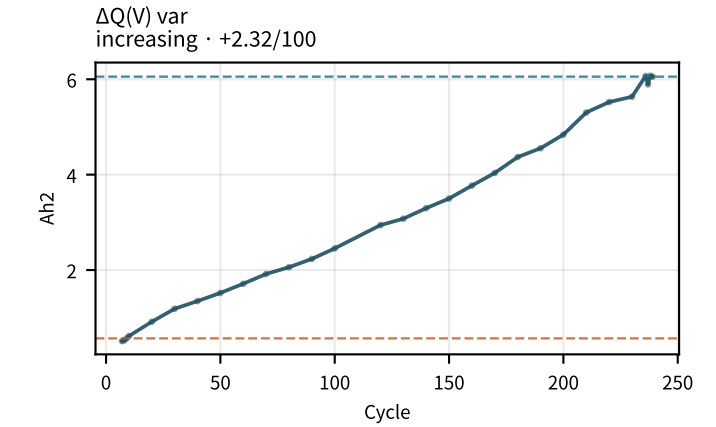
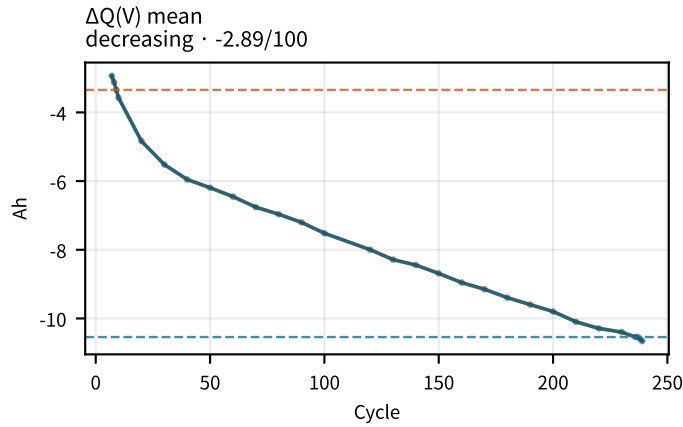
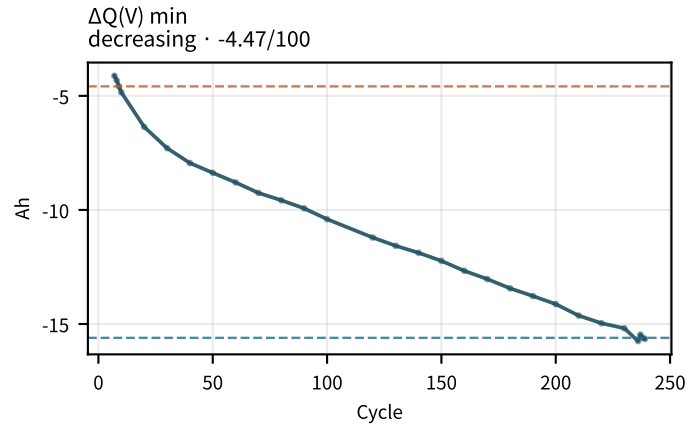
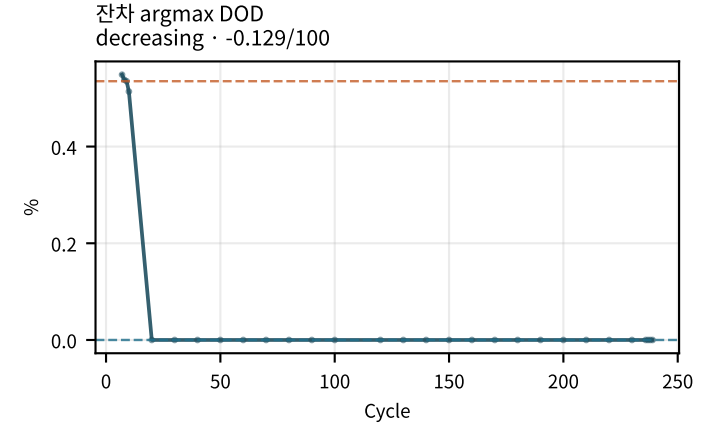
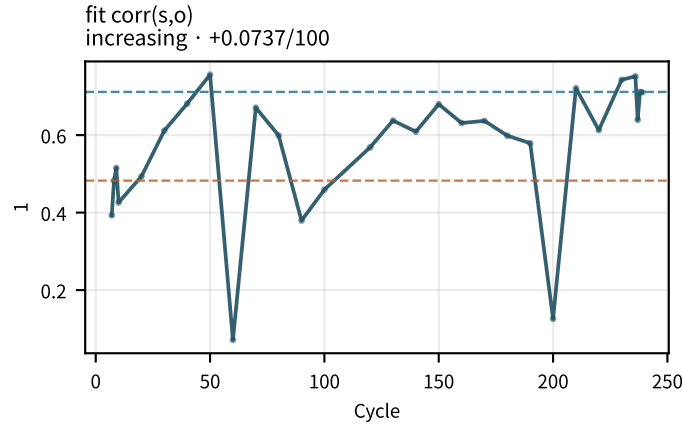
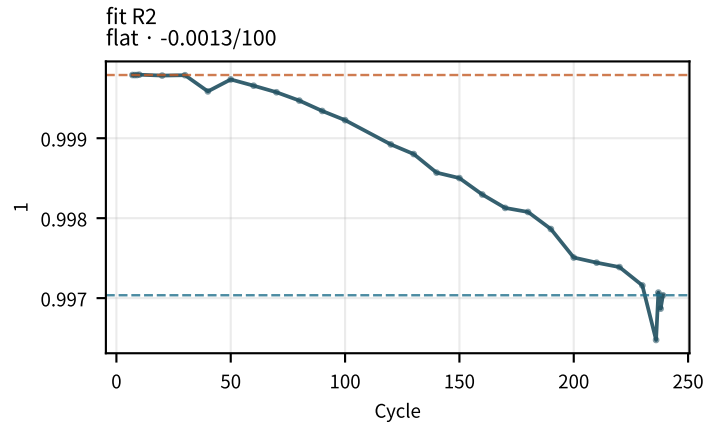
fit 잔차 max
increasing · $+42.6/100$



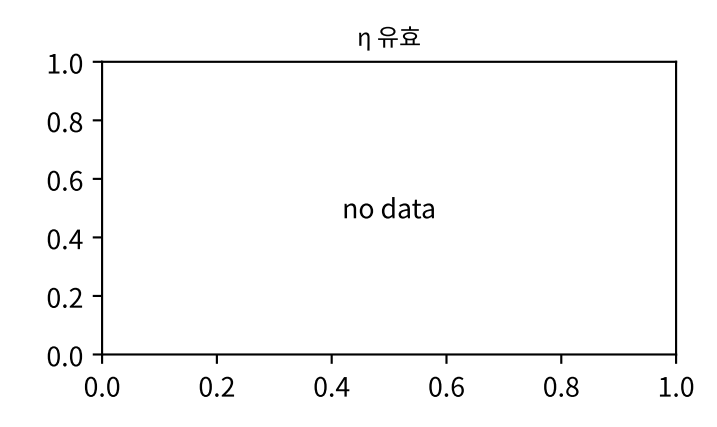
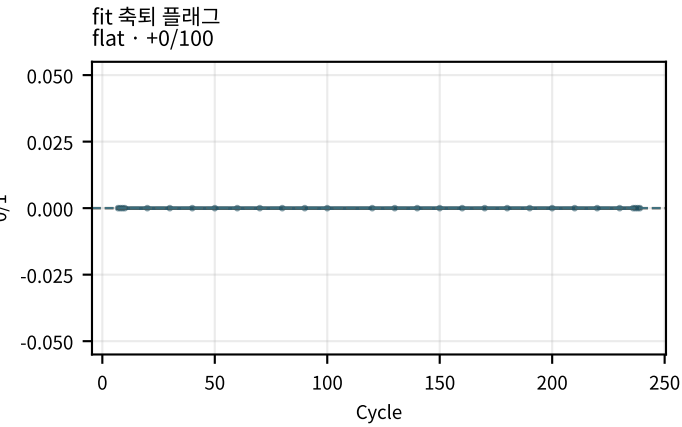
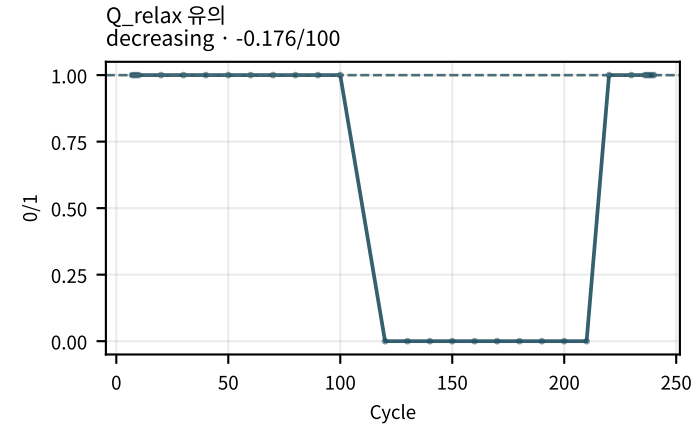
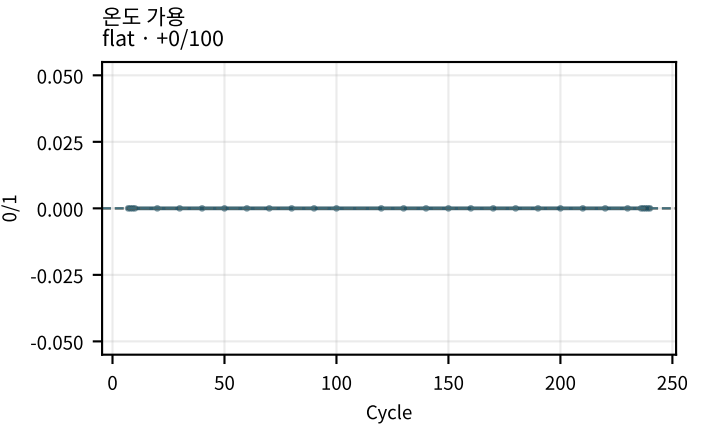
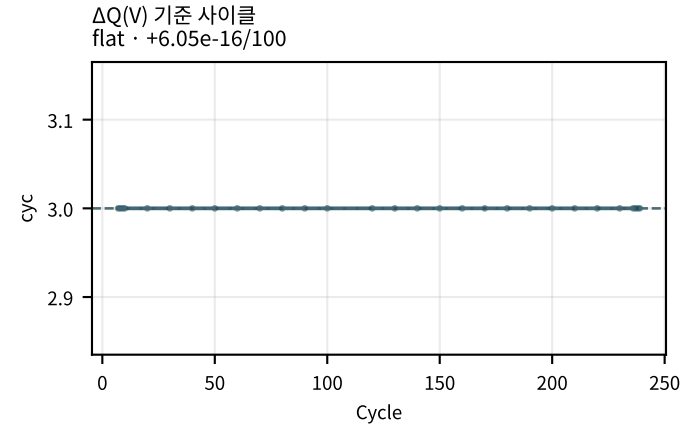
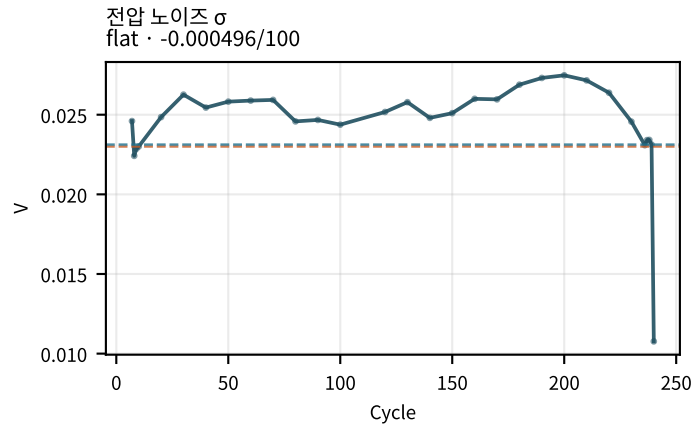
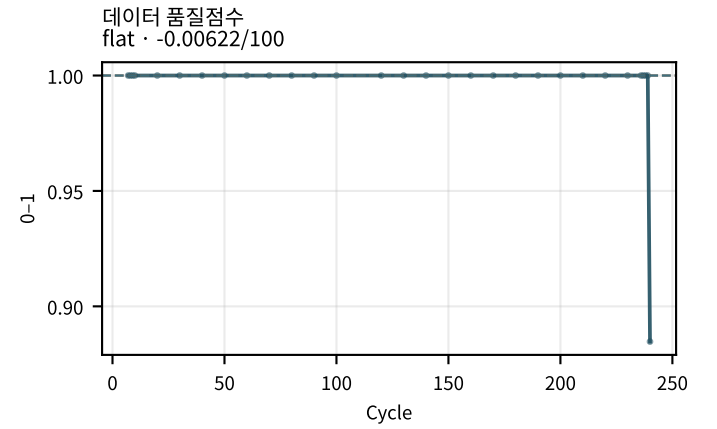
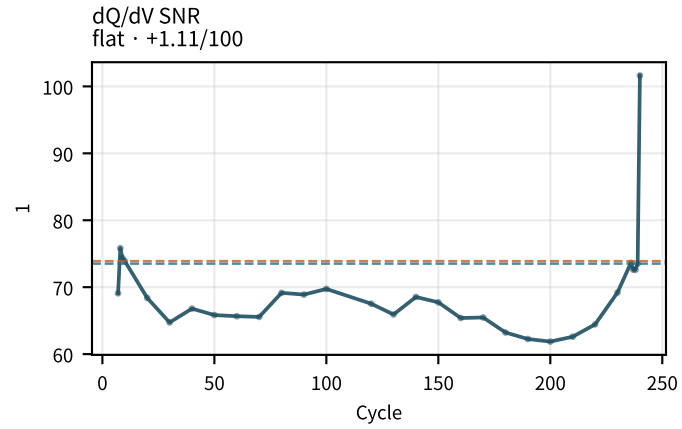
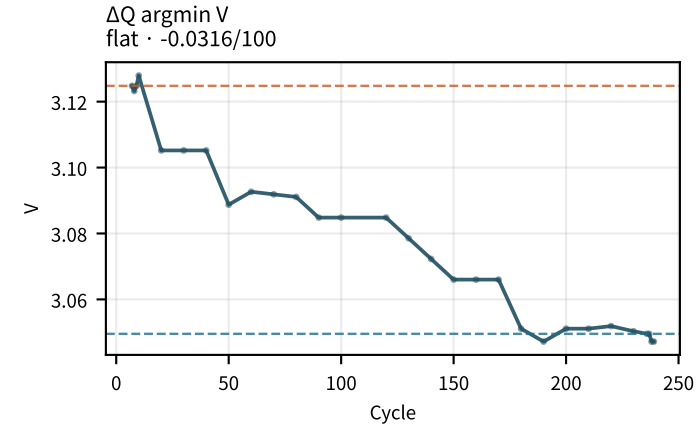
잔차 argmax SOC
flat · $+0.129/100$



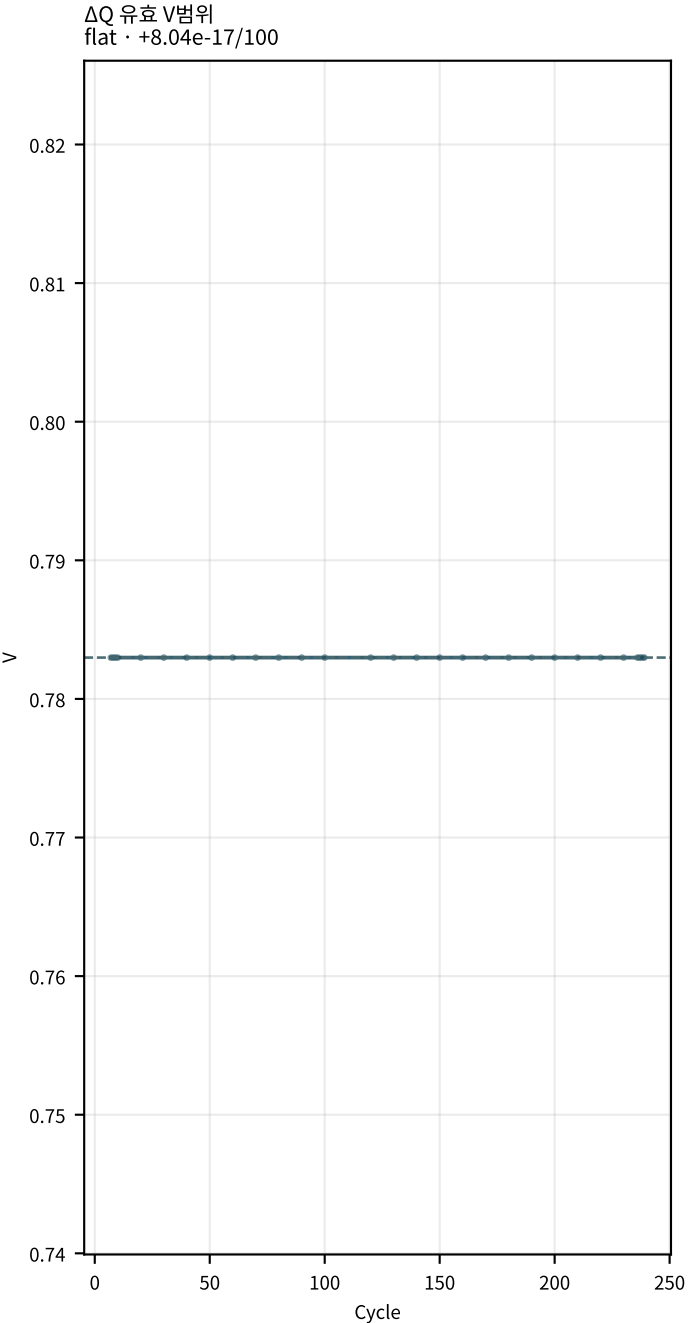
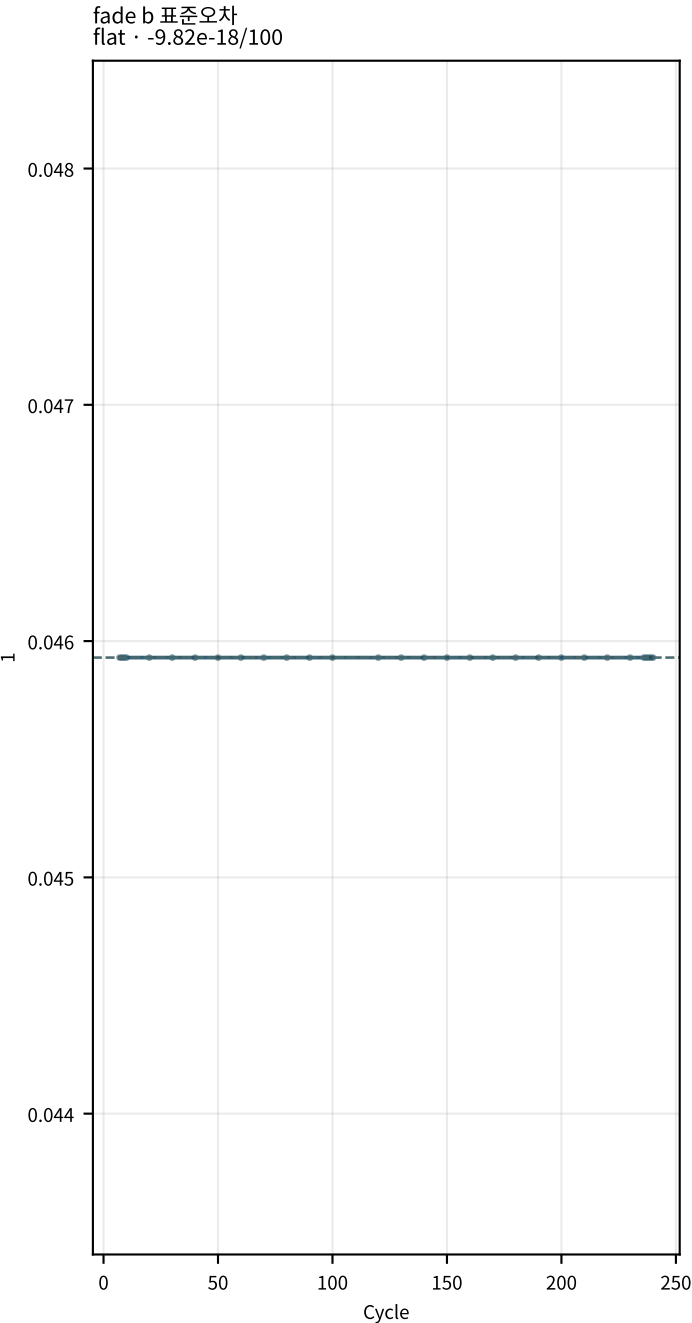
M02Ch103 · 곡선 fit · $\Delta Q(V)$



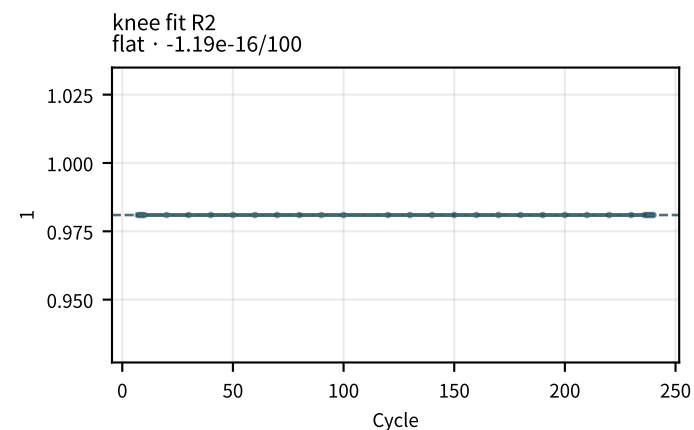
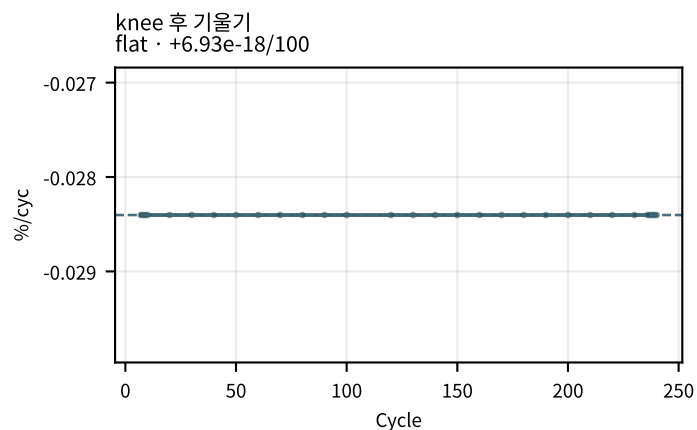
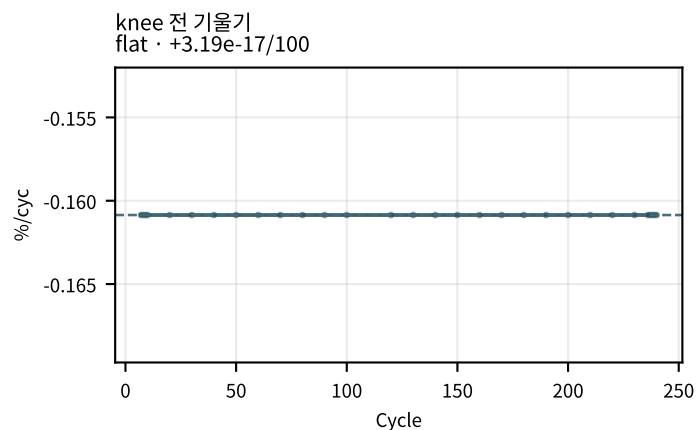
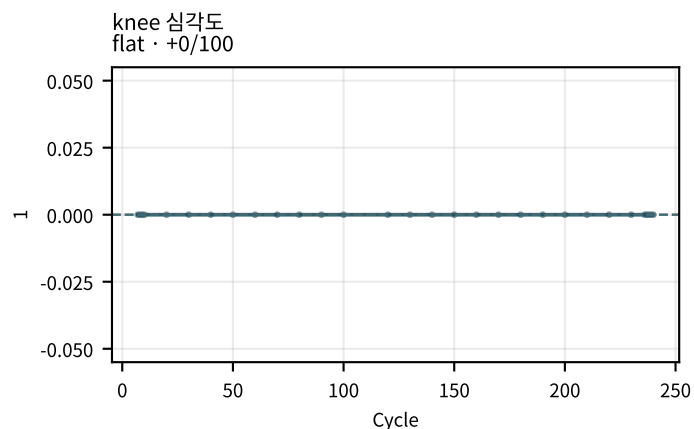
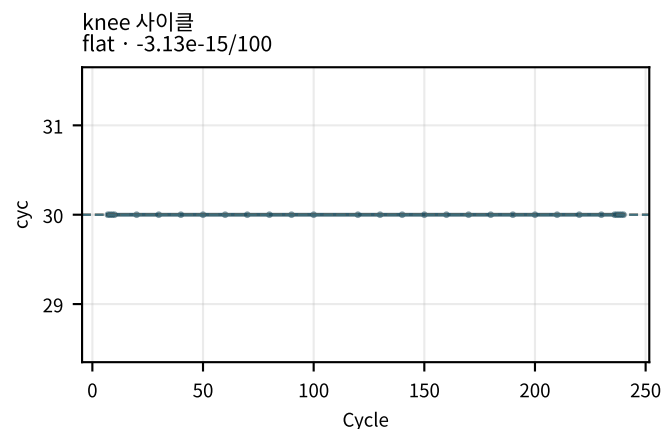
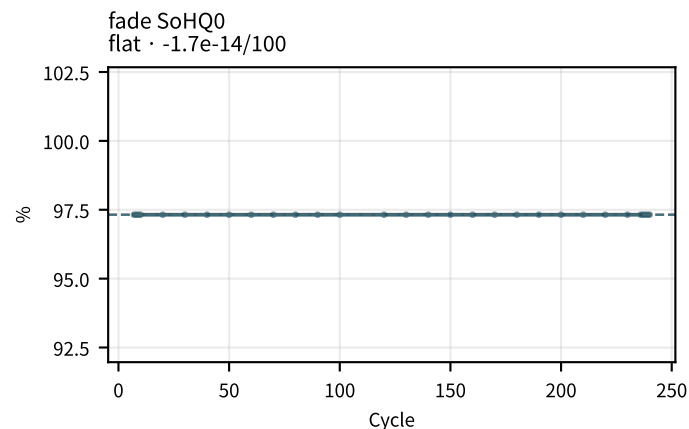
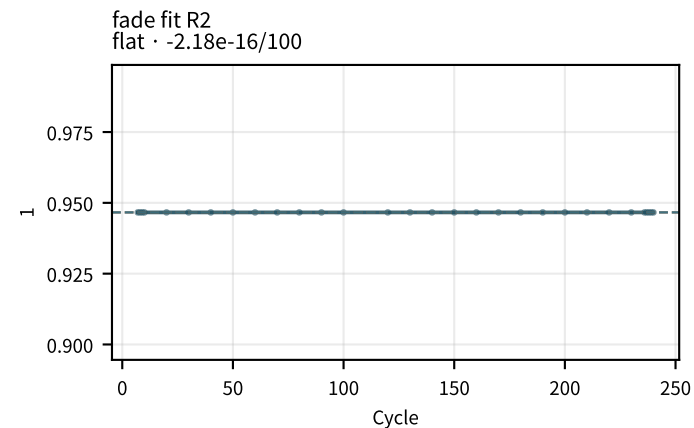
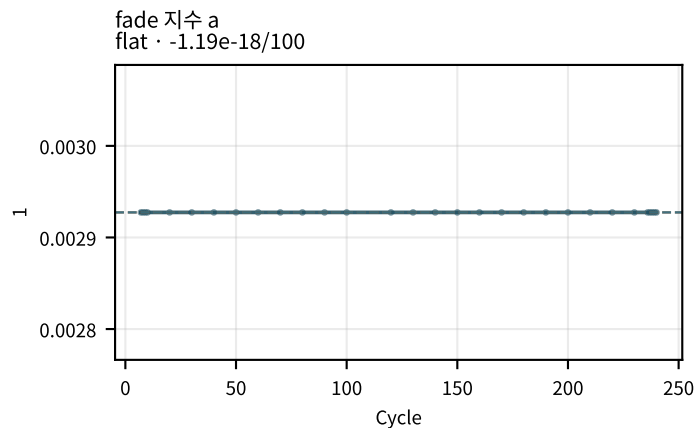
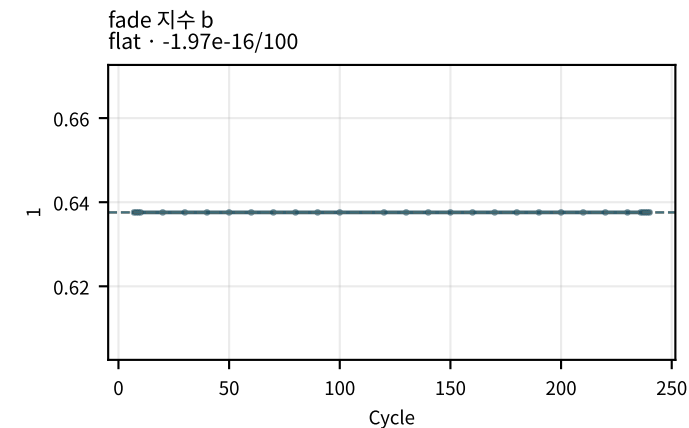
M02Ch103 · 곡선 fit · $\Delta Q(V)$



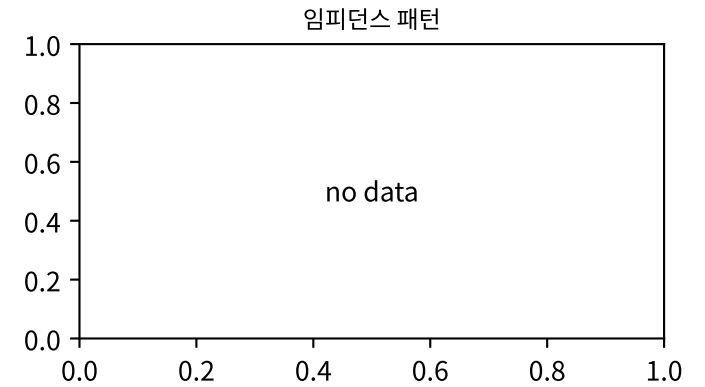
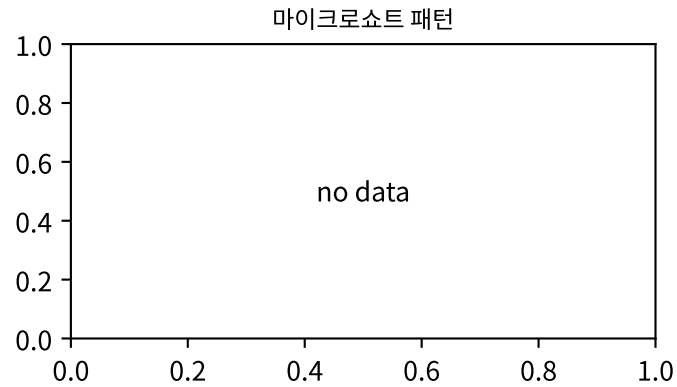
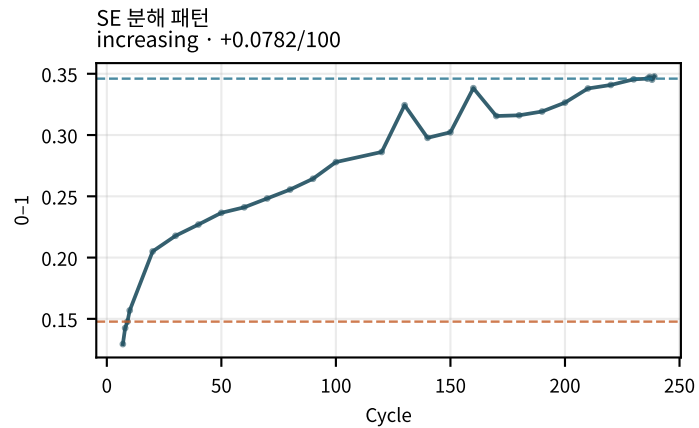
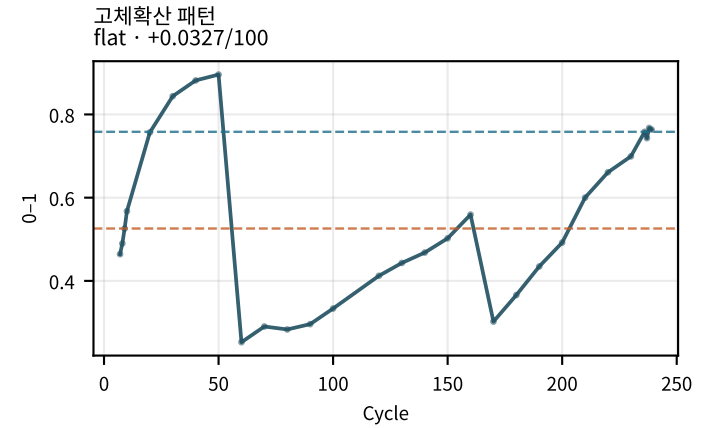
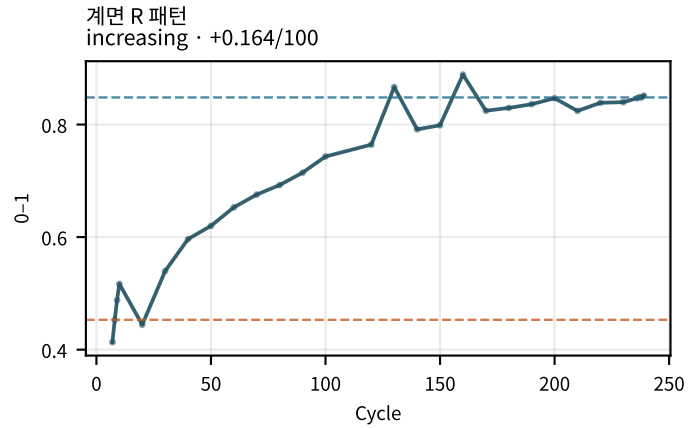
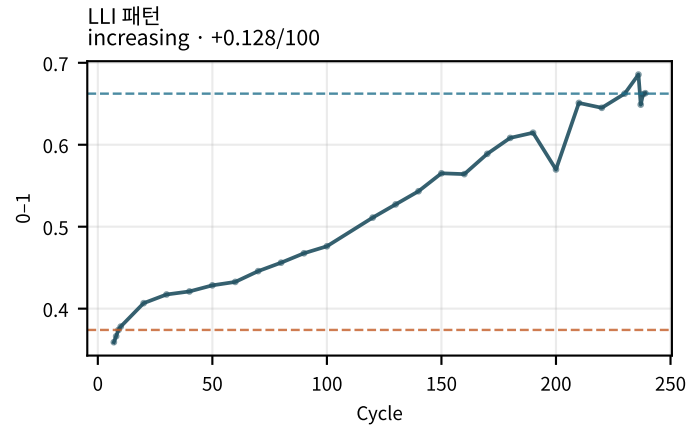
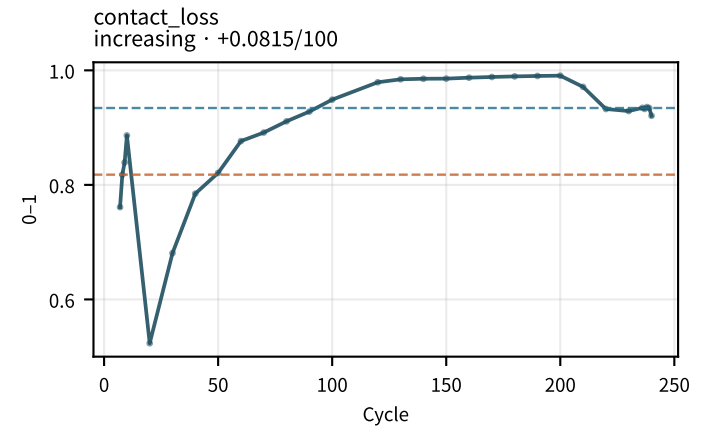
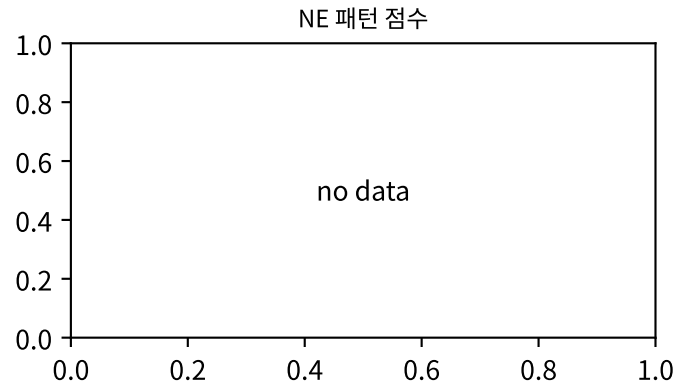
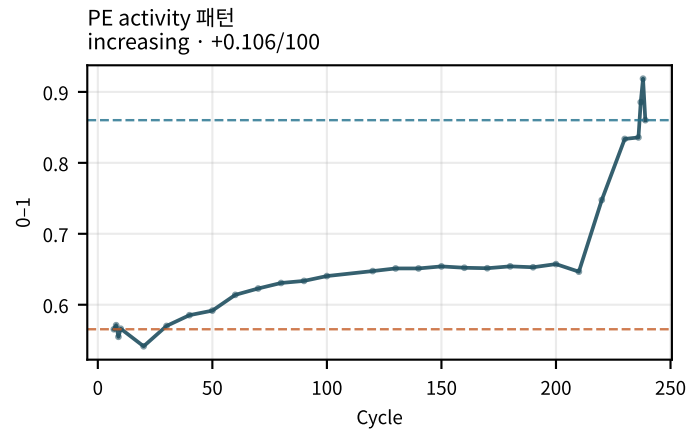
M02Ch103 · 곡선 fit · ΔQ(V)



M02Ch103 · fade · knee

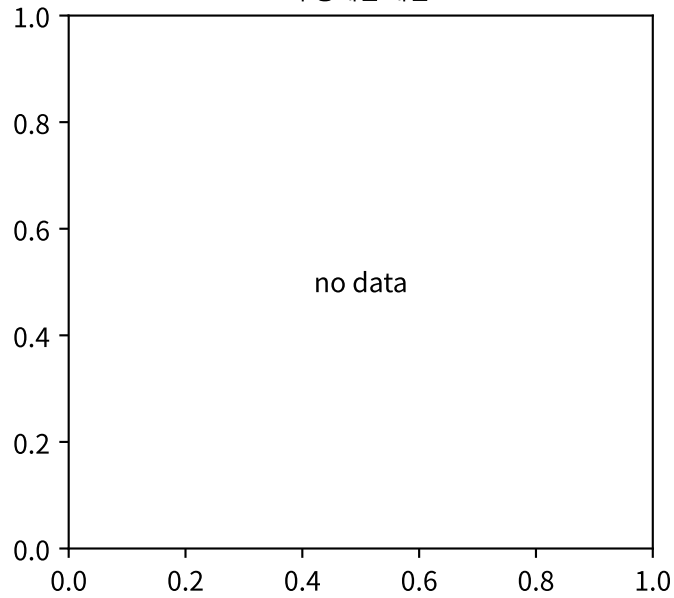


M02Ch103 · 열화 패턴 점수

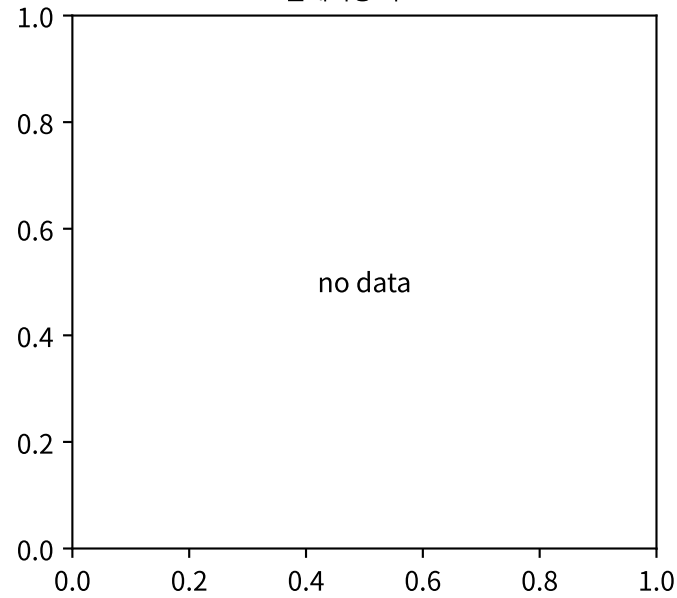


M02Ch103 · 열화 패턴 점수

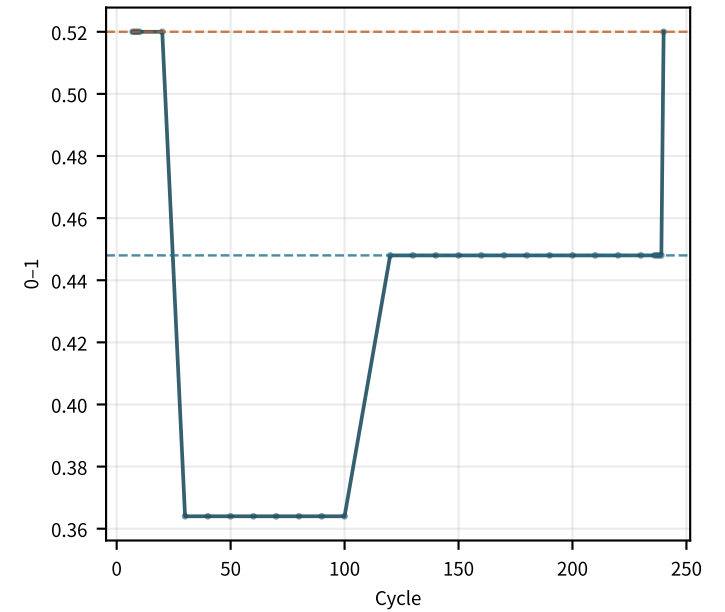
수송제한 패턴



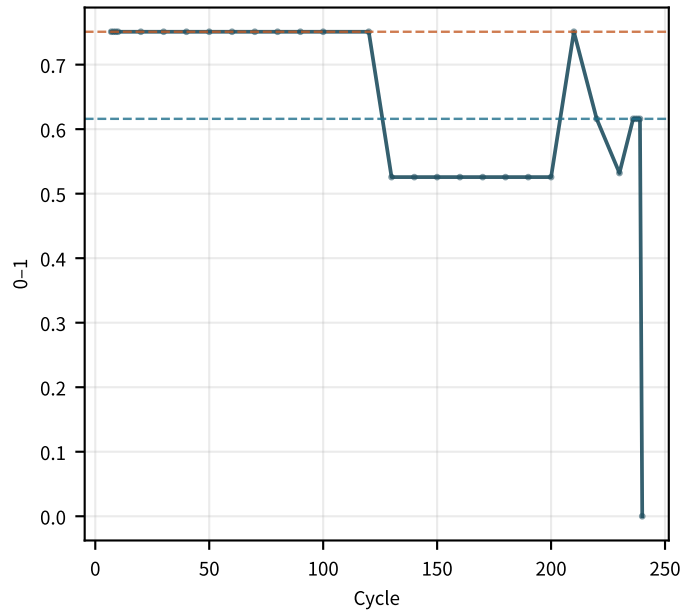
플레이팅 리스크



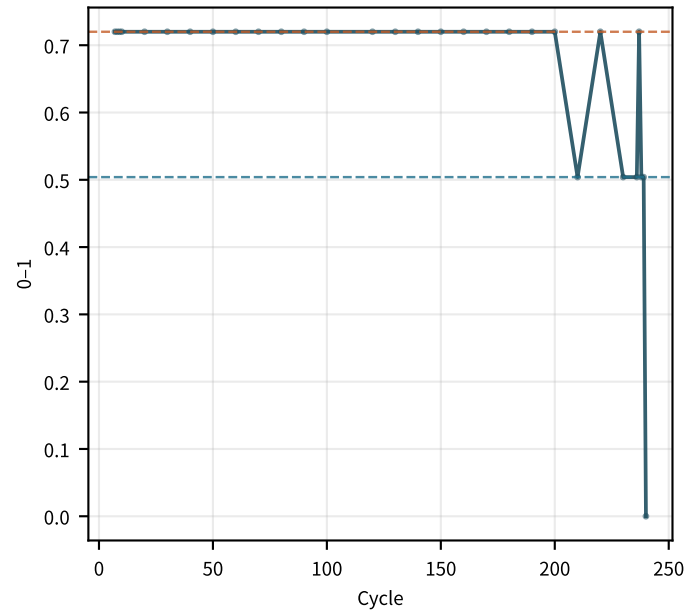
contact_loss 신뢰도
flat · +0.00408/100



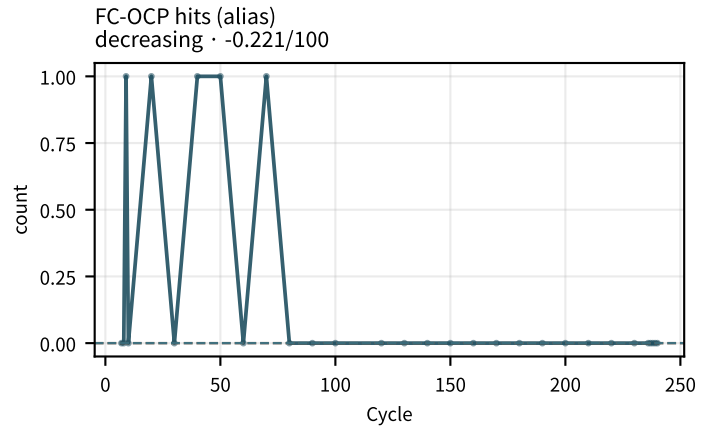
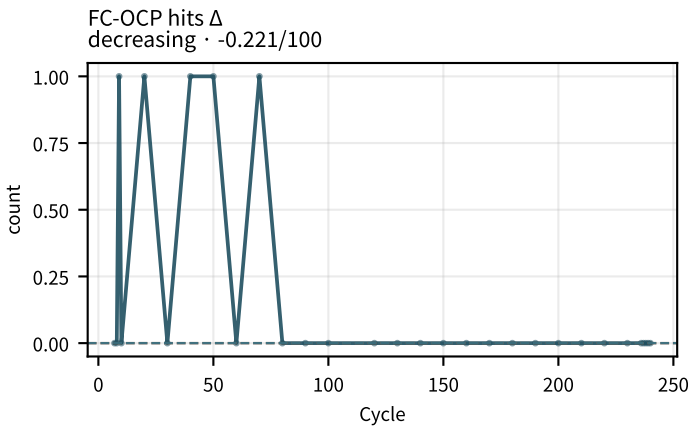
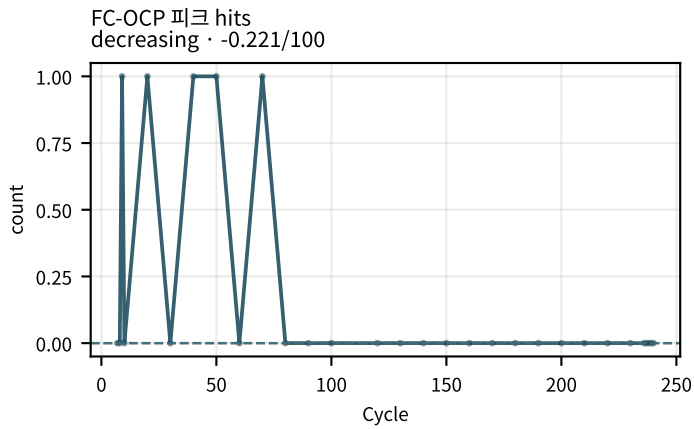
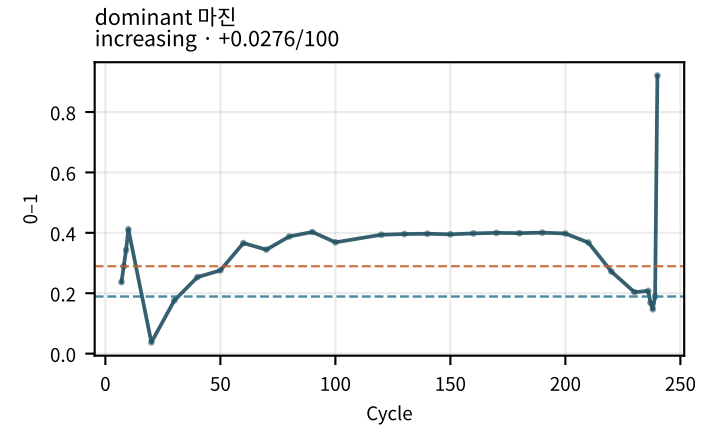
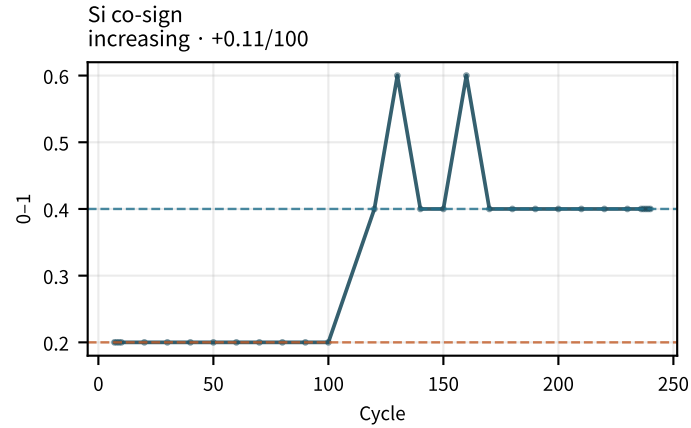
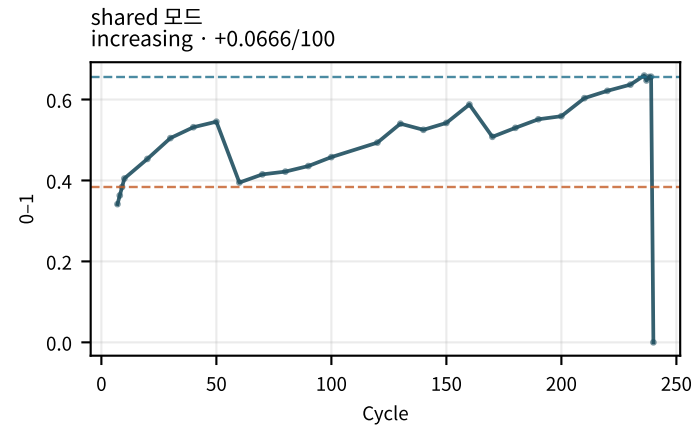
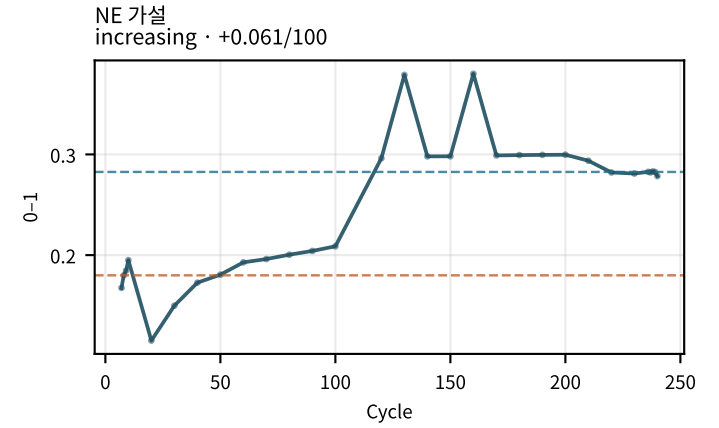
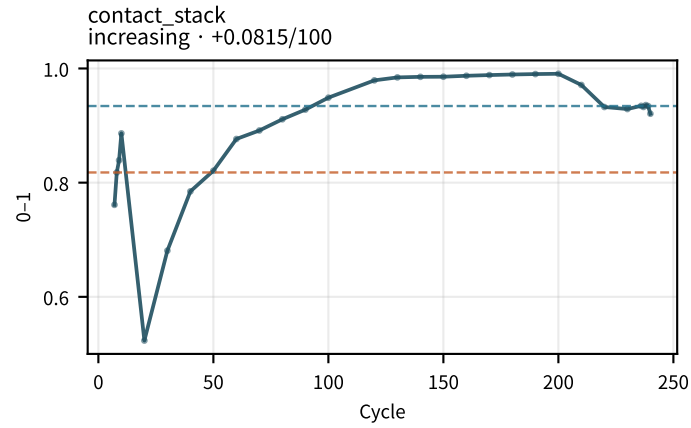
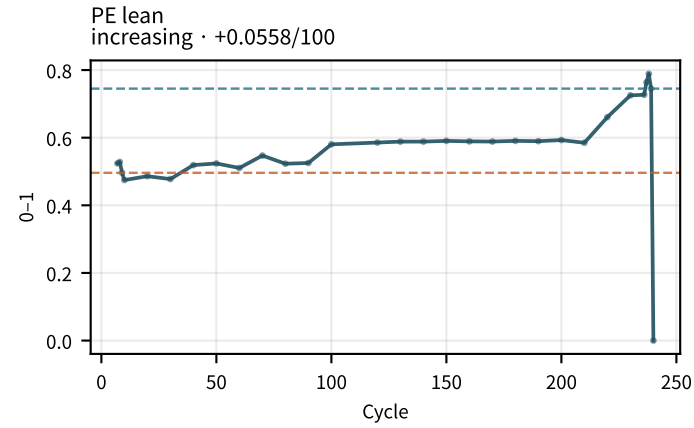
LAM_PE 신뢰도
decreasing · -0.117/100



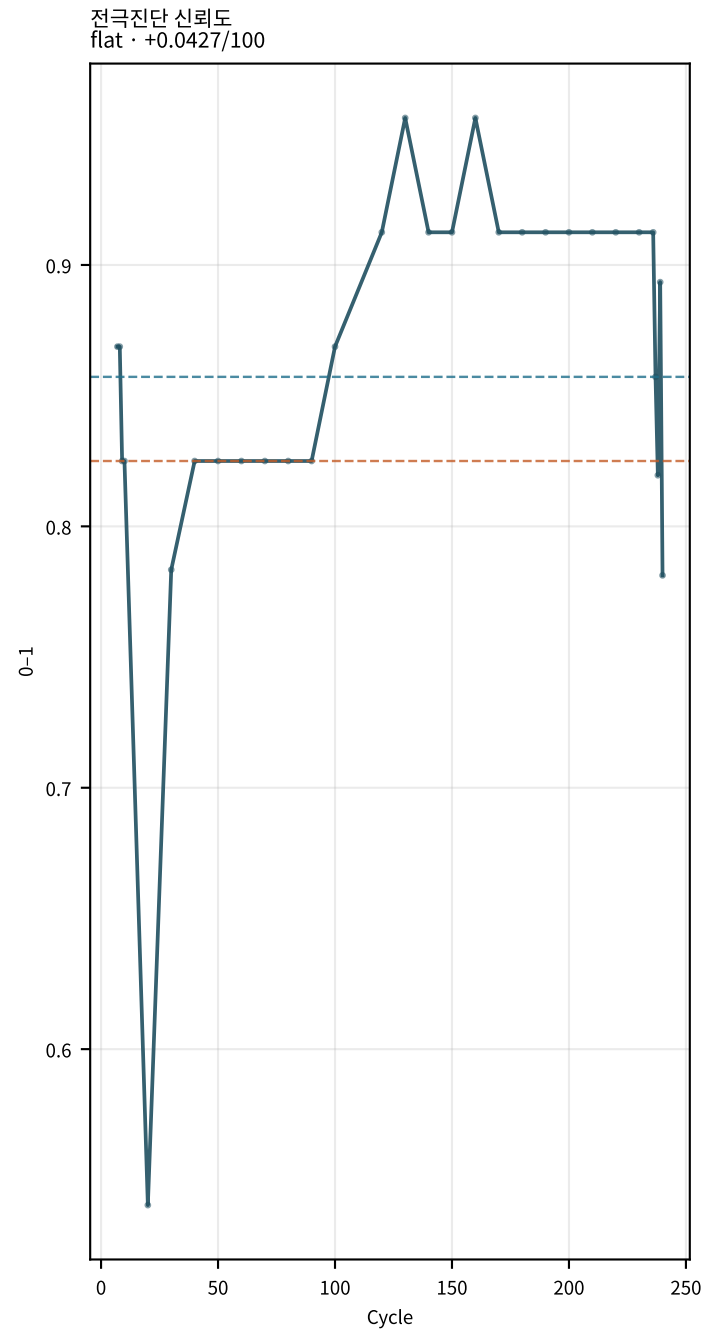
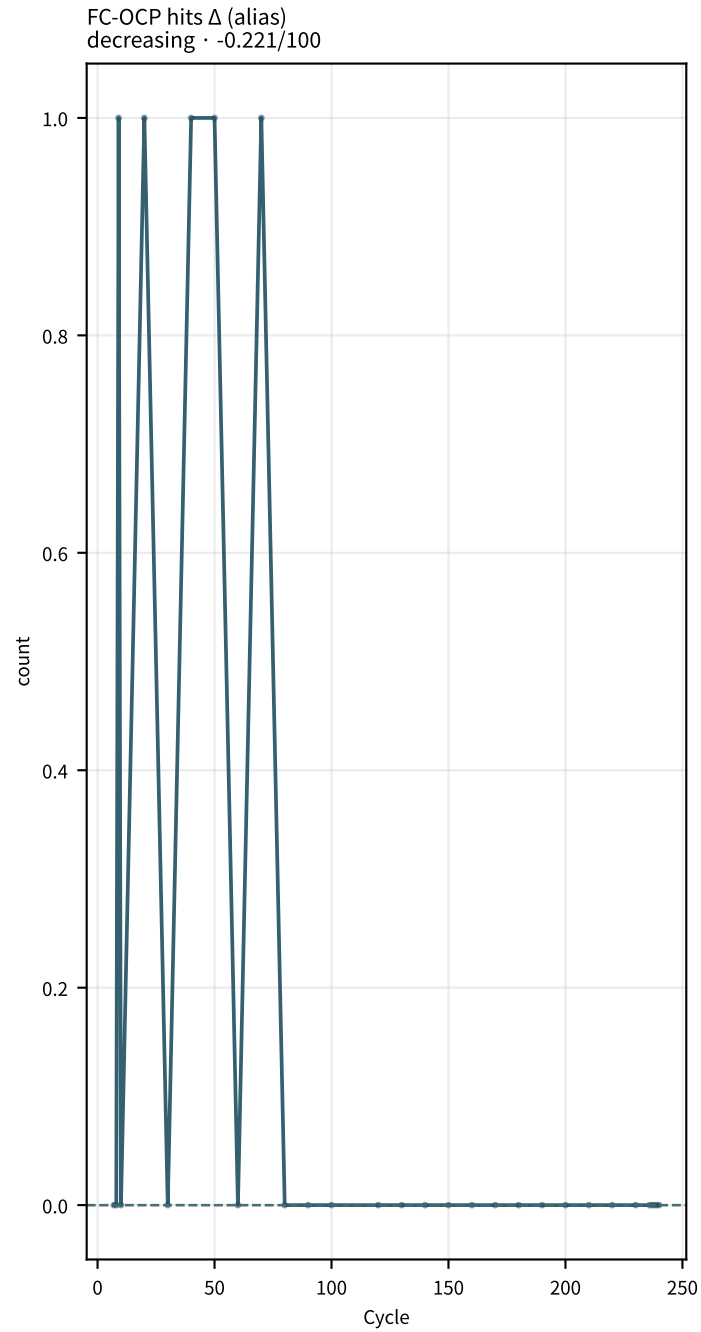
LLI 신뢰도
decreasing · -0.0922/100



M02Ch103 · 전극 lean 가설



M02Ch103 · 전극 lean 가설



지표 설명 · 트렌드 (M02Ch103)

충전 전압 컷오프 (chg_V_cutoff)

충전 종료 전압. | 계산: charge V max/cutoff.

트렌드: 충전 전압 컷오프: early=4.2 → late=4.2 ($\Delta=+0.0001467$, $+7.151\text{e-}05\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전 전압 컷오프 (dchg_V_cutoff)

방전 종료 전압. | 계산: discharge V min/cutoff.

트렌드: 방전 전압 컷오프: early=2.5 → late=2.5 ($\Delta=+3.28\text{e-}05$, $+9.062\text{e-}06\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

충전 전류 컷오프 (chg_I_cutoff)

CV 종료 전류. | 계산: charge I cutoff.

트렌드: 충전 전류 컷오프: early=34.2 → late=34.66 ($\Delta=+0.46$, $-0.1395\text{A}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

충전 평균 온도 (chg_temp_avg)

충전 구간 평균 온도. | 계산: mean Temp on charge.

트렌드: 충전 평균 온도: early=0 → late=0 ($\Delta=+0$, $+0\text{C}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전 평균 온도 (dchg_temp_avg)

방전 구간 평균 온도. | 계산: mean Temp on discharge.

트렌드: 방전 평균 온도: early=0 → late=0 ($\Delta=+0$, $+0\text{C}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

사이클 소요시간 (cycle_duration_h)

한 사이클 벽시계 시간. | 계산: total step time sum.

트렌드: 사이클 소요시간: early=5.113 → late=4.91 ($\Delta=-0.2031$, $-0.1876\text{h}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

용량 유지율 (SoHQ)

기준 대비 방전용량. | 계산: $Q_dchg/Q_base \times 100$.

트렌드: 용량 유지율: early=97.08 → late=87.42 ($\Delta=-9.659$, -3.534%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

방전 용량 (dchgCapa)

사이클 방전용량. | 계산: max discharge capacity.

트렌드: 방전 용량: early=69.7 → late=62.77 ($\Delta=-6.935$, -2.538Ah/100cyc) → flat · stable | aging 기대=decrease | vs=stable

충전 용량 (chgCapa)

사이클 충전용량. | 계산: max charge capacity.

트렌드: 충전 용량: early=64.41 → late=53.47 ($\Delta=-10.95$, -4.119Ah/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

CC 충전용량 (chgCCcapa)

CC 구간 용량. | 계산: CC capacity.

트렌드: CC 충전용량: early=64.1 → late=53.2 ($\Delta=-10.89$, -4.169Ah/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

CV 충전용량 (chgCVcapa)

CV 구간 용량. | 계산: signal/column CV Ah.

트렌드: CV 충전용량: early=0 → late=0 ($\Delta=+0$, +0.06236Ah/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

CC 용량비 (chgCapa_CCratio)

CC/(CC+CV). | 계산: $chgCCcapa/chgCapa$.

트렌드: CC 용량비: early=100 → late=100 ($\Delta=+0$, -0.10261/100cyc) → flat · stable | aging 기대=decrease | vs=stable

지표 설명 · 트렌드 (M02Ch103)

CC비 (정규화) (chgCapa_CCratio_norm)

기준 정규화 CC비. | 계산: CCratio / baseline.

트렌드: CC비 (정규화): early=93.45 → late=93.45 ($\Delta=+0$, n/a) → flat · stable | aging 기대=decrease | vs=stable

CC비 Δ (delta_chgCapa_CCratio)

기준 대비 CC비 변화. | 계산: delta abs.

트렌드: CC비 Δ : early=0 → late=0 ($\Delta=+0$, -0.10261/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

CV 시간 (chgCVtime)

CV 지속시간. | 계산: CV step duration.

트렌드: CV 시간: early=0 → late=0 ($\Delta=+0$, +8.492s/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

CV 시정수 (tau_CV)

CV 전류 감쇠 τ . | 계산: I(t) exp fit.

트렌드: CV 시정수: early=755.9 → late=755.9 ($\Delta=+0$, n/a) → flat · stable | aging 기대=increase | vs=stable

CV Q @Tref (Q_CV_at_Tref)

온도 보정 CV 용량. | 계산: CV Q at ref T.

트렌드: CV Q @Tref: early=4.003 → late=4.003 ($\Delta=+0$, n/a) → flat · context | aging 기대=either | vs=context

쿨롱 효율 (CE)

Q_dchg/Q_chg. | 계산: dchg/chg*100.

트렌드: 쿨롱 효율: early=107.9 → late=117.3 ($\Delta=+9.409$, +3.596%/100cyc) → flat · stable | aging 기대=decrease | vs=stable

지표 설명 · 트렌드 (M02Ch103)

가역 CE (CE_rev)

가역 쿨롱 효율 proxy. | 계산: rev CE extract.

트렌드: 가역 CE: early=92.18 → late=85.17 ($\Delta=-7.01$, $-2.133\%/100\text{cyc}$) → flat · stable | aging 기대=decrease | vs=stable

국소 CE(20) (CE_local_20)

최근 20사이클 국소 CE. | 계산: rolling CE.

트렌드: 국소 CE(20): early=528.7 → late=273.2 ($\Delta=-255.4$, $-85.37\%/100\text{cyc}$) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

쿨롱 비효율 (CI)

100-CE. | 계산: 100-CE.

트렌드: 쿨롱 비효율: early=-7.902 → late=-17.31 ($\Delta=-9.409$, $-3.596\%/100\text{cyc}$) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

CI /시간 (CI_per_hour)

시간당 쿨롱 손실. | 계산: CI / cycle_duration_h.

트렌드: CI /시간: early=-1.548 → late=-3.525 ($\Delta=-1.977$, $-0.7513\%/h/100\text{cyc}$) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

전압 효율 (VE)

에너지 전압효율. | 계산: E_dchg/E_chg.

트렌드: 전압 효율: early=0.8963 → late=0.8602 ($\Delta=-0.03612$, $-0.014771/100\text{cyc}$) → flat · stable | aging 기대=decrease | vs=stable

에너지 효율 (EE)

충방전 에너지비. | 계산: E_dchg/E_chg.

트렌드: 에너지 효율: early=0.9658 → late=1.009 ($\Delta=+0.04291$, $+0.014881/100\text{cyc}$) → flat · stable | aging 기대=decrease | vs=stable

지표 설명 · 트렌드 (M02Ch103)

충전 에너지 (chg_E)

충전 Wh. | 계산: $\int VI \, dt$ charge.

트렌드: 충전 에너지: early=243.6 → late=204.5 ($\Delta=-39.03$, -14.6Wh/100cyc) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 에너지 (dchg_E)

방전 Wh. | 계산: $\int VI \, dt$ discharge.

트렌드: 방전 에너지: early=236.1 → late=206.4 ($\Delta=-29.65$, -11.11Wh/100cyc) → flat · stable | aging 기대=decrease | vs=stable

에너지 손실 (dE)

충전-방전 에너지. | 계산: chg_E-dchg_E.

트렌드: 에너지 손실: early=8.32 → late=-1.791 ($\Delta=-10.11$, -3.476Wh/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

완화 용량 (Q_relax)

휴지 회복 용량. | 계산: DCIR block ΔQ .

트렌드: 완화 용량: early=-0.369 → late=0.07 ($\Delta=+0.439$, +0.2251Ah/100cyc) → increasing · context | aging 기대=either | vs=context

완화 용량 % (Q_relax_pct)

회복 용량 비율. | 계산: $Q_relax/Q \cdot 100$.

트렌드: 완화 용량 %: early=-0.5139 → late=0.106 ($\Delta=+0.6199$, +0.3169%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

dSoHQ/dN (dSoHQ_dN)

용량 유지율 순간 기울기. | 계산: diff SoHQ.

트렌드: dSoHQ/dN: early=-0.1665 → late=-0.02386 ($\Delta=+0.1426$, +0.1581%/cyc/100cyc) → increasing · opposite_aging | aging 기대=decrease | vs=opposite_aging

지표 설명 · 트렌드 (M02Ch103)

d2SoHQ (d2SoHQ)

SoHQ 2차 미분. | 계산: diff dSoHQ.

트렌드: d2SoHQ: early=-0.000978 → late=-0.07146 (Δ =-0.07049, -0.04825%/cyc2/100cyc) → decreasing · context | aging 기대=either | vs=context

충전후 휴지 초기 V (EoC_restV_init)

충전후 rest 시작 전압. | 계산: rest step 첫 샘플.

트렌드: 충전후 휴지 초기 V: early=4.2 → late=4.2 (Δ =+0.0001249, +5.205e-05V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 60s V (EoC_restV_60s)

충전후 rest 60초 전압. | 계산: rest $\approx$ 60 s 보간.

트렌드: 충전후 휴지 60s V: early=4.188 → late=4.183 (Δ =-0.005418, -0.002028V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 30분 V (EoC_restV_30m)

충전후 rest 30분 전압 (OCV에 가까움). | 계산: rest $\approx$ 1800 s 보간.

트렌드: 충전후 휴지 30분 V: early=4.174 → late=4.164 (Δ =-0.009361, -0.003801V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 종료 V (EoC_restV_end)

충전후 rest 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 충전후 휴지 종료 V: early=4.174 → late=4.164 (Δ =-0.009361, -0.003801V/100cyc) → flat · context | aging 기대=either | vs=context

충전후 휴지 완화량 (EoC_restV_relax)

충전후 end-init 완화. | 계산: EoC_restV_end – EoC_restV_init.

트렌드: 충전후 휴지 완화량: early=-0.02615 → late=-0.03556 (Δ =-0.009415, -0.003853V/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

충전후 60s 완화량 (EoC_restV_relax_60s)

충전후 60s-init. | 계산: $\text{EoC_restV_60s} - \text{EoC_restV_init}$.

트렌드: 충전후 60s 완화량: early=-0.01144 → late=-0.0169 (Δ =-0.005465, -0.00208V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전후 휴지 완화 τ (EoC_restV_tau)

충전후 rest 완화 시정수 proxy. | 계산: rest V(t) 완화 fit τ .

트렌드: 충전후 휴지 완화 τ : early=519 → late=507.9 (Δ =-11.08, -1.812s/100cyc) → flat · context | aging 기대=either | vs=context

충전후 60s V Δ vs기준 (delta_EoC_restV_60s)

기준 사이클 대비 EoC_restV_60s 이동. | 계산: $\text{EoC_restV_60s}(\text{cycle}) - \text{baseline}$.

트렌드: 충전후 60s V Δ vs기준: early=-0.000491 → late=-0.005909 (Δ =-0.005418, -0.002028V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전후 30분 V Δ vs기준 (delta_EoC_restV_30m)

기준 사이클 대비 EoC_restV_30m 이동. | 계산: $\text{EoC_restV_30m}(\text{cycle}) - \text{baseline}$.

트렌드: 충전후 30분 V Δ vs기준: early=-0.0005831 → late=-0.009945 (Δ =-0.009361, -0.003801V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전후 종료 V Δ vs기준 (delta_EoC_restV_end)

기준 사이클 대비 EoC_restV_end 이동. | 계산: $\text{EoC_restV_end}(\text{cycle}) - \text{baseline}$.

트렌드: 충전후 종료 V Δ vs기준: early=-0.0005831 → late=-0.009945 (Δ =-0.009361, -0.003801V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전후 완화 τ Δ vs기준 (delta_EoC_restV_tau)

기준 사이클 대비 EoC_restV_tau 이동. | 계산: $\text{EoC_restV_tau}(\text{cycle}) - \text{baseline}$.

트렌드: 충전후 완화 τ Δ vs기준: early=15.81 → late=4.731 (Δ =-11.08, -1.812s/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전후 휴지 초기 V (EoD_restV_init)

방전후 rest 시작 전압. | 계산: rest step 첫 샘플.

트렌드: 방전후 휴지 초기 V: early=2.5 → late=2.5 ($\Delta=+3.28\text{e-}05$, $+9.062\text{e-}06\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 60s V (EoD_restV_60s)

방전후 rest 60초 전압. | 계산: rest ≈ 60 s 보간.

트렌드: 방전후 휴지 60s V: early=2.907 → late=2.986 ($\Delta=+0.07914$, $+0.035\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 30분 V (EoD_restV_30m)

방전후 rest 30분 전압 (OCV에 가까움). | 계산: rest ≈ 1800 s 보간.

트렌드: 방전후 휴지 30분 V: early=3.035 → late=3.106 ($\Delta=+0.07055$, $+0.02831\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 종료 V (EoD_restV_end)

방전후 rest 마지막 전압. | 계산: rest step 끝 샘플.

트렌드: 방전후 휴지 종료 V: early=3.035 → late=3.106 ($\Delta=+0.07055$, $+0.02831\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 휴지 완화량 (EoD_restV_relax)

방전후 end-init 완화. | 계산: EoD_restV_end – EoD_restV_init.

트렌드: 방전후 휴지 완화량: early=0.5351 → late=0.6056 ($\Delta=+0.07057$, $+0.0283\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

방전후 60s 완화량 (EoD_restV_relax_60s)

방전후 60s-init. | 계산: EoD_restV_60s – EoD_restV_init.

트렌드: 방전후 60s 완화량: early=0.4071 → late=0.4862 ($\Delta=+0.07908$, $+0.03499\text{V}/100\text{cyc}$) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전후 휴지 완화 τ (EoD_restV_tau)

방전후 rest 완화 시정수 proxy. | 계산: rest V(t) 완화 fit τ .

트렌드: 방전후 휴지 완화 τ : early=369.3 $\rightarrow$ late=291 (Δ =-78.3, -37.29s/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

방전후 60s V Δ vs기준 (delta_EoD_restV_60s)

기준 사이클 대비 EoD_restV_60s 이동. | 계산: EoD_restV_60s(cycle) - baseline.

트렌드: 방전후 60s V Δ vs기준: early=0.05819 $\rightarrow$ late=0.1373 (Δ =+0.07914, +0.035V/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

방전후 30분 V Δ vs기준 (delta_EoD_restV_30m)

기준 사이클 대비 EoD_restV_30m 이동. | 계산: EoD_restV_30m(cycle) - baseline.

트렌드: 방전후 30분 V Δ vs기준: early=0.04166 $\rightarrow$ late=0.1122 (Δ =+0.07055, +0.02831V/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

방전후 종료 V Δ vs기준 (delta_EoD_restV_end)

기준 사이클 대비 EoD_restV_end 이동. | 계산: EoD_restV_end(cycle) - baseline.

트렌드: 방전후 종료 V Δ vs기준: early=0.04166 $\rightarrow$ late=0.1122 (Δ =+0.07055, +0.02831V/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

방전후 완화 τ Δ vs기준 (delta_EoD_restV_tau)

기준 사이클 대비 EoD_restV_tau 이동. | 계산: EoD_restV_tau(cycle) - baseline.

트렌드: 방전후 완화 τ Δ vs기준: early=9.884 $\rightarrow$ late=-68.41 (Δ =-78.3, -37.29s/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

EoC 방전 10s DCIR (EoC_dchgR_10s)

EoC 방전 시작 후 10s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V_0 - V(10s)|/|I| \cdot 1000$.

트렌드: EoC 방전 10s DCIR: early=0.1614 $\rightarrow$ late=0.1868 (Δ =+0.02544, +0.01702m Ω /100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

EOC 방전 10s 증가% (EoC_dchgR_10s_inc)

기준 대비 EoC_dchgR_10s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoC 방전 10s 증가%: early=28.76 → late=49.05 (Δ =+20.29, +13.58%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 30s DCIR (EoC_dchgR_30s)

EOC 방전 시작 후 30s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(30s)|/|I| \cdot 1000$.

트렌드: EoC 방전 30s DCIR: early=0.4343 → late=0.5117 (Δ =+0.0774, +0.04427m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 30s 증가% (EoC_dchgR_30s_inc)

기준 대비 EoC_dchgR_30s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoC 방전 30s 증가%: early=22.07 → late=43.82 (Δ =+21.75, +12.44%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 60s DCIR (EoC_dchgR_60s)

EOC 방전 시작 후 60s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(60s)|/|I| \cdot 1000$.

트렌드: EoC 방전 60s DCIR: early=0.757 → late=0.8939 (Δ =+0.137, +0.0728m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOC 방전 60s 증가% (EoC_dchgR_60s_inc)

기준 대비 EoC_dchgR_60s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoC 방전 60s 증가%: early=17.67 → late=38.96 (Δ =+21.29, +11.32%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EOd 충전 10s DCIR (EoD_chgR_10s)

EOd 충전 시작 후 10s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(10s)|/|I| \cdot 1000$.

트렌드: EoD 충전 10s DCIR: early=0.2829 → late=0.3339 (Δ =+0.05099, +0.01935m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

EoD 충전 10s 증가% (EoD_chgR_10s_inc)

기준 대비 EoD_chgR_10s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoD 충전 10s 증가%: early=0.4935 → late=18.6 (Δ =+18.11, +6.874%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD 충전 30s DCIR (EoD_chgR_30s)

EoD 충전 시작 후 30s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(30s)|/|I| \cdot 1000$.

트렌드: EoD 충전 30s DCIR: early=0.7654 → late=0.8268 (Δ =+0.06136, +0.01948m Ω /100cyc) → flat · stable | aging 기대=increase | vs=stable

EoD 충전 30s 증가% (EoD_chgR_30s_inc)

기준 대비 EoD_chgR_30s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoD 충전 30s 증가%: early=-3.036 → late=4.738 (Δ =+7.774, +2.468%/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD 충전 60s DCIR (EoD_chgR_60s)

EoD 충전 시작 후 60s 시점 $\Delta V/I$. Rct·확산 포함 (순수 R Ω 아님). | 계산: $|V0-V(60s)|/|I| \cdot 1000$.

트렌드: EoD 충전 60s DCIR: early=1.355 → late=1.333 (Δ =-0.02159, -0.02573m Ω /100cyc) → flat · stable | aging 기대=increase | vs=stable

EoD 충전 60s 증가% (EoD_chgR_60s_inc)

기준 대비 EoD_chgR_60s 상대 증가율. | 계산: $100 \cdot (R/R0 - 1)$.

트렌드: EoD 충전 60s 증가%: early=-6.437 → late=-7.927 (Δ =-1.49, -1.777%/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

EoC R10/R60 (EoC_dchgR_10_60_ratio)

10s/60s 비. 초기 응답 비중. | 계산: EoC_dchgR_10s / EoC_dchgR_60s.

트렌드: EoC R10/R60: early=0.2113 → late=0.2078 (Δ =-0.003425, +0.001961/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

EoD R10/R60 (EoD_chgR_10_60_ratio)

10s/60s 비. | 계산: EoD_chgR_10s / EoD_chgR_60s.

트렌드: EoD R10/R60: early=0.2088 → late=0.2509 (Δ =+0.04206, +0.018091/100cyc) → increasing · context | aging 기대=either | vs=context

EoC R10s @25C (EoC_dchgR_10s_T25)

온도 보정 10s DCIR. | 계산: Arrhenius-ish correct_r_to_25c.

트렌드: EoC R10s @25C: early=0.07712 → late=0.08928 (Δ =+0.01216, +0.008134m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoD R10s @25C (EoD_chgR_10s_T25)

온도 보정 10s DCIR. | 계산: correct_r_to_25c.

트렌드: EoD R10s @25C: early=0.1352 → late=0.1596 (Δ =+0.02436, +0.009247m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

EoC R10s 성장/50cyc (EoC_dchgR_10s_growth_50)

롤링 기울기 (50사이클). | 계산: rolling slope of EoC_dchgR_10s.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R Ω (SOC80) (R_ohmic_soc80)

$\sqrt{t}$ 외삽 t→0 절편 (초기 비음 포함 proxy). | 계산: DCIR early $\sqrt{t}$ intercept.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct (SOC80) (R_ct_soc80)

중간 잔차 지수항. Cdl 미분리. | 계산: resid exp-sat fit.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

τ_{ct} (SOC80) (tau_ct_soc80)

Rct 시정수 (fit seed 2s). | 계산: curve_fit τ .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

A_diff (SOC80) (A_diff_soc80)

후반 $\sqrt{t}$ 확산 계수. | 계산: $t \in [10, 30]$ R vs $\sqrt{t}$ slope.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R30s 총 (SOC80) (R_30s_total_soc80)

펄스 30초 총 DCIR. | 계산: R($t \approx 30s$).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R Ω 분율 (SOC80) (R_ohmic_frac_soc80)

R Ω / R30s. | 계산: R_ohmic / R_30s_total.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct분율 (SOC80) (R_ct_frac_soc80)

Rct / R30s. | 계산: R_ct / R_30s_total.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Rdiff분율 (SOC80) (R_diff_frac_soc80)

$A\sqrt{30}$ / R30s. | 계산: A_diff*sqrt(30)/R30s.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

회복 τ_1 (SOC80) (R_recovery_tau1_soc80)

펄스 후 빠른 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 τ_2 (SOC80) (R_recovery_tau2_soc80)

펄스 후 느린 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} est (SOC80) (V_inf_est_soc80)

회복 fit 무한시간 전압. | 계산: recovery V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} rest (SOC80) (V_inf_rest_soc80)

rest 기반 V_{∞} . | 계산: rest asymptote.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 (SOC80) (self_discharge_rate_soc80)

휴지 중 전압 강하율. | 계산: dV/dt on long rest.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

DCIR fit R2 (SOC80) (dcir_fit_r2_soc80)

R(t) 3성분 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

DCIR fit RMSE (SOC80) (dcir_fit_rmse_soc80)

상대 RMSE. | 계산: rmse/meanR .

트렌드: 데이터 부족 | aging 기대=decrease | vs=n/a

펄스 전류 (SOC80) (pulse_current_A_soc80)

DCIR 펄스 || 중앙값. | 계산: median || on pulse.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$t \leq 1s$ 샘플수 (SOC80) (n_t_le_1s_soc80)

early 샘플 수 (R Ω 외삽 품질). | 계산: count $t \leq 1$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 amp비 (SOC80) (relax_amp_ratio_soc80)

느린/전체 회복 진폭비. | 계산: $|b_2|/(|b_1|+|b_2|)$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

완화 완성도 (SOC80) (relax_completeness_soc80)

rest 완화 충분성. | 계산: relax completeness.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 fit R2 (SOC80) (recovery_fit_r2_soc80)

two-exp 회복 fit 품질. | 계산: r^2 .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

DCIR fit 유효 (SOC80) (dcir_fit_valid_soc80)

rmse/r2/cond 게이트 통과. | 계산: dcir_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

DCIR fit 조건수 (SOC80) (dcir_fit_cond_soc80)

공분산 조건수 (축퇴 지표). | 계산: cond(pcov).

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 fit 유효 (SOC80) (sd_fit_valid_soc80)

self-discharge fit 게이트. | 계산: sd_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

펄스 포인트수 (SOC80) (n_points_soc80)

DCIR 펄스 샘플 수. | 계산: n_points.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R Ω (SOC50) (R_ohmic_soc50)

$\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편 (초기 비음 포함 proxy). | 계산: DCIR early $\sqrt{t}$ intercept.

트렌드: R Ω (SOC50): early=1.362 $\rightarrow$ late=2.366 (Δ =+1.004, +0.5021m Ω /100cyc) $\rightarrow$ increasing · matches_aging | aging
기대=increase | vs=matches_aging

R_{ct} (SOC50) (R_ct_soc50)

중간 잔차 지수항. Cdl 미분리. | 계산: resid exp-sat fit.

트렌드: R_{ct} (SOC50): early=0.7175 $\rightarrow$ late=0.9128 (Δ =+0.1953, +0.08828m Ω /100cyc) $\rightarrow$ increasing · matches_aging |
aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

τ_{ct} (SOC50) (tau_ct_soc50)

Rct 시정수 (fit seed 2s). | 계산: curve_fit τ .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

A_diff (SOC50) (A_diff_soc50)

후반 $\sqrt{t}$ 확산 계수. | 계산: $t \in [10, 30]$ R vs $\sqrt{t}$ slope.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R30s 총 (SOC50) (R_30s_total_soc50)

펄스 30초 총 DCIR. | 계산: R($t \approx 30s$).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R Ω 분율 (SOC50) (R_ohmic_frac_soc50)

R Ω / R30s. | 계산: R_ohmic / R_30s_total.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct분율 (SOC50) (R_ct_frac_soc50)

Rct / R30s. | 계산: R_ct / R_30s_total.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Rdiff분율 (SOC50) (R_diff_frac_soc50)

A $\sqrt{30}$ / R30s. | 계산: A_diff*sqrt(30)/R30s.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

회복 τ_1 (SOC50) (R_recovery_tau1_soc50)

펄스 후 빠른 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 τ_2 (SOC50) (R_recovery_tau2_soc50)

펄스 후 느린 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} est (SOC50) (V_inf_est_soc50)

회복 fit 무한시간 전압. | 계산: recovery V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} rest (SOC50) (V_inf_rest_soc50)

rest 기반 V_{∞} . | 계산: rest asymptote.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 (SOC50) (self_discharge_rate_soc50)

휴지 중 전압 강하율. | 계산: dV/dt on long rest.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

DCIR fit R2 (SOC50) (dcir_fit_r2_soc50)

R(t) 3성분 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

DCIR fit RMSE (SOC50) (dcir_fit_rmse_soc50)

상대 RMSE. | 계산: rmse/meanR .

트렌드: 데이터 부족 | aging 기대=decrease | vs=n/a

펄스 전류 (SOC50) (pulse_current_A_soc50)

DCIR 펄스 || 중앙값. | 계산: median || on pulse.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$t \leq 1s$ 샘플수 (SOC50) (n_t_le_1s_soc50)

early 샘플 수 (R Ω 외삽 품질). | 계산: count $t \leq 1$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 amp비 (SOC50) (relax_amp_ratio_soc50)

느린/전체 회복 진폭비. | 계산: $|b_2|/(|b_1|+|b_2|)$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

완화 완성도 (SOC50) (relax_completeness_soc50)

rest 완화 충분성. | 계산: relax completeness.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 fit R2 (SOC50) (recovery_fit_r2_soc50)

two-exp 회복 fit 품질. | 계산: r^2 .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

DCIR fit 유효 (SOC50) (dcir_fit_valid_soc50)

rmse/r2/cond 게이트 통과. | 계산: dcir_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

DCIR fit 조건수 (SOC50) (dcir_fit_cond_soc50)

공분산 조건수 (축퇴 지표). | 계산: cond(pcov).

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 fit 유효 (SOC50) (sd_fit_valid_soc50)

self-discharge fit 게이트. | 계산: sd_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

펄스 포인트수 (SOC50) (n_points_soc50)

DCIR 펄스 샘플 수. | 계산: n_points.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$R\Omega$ (SOC20) (R_ohmic_soc20)

$\sqrt{t}$ 외삽 $t \rightarrow 0$ 절편 (초기 비음 포함 proxy). | 계산: DCIR early $\sqrt{t}$ intercept.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R_{ct} (SOC20) (R_ct_soc20)

중간 잔차 지수항. Cdl 미분리. | 계산: resid exp-sat fit.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

τ_{ct} (SOC20) (τ_{ct_soc20})

Rct 시정수 (fit seed 2s). | 계산: $\text{curve_fit } \tau$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

A_{diff} (SOC20) (A_{diff_soc20})

후반 $\sqrt{t}$ 확산 계수. | 계산: $t \in [10, 30]$ R vs $\sqrt{t}$ slope.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

R30s 총 (SOC20) ($R_{30s_total_soc20}$)

펄스 30초 총 DCIR. | 계산: $R(t \approx 30s)$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

$R\Omega$ 분율 (SOC20) ($R_{ohmic_frac_soc20}$)

$R\Omega$ / R30s. | 계산: $R_{ohmic} / R_{30s_total}$.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Rct분율 (SOC20) ($R_{ct_frac_soc20}$)

Rct / R30s. | 계산: R_{ct} / R_{30s_total} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Rdiff분율 (SOC20) ($R_{diff_frac_soc20}$)

$A\sqrt{30}$ / R30s. | 계산: $A_{diff} \cdot \sqrt{30} / R_{30s}$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

회복 τ_1 (SOC20) (R_recovery_tau1_soc20)

펄스 후 빠른 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 τ_2 (SOC20) (R_recovery_tau2_soc20)

펄스 후 느린 회복 시정수. | 계산: two-exp recovery fit.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} est (SOC20) (V_inf_est_soc20)

회복 fit 무한시간 전압. | 계산: recovery V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

V_{∞} rest (SOC20) (V_inf_rest_soc20)

rest 기반 V_{∞} . | 계산: rest asymptote.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 (SOC20) (self_discharge_rate_soc20)

휴지 중 전압 강하율. | 계산: dV/dt on long rest.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

DCIR fit R2 (SOC20) (dcir_fit_r2_soc20)

R(t) 3성분 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

DCIR fit RMSE (SOC20) (dcir_fit_rmse_soc20)

상대 RMSE. | 계산: rmse/meanR .

트렌드: 데이터 부족 | aging 기대=decrease | vs=n/a

펄스 전류 (SOC20) (pulse_current_A_soc20)

DCIR 펄스 || 중앙값. | 계산: median || on pulse.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

$t \leq 1s$ 샘플수 (SOC20) (n_t_le_1s_soc20)

early 샘플 수 (R Ω 외삽 품질). | 계산: count $t \leq 1$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 amp비 (SOC20) (relax_amp_ratio_soc20)

느린/전체 회복 진폭비. | 계산: $|b_2|/(|b_1|+|b_2|)$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

완화 완성도 (SOC20) (relax_completeness_soc20)

rest 완화 충분성. | 계산: relax completeness.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

회복 fit R2 (SOC20) (recovery_fit_r2_soc20)

two-exp 회복 fit 품질. | 계산: r^2 .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

DCIR fit 유효 (SOC20) (dcir_fit_valid_soc20)

rmse/r2/cond 게이트 통과. | 계산: dcir_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

DCIR fit 조건수 (SOC20) (dcir_fit_cond_soc20)

공분산 조건수 (축퇴 지표). | 계산: cond(pcov).

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

자기방전 fit 유효 (SOC20) (sd_fit_valid_soc20)

self-discharge fit 게이트. | 계산: sd_fit_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

펄스 포인트수 (SOC20) (n_points_soc20)

DCIR 펄스 샘플 수. | 계산: n_points.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

기계/화학 비 (mech_vs_chem_ratio)

R Ω /R_{ct} 상대 비중. | 계산: R_ohmic_soc50 / R_ct_soc50.

트렌드: 기계/화학 비: early=1.898 → late=2.592 (Δ =+0.6938, +0.37961/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

R Ω 성장/100cyc (R_ohmic_growth_100)

기준 대비 R Ω 성장률 (레벨과 별개). | 계산: (R-R0)/((N-N0)/100).

트렌드: R Ω 성장/100cyc: early=0.7711 → late=0.4781 (Δ =-0.293, -0.2952m Ω /100cyc/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

Rct 성장/100cyc (R_ct_growth_100)

기준 대비 Rct 성장률. | 계산: $(Rct-Rct0)/((N-N0)/100)$.

트렌드: Rct 성장/100cyc: early=0.09905 → late=0.09302 ($\Delta=-0.006022$, $-0.006067m\Omega/100cyc/100cyc$) → decreasing · context | aging 기대=either | vs=context

R30s 20/50 (R_ratio_20_50)

SOC20 vs 50 총저항 비. | 계산: $R30s_{20} / R30s_{50}$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R30s 80/50 (R_ratio_80_50)

SOC80 vs 50 총저항 비. | 계산: $R30s_{80} / R30s_{50}$.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R-SOC 기울기 (R_SOC_slope)

SOC20/50/80 R30s 선형 기울기. | 계산: linreg R vs SOC.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

R-SOC 곡률 (R_SOC_curvature)

3점 이차 계수. | 계산: polyfit deg2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

RΩ SOC50 (ff) (R_ohmic_soc50_ff)

DCIR 블록 forward-fill 값. | 계산: block stamp + ffill.

트렌드: RΩ SOC50 (ff): early=1.362 → late=2.366 ($\Delta=+1.004$, $+0.5021m\Omega/100cyc$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

Rct SOC50 (ff) (R_ct_soc50_ff)

DCIR 블록 forward-fill 값. | 계산: block stamp + ffill.

트렌드: Rct SOC50 (ff): early=0.7175 → late=0.9128 (Δ =+0.1953, +0.08828m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전 평균 V (chg_V_avg)

충전 평균 전압. | 계산: mean V charge.

트렌드: 충전 평균 V: early=3.798 → late=3.845 (Δ =+0.04697, +0.01843V/100cyc) → flat · context | aging 기대=either | vs=context

방전 평균 V (dchg_V_avg)

방전 평균 전압. | 계산: mean V discharge.

트렌드: 방전 평균 V: early=3.397 → late=3.305 (Δ =-0.09147, -0.03913V/100cyc) → flat · context | aging 기대=either | vs=context

충전 평균V Δ (delta_chg_V_avg)

기준 대비. | 계산: delta.

트렌드: 충전 평균V Δ : early=0.03835 → late=0.08532 (Δ =+0.04697, +0.01843V/100cyc) → increasing · context | aging 기대=either | vs=context

방전 평균V Δ (delta_dchg_V_avg)

기준 대비. | 계산: delta.

트렌드: 방전 평균V Δ : early=-0.03798 → late=-0.1294 (Δ =-0.09147, -0.03913V/100cyc) → decreasing · context | aging 기대=either | vs=context

충전 IR drop proxy (chg_ir_drop_proxy)

충전 초반 IR 강하 proxy. | 계산: early Δ V.

트렌드: 충전 IR drop proxy: early=0.04018 → late=0.0401 (Δ =-7.45e-05, -1.465e-05V/100cyc) → flat · stable | aging 기대=increase | vs=stable

지표 설명 · 트렌드 (M02Ch103)

방전 IR drop proxy (dchg_ir_drop_proxy)

방전 초반 IR 강하 proxy. | 계산: early ΔV .

트렌드: 방전 IR drop proxy: early=0.0401 $\rightarrow$ late=0.0401 ($\Delta=-1.7\text{e-}06$, +0.0004881V/100cyc) $\rightarrow$ flat · stable | aging
기대=increase | vs=stable

히스테리시스 면적 (hyst_area)

전체 충방전 히스테리시스. | 계산: $\oint (V_{\text{chg}} - V_{\text{dchg}}) dQ$.

트렌드: 히스테리시스 면적: early=0.5569 $\rightarrow$ late=0.6104 ($\Delta=+0.05352$, +0.02366V/100cyc) $\rightarrow$ flat · stable | aging
기대=increase | vs=stable

저SOC 히스테리시스 (hyst_area_low)

저SOC 밴드. Si chemo-mech. | 계산: band integral.

트렌드: 저SOC 히스테리시스: early=0.0766 $\rightarrow$ late=0.0465 ($\Delta=-0.0301$, -0.01222V/100cyc) $\rightarrow$ decreasing · opposite_aging |
aging 기대=increase | vs=opposite_aging

중SOC 히스테리시스 (hyst_area_mid)

중SOC 밴드. | 계산: band integral.

트렌드: 중SOC 히스테리시스: early=0.2613 $\rightarrow$ late=0.3211 ($\Delta=+0.05974$, +0.02561V/100cyc) $\rightarrow$ increasing · context | aging
기대=either | vs=context

고SOC 히스테리시스 (hyst_area_high)

고SOC 밴드. PE 보조. | 계산: band integral.

트렌드: 고SOC 히스테리시스: early=0.2168 $\rightarrow$ late=0.2401 ($\Delta=+0.02337$, +0.01028V/100cyc) $\rightarrow$ flat · stable | aging
기대=increase | vs=stable

히스테리시스 저SOC분율 (hyst_frac_low)

low/total. | 계산: hyst_low/hyst.

트렌드: 히스테리시스 저SOC분율: early=0.1375 $\rightarrow$ late=0.07631 ($\Delta=-0.06123$, -0.02511/100cyc) $\rightarrow$ decreasing · context | aging
기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

히스테리시스 고SOC분율 (hyst_frac_high)

high/total. | 계산: hyst_high/hyst.

트렌드: 히스테리시스 고SOC분율: early=0.3891 → late=0.3931 (Δ =+0.003984, +0.0017611/100cyc) → flat · context | aging 기대=either | vs=context

최대 히스테리시스 dV (hyst_max_dV)

최대 총방전 전압차. | 계산: max|Vchg-Vdchg|.

트렌드: 최대 히스테리시스 dV: early=1.548 → late=1.619 (Δ =+0.07142, +0.03149V/100cyc) → flat · stable | aging 기대=increase | vs=stable

max dV 저SOC (hyst_max_dV_low)

저SOC 최대 dV. | 계산: band max.

트렌드: max dV 저SOC: early=0.6514 → late=0.4936 (Δ =-0.1578, -0.06261V/100cyc) → decreasing · opposite_aging | aging 기대=increase | vs=opposite_aging

max dV 중SOC (hyst_max_dV_mid)

중SOC 최대 dV. | 계산: band max.

트렌드: max dV 중SOC: early=0.9765 → late=1.114 (Δ =+0.1378, +0.06025V/100cyc) → increasing · context | aging 기대=either | vs=context

max dV 고SOC (hyst_max_dV_high)

고SOC 최대 dV. | 계산: band max.

트렌드: max dV 고SOC: early=1.548 → late=1.619 (Δ =+0.07142, +0.03149V/100cyc) → flat · context | aging 기대=either | vs=context

히스테리시스 Δ (delta_hyst_area)

기준 대비 면적. | 계산: delta.

트렌드: 히스테리시스 Δ: early=0.01062 → late=0.06414 (Δ =+0.05352, +0.02366V/100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

max dV Δ (delta_hyst_max_dV)

기준 대비. | 계산: delta.

트렌드: max dV Δ : early=0.01247 $\rightarrow$ late=0.08389 (Δ =+0.07142, +0.03149V/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전 플레토 V (chg_plateau_V)

충전 플레토 전압. | 계산: plateau detect.

트렌드: 충전 플레토 V: early=3.732 $\rightarrow$ late=3.778 (Δ =+0.04597, +0.01691V/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

충전 플레토 폭 (chg_plateau_width)

충전 플레토 Q 폭. | 계산: plateau width.

트렌드: 충전 플레토 폭: early=14.64 $\rightarrow$ late=14.28 (Δ =-0.3585, +0.1708Q-units/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 플레토 V (dchg_plateau_V)

방전 플레토 전압. | 계산: plateau detect.

트렌드: 방전 플레토 V: early=3.169 $\rightarrow$ late=3.131 (Δ =-0.03847, -0.01613V/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 플레토 폭 (dchg_plateau_width)

방전 플레토 Q 폭. | 계산: plateau width.

트렌드: 방전 플레토 폭: early=16.85 $\rightarrow$ late=14.17 (Δ =-2.681, -1.363Q-units/100cyc) $\rightarrow$ decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 플레토 ΔV (delta_dchg_plateau_V)

기준 대비 이동. | 계산: delta plateau V.

트렌드: 방전 플레토 ΔV : early=-0.0323 $\rightarrow$ late=-0.07078 (Δ =-0.03847, -0.01613V/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전 컷오프 마진 (dchg_V_cutoff_margin)

컷오프까지 여유. | 계산: Vmin-margin.

트렌드: 방전 컷오프 마진: early=0.4038 → late=0.2673 ($\Delta=-0.1365$, $-0.0606\text{V}/100\text{cyc}$) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

컷오프 마진 Δ (delta_dchg_V_cutoff_margin)

기준 대비. | 계산: delta.

트렌드: 컷오프 마진 Δ : early=-0.03639 → late=-0.1729 ($\Delta=-0.1365$, $-0.0606\text{V}/100\text{cyc}$) → decreasing · matches_aging | aging 기대=decrease | vs=matches_aging

방전 형상 DTW (dchg_shape_DTW)

기준 곡선 DTW 거리. | 계산: DTW vs baseline.

트렌드: 방전 형상 DTW: early=0.003829 → late=0.007357 ($\Delta=+0.003528$, $+0.0015351/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

DTW Δ (delta_dchg_shape_DTW)

기준 대비 DTW. | 계산: delta.

트렌드: DTW Δ : early=0.002337 → late=0.005865 ($\Delta=+0.003528$, $+0.0015351/100\text{cyc}$) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

충전 dQ/dV 피크1 V (chg_dQdV_peak1_V)

충전 IC 1번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크1 V: early=3.715 → late=3.764 ($\Delta=+0.04926$, $+0.07757\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크1 높이 (chg_dQdV_peak1)

충전 IC 1번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크1 높이: early=102 → late=99.59 ($\Delta=-2.451$, $+1640\text{Ah}/\text{V}/100\text{cyc}$) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전 dQ/dV 피크1 V (dchg_dQdV_peak1_V)

방전 IC 1번 피크 전압. | 계산: SG + find_peaks.

트렌드: 방전 dQ/dV 피크1 V: early=3.165 → late=3.11 (Δ=-0.05533, -0.02183V/100cyc) → flat · context | aging 기대=either | vs=context

방전 dQ/dV 피크1 높이 (dchg_dQdV_peak1)

방전 IC 1번 피크 높이. | 계산: peak height.

트렌드: 방전 dQ/dV 피크1 높이: early=-70.17 → late=-52.85 (Δ=+17.33, +7.317Ah/V/100cyc) → increasing · context | aging 기대=either | vs=context

충전 dV/dQ 피크1 Q (chg_dVdQ_peak1_Q)

충전 dV/dQ 1번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 충전 dV/dQ 피크1 Q: early=56.52 → late=49.69 (Δ=-6.83, -2.563Ah/100cyc) → flat · context | aging 기대=either | vs=context

충전 dV/dQ 피크1 높이 (chg_dVdQ_peak1)

충전 dV/dQ 1번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 충전 dV/dQ 피크1 높이: early=0.0153 → late=0.0164 (Δ=+0.001109, +0.0004287V/Ah/100cyc) → flat · context | aging 기대=either | vs=context

방전 dV/dQ 피크1 Q (dchg_dVdQ_peak1_Q)

방전 dV/dQ 1번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 방전 dV/dQ 피크1 Q: early=12.54 → late=9.178 (Δ=-3.358, +0.2631Ah/100cyc) → flat · context | aging 기대=either | vs=context

방전 dV/dQ 피크1 높이 (dchg_dVdQ_peak1)

방전 dV/dQ 1번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 방전 dV/dQ 피크1 높이: early=-0.02331 → late=-0.02206 (Δ=+0.001243, +0.0006956V/Ah/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

충전 dQ/dV 피크2 V (chg_dQdV_peak2_V)

충전 IC 2번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크2 V: early=3.967 → late=4.002 (Δ =+0.03528, +0.05421V/100cyc) → flat · context | aging
기대=either | vs=context

충전 dQ/dV 피크2 높이 (chg_dQdV_peak2)

충전 IC 2번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크2 높이: early=99.8 → late=81.02 (Δ =-18.77, -3.899Ah/V/100cyc) → flat · context | aging
기대=either | vs=context

방전 dQ/dV 피크2 V (dchg_dQdV_peak2_V)

방전 IC 2번 피크 전압. | 계산: SG + find_peaks.

트렌드: 방전 dQ/dV 피크2 V: early=3.639 → late=3.58 (Δ =-0.05918, -0.04113V/100cyc) → flat · context | aging
기대=either | vs=context

방전 dQ/dV 피크2 높이 (dchg_dQdV_peak2)

방전 IC 2번 피크 높이. | 계산: peak height.

트렌드: 방전 dQ/dV 피크2 높이: early=-54.17 → late=-48.35 (Δ =+5.829, +1.858Ah/V/100cyc) → flat · context | aging
기대=either | vs=context

충전 dV/dQ 피크2 Q (chg_dVdQ_peak2_Q)

충전 dV/dQ 2번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크2 높이 (chg_dVdQ_peak2)

충전 dV/dQ 2번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

방전 dV/dQ 피크2 Q (dchg_dVdQ_peak2_Q)

방전 dV/dQ 2번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 방전 dV/dQ 피크2 Q: early=28.95 → late=28.28 (Δ =-0.6713, -1.905Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dV/dQ 피크2 높이 (dchg_dVdQ_peak2)

방전 dV/dQ 2번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 방전 dV/dQ 피크2 높이: early=-0.02133 → late=-0.02146 (Δ =-0.0001316, -0.0004902V/Ah/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크3 V (chg_dQdV_peak3_V)

충전 IC 3번 피크 전압. | 계산: SG + find_peaks.

트렌드: 충전 dQ/dV 피크3 V: early=3.964 → late=3.964 (Δ =+0, +0.01345V/100cyc) → flat · context | aging 기대=either | vs=context

충전 dQ/dV 피크3 높이 (chg_dQdV_peak3)

충전 IC 3번 피크 높이. | 계산: peak prominence/height.

트렌드: 충전 dQ/dV 피크3 높이: early=83.79 → late=83.79 (Δ =+0, -4.882Ah/V/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dQ/dV 피크3 V (dchg_dQdV_peak3_V)

방전 IC 3번 피크 전압. | 계산: SG + find_peaks.

트렌드: 방전 dQ/dV 피크3 V: early=3.977 → late=3.829 (Δ =-0.148, -0.06948V/100cyc) → flat · context | aging 기대=either | vs=context

방전 dQ/dV 피크3 높이 (dchg_dQdV_peak3)

방전 IC 3번 피크 높이. | 계산: peak height.

트렌드: 방전 dQ/dV 피크3 높이: early=-63.72 → late=-47.5 (Δ =+16.22, +6.429Ah/V/100cyc) → increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

충전 dV/dQ 피크3 Q (chg_dVdQ_peak3_Q)

충전 dV/dQ 3번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크3 높이 (chg_dVdQ_peak3)

충전 dV/dQ 3번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크3 Q (dchg_dVdQ_peak3_Q)

방전 dV/dQ 3번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크3 높이 (dchg_dVdQ_peak3)

방전 dV/dQ 3번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크4 V (chg_dQdV_peak4_V)

충전 IC 4번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크4 높이 (chg_dQdV_peak4)

충전 IC 4번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

방전 dQ/dV 피크4 V (dchg_dQdV_peak4_V)

방전 IC 4번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크4 높이 (dchg_dQdV_peak4)

방전 IC 4번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크4 Q (chg_dVdQ_peak4_Q)

충전 dV/dQ 4번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크4 높이 (chg_dVdQ_peak4)

충전 dV/dQ 4번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크4 Q (dchg_dVdQ_peak4_Q)

방전 dV/dQ 4번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크4 높이 (dchg_dVdQ_peak4)

방전 dV/dQ 4번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

충전 dQ/dV 피크5 V (chg_dQdV_peak5_V)

충전 IC 5번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크5 높이 (chg_dQdV_peak5)

충전 IC 5번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크5 V (dchg_dQdV_peak5_V)

방전 IC 5번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크5 높이 (dchg_dQdV_peak5)

방전 IC 5번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크5 Q (chg_dVdQ_peak5_Q)

충전 dV/dQ 5번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크5 높이 (chg_dVdQ_peak5)

충전 dV/dQ 5번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

방전 dV/dQ 피크5 Q (dchg_dVdQ_peak5_Q)

방전 dV/dQ 5번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크5 높이 (dchg_dVdQ_peak5)

방전 dV/dQ 5번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크6 V (chg_dQdV_peak6_V)

충전 IC 6번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크6 높이 (chg_dQdV_peak6)

충전 IC 6번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크6 V (dchg_dQdV_peak6_V)

방전 IC 6번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크6 높이 (dchg_dQdV_peak6)

방전 IC 6번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

충전 dV/dQ 피크6 Q (chg_dVdQ_peak6_Q)

충전 dV/dQ 6번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크6 높이 (chg_dVdQ_peak6)

충전 dV/dQ 6번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크6 Q (dchg_dVdQ_peak6_Q)

방전 dV/dQ 6번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크6 높이 (dchg_dVdQ_peak6)

방전 dV/dQ 6번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크7 V (chg_dQdV_peak7_V)

충전 IC 7번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크7 높이 (chg_dQdV_peak7)

충전 IC 7번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

방전 dQ/dV 피크7 V (dchg_dQdV_peak7_V)

방전 IC 7번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크7 높이 (dchg_dQdV_peak7)

방전 IC 7번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크7 Q (chg_dVdQ_peak7_Q)

충전 dV/dQ 7번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크7 높이 (chg_dVdQ_peak7)

충전 dV/dQ 7번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크7 Q (dchg_dVdQ_peak7_Q)

방전 dV/dQ 7번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크7 높이 (dchg_dVdQ_peak7)

방전 dV/dQ 7번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

충전 dQ/dV 피크8 V (chg_dQdV_peak8_V)

충전 IC 8번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dQ/dV 피크8 높이 (chg_dQdV_peak8)

충전 IC 8번 피크 높이. | 계산: peak prominence/height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크8 V (dchg_dQdV_peak8_V)

방전 IC 8번 피크 전압. | 계산: SG + find_peaks.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dQ/dV 피크8 높이 (dchg_dQdV_peak8)

방전 IC 8번 피크 높이. | 계산: peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크8 Q (chg_dVdQ_peak8_Q)

충전 dV/dQ 8번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 dV/dQ 피크8 높이 (chg_dVdQ_peak8)

충전 dV/dQ 8번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

방전 dV/dQ 피크8 Q (dchg_dVdQ_peak8_Q)

방전 dV/dQ 8번 피크 용량 위치. | 계산: dV/dQ peak Q.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

방전 dV/dQ 피크8 높이 (dchg_dVdQ_peak8)

방전 dV/dQ 8번 피크 높이. | 계산: dV/dQ peak height.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

충전 피크1 ΔV (delta_chg_dQdV_peak1_V)

기준 대비 충전 dQ/dV 피크1 이동. | 계산: delta_chg_dQdV_peak1_V

트렌드: 충전 피크1 ΔV: early=0.2809 → late=0.3301 (Δ=+0.04926, +0.07757V/100cyc) → increasing · context | aging 기대=either | vs=context

방전 피크1 ΔV (delta_dchg_dQdV_peak1_V)

기준 대비 방전 dQ/dV 피크1 이동. | 계산: delta_dchg_dQdV_peak1_V

트렌드: 방전 피크1 ΔV: early=-0.02173 → late=-0.07706 (Δ=-0.05533, -0.02183V/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dV/dQ @SOC0 (dchg_dVdQ_SOC0)

저SOC cliff dV/dQ. | 계산: dchg_dVdQ_SOC0

트렌드: 방전 dV/dQ @SOC0: early=0.103 → late=0.05649 (Δ=-0.04656, -0.0216V/Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

방전 dV/dQ @SOC5 (dchg_dVdQ_SOC5)

SOC≈5% dV/dQ. | 계산: dchg_dVdQ_SOC5

트렌드: 방전 dV/dQ @SOC5: early=0.04471 → late=0.03891 (Δ=-0.005793, -0.002904V/Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전 dV/dQ @SOC10 (dchg_dVdQ_SOC10)

SOC $\approx$ 10% dV/dQ. | 계산: dchg_dVdQ_SOC10

트렌드: 방전 dV/dQ @SOC10: early=0.02643 $\rightarrow$ late=0.02972 (Δ =+0.003291, +0.001112V/Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 dV/dQ @mid (dchg_dVdQ_SOCmid)

중SOC dV/dQ. | 계산: dchg_dVdQ_SOCmid

트렌드: 방전 dV/dQ @mid: early=0.019 $\rightarrow$ late=0.02103 (Δ =+0.002033, +0.0009221V/Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 cliff Q (dchg_dVdQ_SOC0_Q)

SOC0 dV/dQ 위치 Q. | 계산: dchg_dVdQ_SOC0_Q

트렌드: 방전 cliff Q: early=69.5 $\rightarrow$ late=62.47 (Δ =-7.031, -2.55Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

방전 cliff 폭 (dchg_dVdQ_SOC0_cliff_width)

저SOC cliff 폭. | 계산: dchg_dVdQ_SOC0_cliff_width

트렌드: 방전 cliff 폭: early=4.736 $\rightarrow$ late=2.627 (Δ =-2.108, -0.9554Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

cliff/mid 비 (dchg_dVdQ_SOC0_to_mid_ratio)

SOC0/mid dV/dQ 비. | 계산: dchg_dVdQ_SOC0_to_mid_ratio

트렌드: cliff/mid 비: early=5.423 $\rightarrow$ late=2.682 (Δ =-2.741, -1.2711/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

충전 dV/dQ @100 (chg_dVdQ_SOC100)

만충 부근 dV/dQ. | 계산: chg_dVdQ_SOC100

트렌드: 충전 dV/dQ @100: early=0.007272 $\rightarrow$ late=0.008133 (Δ =+0.000861, +0.0005336V/Ah/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

dV/dQ SOC0 Δ (delta_dchg_dVdQ_SOC0)

기준 대비. | 계산: delta_dchg_dVdQ_SOC0

트렌드: dV/dQ SOC0 Δ: early=-0.003518 → late=-0.05008 (Δ=-0.04656, -0.0216V/Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

dV/dQ SOC5 Δ (delta_dchg_dVdQ_SOC5)

기준 대비. | 계산: delta_dchg_dVdQ_SOC5

트렌드: dV/dQ SOC5 Δ: early=-0.00347 → late=-0.009262 (Δ=-0.005793, -0.002904V/Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

dV/dQ SOC10 Δ (delta_dchg_dVdQ_SOC10)

기준 대비. | 계산: delta_dchg_dVdQ_SOC10

트렌드: dV/dQ SOC10 Δ: early=-0.00169 → late=0.001602 (Δ=+0.003291, +0.001112V/Ah/100cyc) → increasing · context | aging 기대=either | vs=context

dV/dQ mid Δ (delta_dchg_dVdQ_SOCmid)

기준 대비. | 계산: delta_dchg_dVdQ_SOCmid

트렌드: dV/dQ mid Δ: early=0.0004595 → late=0.002493 (Δ=+0.002033, +0.0009221V/Ah/100cyc) → increasing · context | aging 기대=either | vs=context

dV/dQ 100 Δ (delta_chg_dVdQ_SOC100)

기준 대비. | 계산: delta_chg_dVdQ_SOC100

트렌드: dV/dQ 100 Δ: early=-0.003339 → late=-0.002478 (Δ=+0.000861, +0.0005336V/Ah/100cyc) → increasing · context | aging 기대=either | vs=context

cliff 폭 Δ (delta_dchg_dVdQ_SOC0_cliff_width)

기준 대비. | 계산: delta_dchg_dVdQ_SOC0_cliff_width

트렌드: cliff 폭 Δ: early=-0.855 → late=-2.963 (Δ=-2.108, -0.9554Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

cliff/mid Δ (delta_dchg_dVdQ_SOC0_to_mid_ratio)

기준 대비. | 계산: delta_dchg_dVdQ_SOC0_to_mid_ratio

트렌드: cliff/mid Δ : early=-0.3241 $\rightarrow$ late=-3.065 (Δ =-2.741, -1.2711/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

충전 IC 면적합 (chg_dQdV_area_sum)

dQ/dV 피크 면적 합. | 계산: chg_dQdV_area_sum

트렌드: 충전 IC 면적합: early=64.1 $\rightarrow$ late=53.2 (Δ =-10.89, -3.505Ah/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

방전 IC 면적합 (dchg_dQdV_area_sum)

dQ/dV 피크 면적 합. | 계산: dchg_dQdV_area_sum

트렌드: 방전 IC 면적합: early=69.5 $\rightarrow$ late=62.47 (Δ =-7.031, -2.55Ah/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

RCF (RCF)

Q_0.5C / Q_C/3. | 계산: routine/RPT Q.

트렌드: RCF: early=0.9708 $\rightarrow$ late=0.9504 (Δ =-0.02038, -0.0011661/100cyc) $\rightarrow$ flat · stable | aging 기대=decrease | vs=stable

RCF 기울기/100 (RCF_slope_100)

RCF 변화율. | 계산: first-last slope.

트렌드: RCF 기울기/100: early=-0.01762 $\rightarrow$ late=-0.01762 (Δ =+0, -4.244e-181/100cyc/100cyc) $\rightarrow$ flat · stable | aging 기대=decrease | vs=stable

PER (PER)

$\eta/(\Delta I \cdot R)$. | 계산: eta_SOC50/(dl*R).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

η @SOC20 (eta_SOC20)

저SOC 과전위. | 계산: C/3 vs 0.5C.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η @SOC50 (eta_SOC50)

중SOC 과전위. | 계산: C/3 vs 0.5C.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η @SOC80 (eta_SOC80)

고SOC 과전위. | 계산: C/3 vs 0.5C.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η max (eta_max)

최대 과전위. | 계산: max η (SOC).

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η mean (eta_mean)

평균 과전위. | 계산: mean η .

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

η argmax SOC (eta_argmax_SOC)

η 최대 SOC. | 계산: argmax.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

η 저SOC 기울기 (eta_slope_lowSOC)

저SOC η 기울기. | 계산: slope low band.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff scale (Reff_scale)

유효 R 스케일. | 계산: shape fit scale.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

Reff fit R2 (Reff_shape_fit_r2)

Reff 형상 fit 품질. | 계산: r2.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff 잔차 SOC20 (Reff_resid_soc20)

Reff 모델 잔차. | 계산: resid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff 잔차 SOC50 (Reff_resid_soc50)

Reff 모델 잔차. | 계산: resid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

Reff 잔차 SOC80 (Reff_resid_soc80)

Reff 모델 잔차. | 계산: resid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

l_{∞} 정규화 (l_inf_norm)

CV 잔류전류 정규화. | 계산: $l_{\infty} / l_{\text{ref}}$.

트렌드: l_{∞} 정규화: early=0.2833 $\rightarrow$ late=0.2833 ($\Delta=+0$, n/a) $\rightarrow$ flat · context | aging 기대=either | vs=context

펄스 1s 샘플수 (pulse_sample_count_1s)

$t \leq 1s$ 샘플 수. | 계산: quality.

트렌드: 펄스 1s 샘플수: early=5 $\rightarrow$ late=7 ($\Delta=+2$, +0.8399count/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

펄스 전류 안정도 (pulse_current_stability)

$\text{std}(I) / |I|$. | 계산: quality.

트렌드: 펄스 전류 안정도: early=0.03731 $\rightarrow$ late=0.04793 ($\Delta=+0.01062$, +0.0055921/100cyc) $\rightarrow$ increasing · opposite_aging | aging 기대=decrease | vs=opposite_aging

rest 충분성 (rest_sufficiency)

휴지 길이/품질. | 계산: quality.

트렌드: rest 충분성: early=3 $\rightarrow$ late=3 ($\Delta=+0$, -1.678e-171/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

레그 완전성 (leg_completeness)

충방전 레그 완전성. | 계산: quality.

트렌드: 레그 완전성: early=0.9999 $\rightarrow$ late=0.9999 ($\Delta=+7.647e-05$, -0.019181/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

완화 완성도 max (relax_completeness_max)

SOC별 최대 완화 완성도. | 계산: max.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

샘플/mV (samples_per_mV)

전압 해상도 샘플밀도. | 계산: dqdv quality.

트렌드: 샘플/mV: early=0.3577 → late=0.3477 (Δ =-0.01002, -0.0093521/mV/100cyc) → flat · context | aging 기대=either | vs=context

OCV V_{∞} SOC80 (ocv_V_inf_soc80)

고SOC 무한시간 OCV. | 계산: recovery/rest V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV V_{∞} SOC50 (ocv_V_inf_soc50)

중SOC OCV. | 계산: V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV V_{∞} SOC20 (ocv_V_inf_soc20)

저SOC OCV. | 계산: V_{∞} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV spread 20–80 (ocv_spread_20_80)

SOC20–80 OCV 폭. | 계산: V_{80} - V_{20} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV spread 50–80 (ocv_spread_50_80)

SOC50–80 폭. | 계산: V_{80} - V_{50} .

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

OCV spread 20–50 (ocv_spread_20_50)

SOC20–50 폭. | 계산: V50-V20.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV 평행이동 (ocv_parallel_shift)

OCV 곡선 평행 시프트. | 계산: block OCV shift.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV 폭 압축 (ocv_spread_compression)

spread 축소. | 계산: Δ spread.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV80 Δ (delta_ocv_V_inf_soc80)

기준 대비. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV50 Δ (delta_ocv_V_inf_soc50)

기준 대비. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

OCV20 Δ (delta_ocv_V_inf_soc20)

기준 대비. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

OCV spread Δ (delta_ocv_spread_20_80)

기준 대비 폭 변화. | 계산: delta.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

LAM 곡선 proxy (LAM_curve_proxy)

V-Q scale 축소 proxy (절대 LAM% 아님). | 계산: (1-s)*100.

트렌드: LAM 곡선 proxy: early=-3.11 → late=-17.48 (Δ =-14.37, -5.717%/100cyc) → decreasing · context | aging 기대=either | vs=context

LLI 곡선 proxy (LLI_curve_proxy)

Q 오프셋 proxy (절대 LLI% 아님). | 계산: offset/Qmax*100.

트렌드: LLI 곡선 proxy: early=-0.6932 → late=-3.752 (Δ =-3.059, -1.527%/100cyc) → decreasing · context | aging 기대=either | vs=context

R 곡선 proxy (R_curve_proxy)

곡선 fit dR proxy. | 계산: fit_dR.

트렌드: R 곡선 proxy: early=1.345 → late=4.453 (Δ =+3.108, +1.383m Ω /100cyc) → increasing · matches_aging | aging 기대=increase | vs=matches_aging

방전 fit scale (dchg_fit_scale)

기준 대비 scale s. | 계산: 3-param fit.

트렌드: 방전 fit scale: early=1.031 → late=1.175 (Δ =+0.1437, +0.057171/100cyc) → flat · stable | aging 기대=decrease | vs=stable

방전 fit offset (dchg_fit_offset)

Q 오프셋. | 계산: 3-param fit.

트렌드: 방전 fit offset: early=-0.4977 → late=-2.694 (Δ =-2.196, -1.096Ah/100cyc) → decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

방전 fit dR (dchg_fit_dR)

저항 항. | 계산: 3-param fit.

트렌드: 방전 fit dR: early=1.345 → late=4.453 (Δ =+3.108, +1.383m Ω /100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

fit 잔차 RMS (dchg_fit_residual_rms)

잔차 rms. | 계산: RMS(resid).

트렌드: fit 잔차 RMS: early=6.331 → late=23.51 (Δ =+17.18, +8.114mV/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

fit 잔차 max (dchg_fit_residual_max)

잔차 최대. | 계산: max|resid|.

트렌드: fit 잔차 max: early=55.92 → late=152.7 (Δ =+96.79, +42.63mV/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

잔차 argmax SOC (dchg_fit_residual_argmax_SOC)

잔차 최대 SOC (방전 DOD→SOC 변환). | 계산: argmax residual.

트렌드: 잔차 argmax SOC: early=99.46 → late=100 (Δ =+0.5351, +0.1287%/100cyc) → flat · context | aging 기대=either
| vs=context

fit R2 (dchg_fit_r2)

곡선 fit 품질. | 계산: r2.

트렌드: fit R2: early=0.9998 → late=0.997 (Δ =-0.002754, -0.0012961/100cyc) → flat · context | aging 기대=either
| vs=context

fit corr(s,o) (dchg_fit_corr_s_o)

scale-offset 상관 (축퇴 지표). | 계산: corr.

트렌드: fit corr(s,o): early=0.4825 → late=0.7115 (Δ =+0.229, +0.073681/100cyc) → increasing · context | aging
기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

잔차 argmax DOD (dchg_fit_residual_argmax_DOD)

잔차 최대 DOD. | 계산: argmax DOD.

트렌드: 잔차 argmax DOD: early=0.5351 → late=0 (Δ =-0.5351, -0.1287%/100cyc) → decreasing · context | aging
기대=either | vs=context

$\Delta Q(V)$ min (dQV_min)

전압빈 ΔQ 최소. | 계산: histogram.

트렌드: $\Delta Q(V)$ min: early=-4.585 → late=-15.61 (Δ =-11.02, -4.474Ah/100cyc) → decreasing · context | aging
기대=either | vs=context

$\Delta Q(V)$ mean (dQV_mean)

ΔQ 평균. | 계산: mean.

트렌드: $\Delta Q(V)$ mean: early=-3.345 → late=-10.54 (Δ =-7.197, -2.89Ah/100cyc) → decreasing · context | aging
기대=either | vs=context

$\Delta Q(V)$ var (dQV_var)

ΔQ 분산. | 계산: var.

트렌드: $\Delta Q(V)$ var: early=0.5675 → late=6.055 (Δ =+5.487, +2.321Ah²/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

$\Delta Q(V)$ log-var (dQV_log_var)

log10 분산. | 계산: log10(var).

트렌드: $\Delta Q(V)$ log-var: early=-0.2461 → late=0.7821 (Δ =+1.028, +0.40671/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

$\Delta Q(V)$ skew (dQV_skew)

왜도. | 계산: skew.

트렌드: $\Delta Q(V)$ skew: early=-0.139 → late=-0.6781 (Δ =-0.5391, -0.25111/100cyc) → decreasing · context | aging
기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

$\Delta Q(V)$ kurtosis (dQV_kurtosis)

첨도. | 계산: kurtosis.

트렌드: $\Delta Q(V)$ kurtosis: early=-0.9841 $\rightarrow$ late=-0.693 (Δ =+0.2911, +0.088961/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

ΔQ argmin V (dQV_argmin_V)

ΔQ 최소 전압. | 계산: argmin.

트렌드: ΔQ argmin V: early=3.125 $\rightarrow$ late=3.05 (Δ =-0.07524, -0.03163V/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

dQ/dV SNR (dqdv_snr)

IC 신호대잡음. | 계산: snr estimate.

트렌드: dQ/dV SNR: early=73.9 $\rightarrow$ late=73.51 (Δ =-0.3834, +1.1051/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

데이터 품질점수 (quality_score)

추출 품질 종합. | 계산: quality gates.

트렌드: 데이터 품질점수: early=1 $\rightarrow$ late=1 (Δ =+0, -0.0062210-1/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

전압 노이즈 σ (v_noise_sigma)

전압 노이즈 추정. | 계산: noise sigma.

트렌드: 전압 노이즈 σ : early=0.023 $\rightarrow$ late=0.02313 (Δ =+0.0001232, -0.0004959V/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

$\Delta Q(V)$ 기준 사이클 (dQV_ref_cycle)

ΔQ 비교 기준 사이클. | 계산: ref cycle id.

트렌드: $\Delta Q(V)$ 기준 사이클: early=3 $\rightarrow$ late=3 (Δ =+0, +6.051e-16cyc/100cyc) $\rightarrow$ flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

온도 가용 (temperature_available)

Temp 컬럼 유효 여부. | 계산: bool→float.

트렌드: 온도 가용: early=0 → late=0 ($\Delta=+0$, +00/1/100cyc) → flat · context | aging 기대=either | vs=context

Q_relax 유의 (Q_relax_significant)

완화 용량 유의 플래그. | 계산: threshold flag.

트렌드: Q_relax 유의: early=1 → late=1 ($\Delta=+0$, -0.17570/1/100cyc) → decreasing · context | aging 기대=either | vs=context

fit 축퇴 플래그 (dchg_fit_degenerate_flag)

scale bound 포화 등. | 계산: degenerate flag.

트렌드: fit 축퇴 플래그: early=0 → late=0 ($\Delta=+0$, +00/1/100cyc) → flat · context | aging 기대=either | vs=context

η 유효 (eta_valid)

과전위 계산 유효 플래그. | 계산: eta_valid.

트렌드: 데이터 부족 | aging 기대=either | vs=n/a

fade b 표준오차 (fade_exponent_b_se)

fade 지수 표준오차. | 계산: fit se.

트렌드: fade b 표준오차: early=0.04593 → late=0.04593 ($\Delta=+0$, -9.821e-181/100cyc) → flat · context | aging 기대=either | vs=context

ΔQ 유효 V범위 (dQV_valid_V_range)

ΔQ 집계 전압폭. | 계산: Vmax-Vmin used.

트렌드: ΔQ 유효 V범위: early=0.783 → late=0.783 ($\Delta=+0$, +8.036e-17V/100cyc) → flat · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

fade 지수 b (fade_exponent_b)

SoHQ power-law 지수. | 계산: SoHQ fit.

트렌드: fade 지수 b: early=0.6376 → late=0.6376 ($\Delta=+0$, -1.968e-161/100cyc) → flat · context | aging 기대=either | vs=context

fade 지수 a (fade_exponent_a)

power-law 계수. | 계산: SoHQ fit.

트렌드: fade 지수 a: early=0.002927 → late=0.002927 ($\Delta=+0$, -1.192e-181/100cyc) → flat · context | aging 기대=either | vs=context

fade fit R2 (fade_fit_r2)

fade 적합도. | 계산: r2.

트렌드: fade fit R2: early=0.9466 → late=0.9466 ($\Delta=+0$, -2.176e-161/100cyc) → flat · context | aging 기대=either | vs=context

fade SoHQ0 (fade_sohq0)

fit 초기 SoHQ. | 계산: intercept.

트렌드: fade SoHQ0: early=97.32 → late=97.32 ($\Delta=+0$, -1.701e-14%/100cyc) → flat · context | aging 기대=either | vs=context

knee 사이클 (knee_cycle_bw)

bilinear knee 위치. | 계산: broken-stick SoHQ.

트렌드: knee 사이클: early=30 → late=30 ($\Delta=+0$, -3.132e-15cyc/100cyc) → flat · context | aging 기대=either | vs=context

knee 심각도 (knee_severity)

전후 기울기 차이. | 계산: slope_after-before.

트렌드: knee 심각도: early=0 → late=0 ($\Delta=+0$, +01/100cyc) → flat · stable | aging 기대=increase | vs=stable

지표 설명 · 트렌드 (M02Ch103)

knee 전 기울기 (knee_slope_before)

knee 이전 fade 기울기. | 계산: bilinear.

트렌드: knee 전 기울기: early=-0.1609 → late=-0.1609 ($\Delta=+0$, +3.194e-17%/cyc/100cyc) → flat · stable | aging
기대=decrease | vs=stable

knee 후 기울기 (knee_slope_after)

knee 이후 fade 기울기. | 계산: bilinear.

트렌드: knee 후 기울기: early=-0.0284 → late=-0.0284 ($\Delta=+0$, +6.934e-18%/cyc/100cyc) → flat · stable | aging
기대=decrease | vs=stable

knee fit R2 (knee_fit_r2)

knee 적합도. | 계산: r2.

트렌드: knee fit R2: early=0.981 → late=0.981 ($\Delta=+0$, -1.193e-161/100cyc) → flat · context | aging
기대=either | vs=context

PE activity 패턴 (LAM_PE_pattern_score)

NCM activity/isolation (절대 LAM% 아님). | 계산: mode_weights LAM_PE.

트렌드: PE activity 패턴: early=0.5654 → late=0.8602 ($\Delta=+0.2947$, +0.10610-1/100cyc) → increasing · matches_aging |
aging 기대=increase | vs=matches_aging

NE 패턴 점수 (LAM_NE_pattern_score)

NE 관련 패턴 (Si-on-Gr에선 보조). | 계산: mode_weights LAM_NE.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

contact_loss (contact_loss_score)

옴/스택/접촉 증거 합. | 계산: RQ growth 등 가중합.

트렌드: contact_loss: early=0.8179 → late=0.9342 ($\Delta=+0.1164$, +0.08150-1/100cyc) → increasing · matches_aging |
aging 기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

LLI 패턴 (LLI_pattern_score)

CE·slippage·offset 기반. | 계산: mode_weights LLI.

트렌드: LLI 패턴: early=0.374 → late=0.6624 (Δ =+0.2883, +0.12770–1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

계면 R 패턴 (interface_R_score)

Rct·VE 등 계면저항. | 계산: mode_weights interface_R.

트렌드: 계면 R 패턴: early=0.453 → late=0.8481 (Δ =+0.3951, +0.16420–1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

고체확산 패턴 (solid_diffusion_score)

A_diff·PER·RCF. | 계산: mode_weights solid_diffusion.

트렌드: 고체확산 패턴: early=0.5259 → late=0.7582 (Δ =+0.2324, +0.032690–1/100cyc) → flat · stable | aging
기대=increase | vs=stable

SE 분해 패턴 (SE_decomposition_score)

CE ↓ · Rct ↑ 등 SE 분해 가설. | 계산: mode_weights SE_decomposition.

트렌드: SE 분해 패턴: early=0.1477 → late=0.346 (Δ =+0.1983, +0.07820–1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

마이크로쇼트 패턴 (microshort_score)

자기방전·CE 기반 soft-short 가설. | 계산: mode_weights microshort.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

임피던스 패턴 (impedance_pattern_score)

총 임피던스 성장 패턴. | 계산: mode score.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

지표 설명 · 트렌드 (M02Ch103)

수송제한 패턴 (transport_limitation_score)

rate/수송 제한 패턴. | 계산: mode score.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

플레이팅 리스크 (plating_risk_score)

Li plating 위험 패턴. | 계산: mode score.

트렌드: 데이터 부족 | aging 기대=increase | vs=n/a

contact_loss 신뢰도 (contact_loss_confidence)

패턴 점수 신뢰도. | 계산: evidence coverage.

트렌드: contact_loss 신뢰도: early=0.52 → late=0.448 ($\Delta=-0.072$, +0.0040770–1/100cyc) → flat · context | aging
기대=either | vs=context

LAM_PE 신뢰도 (LAM_PE_confidence)

패턴 신뢰도. | 계산: confidence.

트렌드: LAM_PE 신뢰도: early=0.751 → late=0.616 ($\Delta=-0.135$, -0.11740–1/100cyc) → decreasing · context | aging
기대=either | vs=context

LLI 신뢰도 (LLI_confidence)

패턴 신뢰도. | 계산: confidence.

트렌드: LLI 신뢰도: early=0.72 → late=0.504 ($\Delta=-0.216$, -0.092220–1/100cyc) → decreasing · context | aging
기대=either | vs=context

PE lean (PE_side_score)

0.75 · LAM_PE + feature + FC-OCP Δ hits. | 계산: electrode_side v1.3.

트렌드: PE lean: early=0.4961 → late=0.7451 ($\Delta=+0.249$, +0.055840–1/100cyc) → increasing · matches_aging | aging
기대=increase | vs=matches_aging

지표 설명 · 트렌드 (M02Ch103)

contact_stack (contact_stack_score)

$\approx$ contact_loss (R-centric). | 계산: clip(contact_loss).

트렌드: contact_stack: early=0.8179 $\rightarrow$ late=0.9342 (Δ =+0.1164, +0.08150-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

NE 가설 (NE_side_score)

contact $\times$ Si co-sign. | 계산: electrode_side.

트렌드: NE 가설: early=0.1799 $\rightarrow$ late=0.2826 (Δ =+0.1027, +0.061030-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

shared 모드 (shared_side_score)

LLI/interface 등 공유 모드 평균. | 계산: shared modes mean.

트렌드: shared 모드: early=0.3838 $\rightarrow$ late=0.6556 (Δ =+0.2718, +0.066590-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

Si co-sign (si_cosign)

저SOC hyst·Q_relax·mech/chem·CV 동시 신호. | 계산: SI_NE_COSIGN boost.

트렌드: Si co-sign: early=0.2 $\rightarrow$ late=0.4 (Δ =+0.2, +0.110-1/100cyc) $\rightarrow$ increasing · matches_aging | aging 기대=increase | vs=matches_aging

dominant 마진 (dominance_margin)

1위-2위 점수차. | 계산: top-second.

트렌드: dominant 마진: early=0.2895 $\rightarrow$ late=0.1891 (Δ =-0.1004, +0.027590-1/100cyc) $\rightarrow$ increasing · context | aging 기대=either | vs=context

FC-OCP 피크 hits (pe_peak_hits)

충전 dQ/dV $\leftrightarrow$ 합성 FC-OCP 매칭 수. | 계산: unique nearest $\pm$ 60mV.

트렌드: FC-OCP 피크 hits: early=0 $\rightarrow$ late=0 (Δ =+0, -0.2208count/100cyc) $\rightarrow$ decreasing · context | aging 기대=either | vs=context

지표 설명 · 트렌드 (M02Ch103)

FC-OCP hits Δ (pe_peak_hits_delta)

기준 대비 hits 증가. | 계산: hits-hits0.

트렌드: FC-OCP hits Δ : early=0 $\rightarrow$ late=0 ($\Delta=+0$, $-0.2208\text{count}/100\text{cyc}$) $\rightarrow$ decreasing · opposite_aging | aging
기대=increase | vs=opposite_aging

FC-OCP hits (alias) (fc_ocp_hits)

pe_peak_hits 별칭. | 계산: same as pe_peak_hits.

트렌드: FC-OCP hits (alias): early=0 $\rightarrow$ late=0 ($\Delta=+0$, $-0.2208\text{count}/100\text{cyc}$) $\rightarrow$ decreasing · context | aging
기대=either | vs=context

FC-OCP hits Δ (alias) (fc_ocp_hits_delta)

pe_peak_hits_delta 별칭. | 계산: same as pe_peak_hits_delta.

트렌드: FC-OCP hits Δ (alias): early=0 $\rightarrow$ late=0 ($\Delta=+0$, $-0.2208\text{count}/100\text{cyc}$) $\rightarrow$ decreasing · opposite_aging |
aging 기대=increase | vs=opposite_aging

전극진단 신뢰도 (electrode_confidence)

coverage·분리·OCP 가용성. | 계산: $0.35\text{cov}+0.35\text{sep}+0.30\text{ocp}$.

트렌드: 전극진단 신뢰도: early=0.825 $\rightarrow$ late=0.8572 ($\Delta=+0.03219$, $+0.042750-1/100\text{cyc}$) $\rightarrow$ flat · context | aging
기대=either | vs=context